# TFPT — Architecture and the $E_8$ Compiler

The two axioms $\{c_3, g_{\text{car}}\}$, the derivation map, and the Coxeter–cyclotomic $D_5 \times A_3 \to E_8$ construction

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**What this document is about**

The architecture layer: how the two axioms $c_3 = \frac{1}{8\pi}$ and $g_{\text{car}} = 5$ build the Coxeter–cyclotomic compiler — the carrier $C^+{=}D_5$, the family geometry $\mathbb{P}^1 \setminus \mu_4{=}A_3$, the $\mu_4$ glue $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$, the EM fixed point $\alpha^{-1}$ (with its ablation), and the whole number alphabet $16, 40, 41, 48, 240, 248$ as carrier traces.

**The TFPT document set — what is treated where**

**Plain language:** TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\text{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and four companions** (plus one standalone working note), best read in order:

0. `introduction` — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). *(entry point)*
1. `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point $\alpha^{-1}$, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.
2. `tfpt_2_standard_model` — the SM in one $\varphi_0$-ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, $Q_+$ cohomology), the worked closures.
3. `tfpt_3_e8_audit_bootstrap` — the seven $E_8$ slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. `tfpt_4_frontier` — honest status of $\eta_B$, $m_p/m_e$, Koide, dark matter, full QG.
5. `tfpt_5_redteam` — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

**Companions: R.** `tfpt_research_contracts` — the open interfaces $(v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}})$ as numbered contracts; **H.** `tfpt_horizon_readouts` — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** `origin_theory` — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation); **S.** `tfpt_safeguards` — the verification discipline: every mechanism that defends a claim against coincidence and numerology (status calculus, no-free-pattern, the over-determination map, the firewall, frozen predictions + null model, the independent Wolfram and Lean paths, the red team).

**Working notes: N1.** `note_e8_gaussian_code` — the Gaussian code bridge: $E_8$ over $\mathbb{Z}[i]$ via Construction A over the extended Hamming code $[8, 4, 4]$, the canonical four-bit quotient $L/(1{+}i)L \cong \mathbb{F}_2^4$, and the $G_{31}$ quartic companion (exact algebra, machine-certified).

**You are reading document 1 (Architecture).**

**TFPT status at a glance — read this card the same way in every document**

**Two structural interfaces, one compiler.** P1 is a primitive reflection-positive seam kernel with unit winding; its normalization is $c_3 = \frac{1}{8\pi}$. From this datum and the five-slot carrier $g_{\text{car}} = 5$ (axiom P2), the discrete compiler $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ($\alpha^{-1} = 137.0359992$) and the scale grammar are read off as fixed points. The numerical readouts $c_3$ and $g_{\text{car}}$ are elementary symmetric readouts of the anchor $a = (1, 1, 2)$. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below. Existing reductions share the four-mark geometry and anchor $a = (1, 1, 2)$; both W-bridge premises are derived at finite level (`v1014/v1015`), modulo the external MMST identification. The additional P1 determinant-response identification is separate. The Architecture chapter "Shared origin" consolidates these scopes (`REC20261105C3`).

**What is closed, and what genuinely remains.** The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

| Interface | Question | Status |
|---|---|---|
| $v_{\text{geo}}$ | the one metrology unit ($= 1/\sqrt{G} = m/\mu$); CLOSED METROLOGY ($\mathbb{R}_+$ torsor; one dim. calibration by dimensional analysis; No-Unit v153/v725) | closed metrology [O] |
| $G_{\text{net}}$ | simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1,1) \rangle \cong (E_8)_1$ | [C] |
| $F_{\text{transfer}}$ | one functor, four typed interfaces (Koide, $\eta_B$, axion, $m_p/m_e$) | [C] |

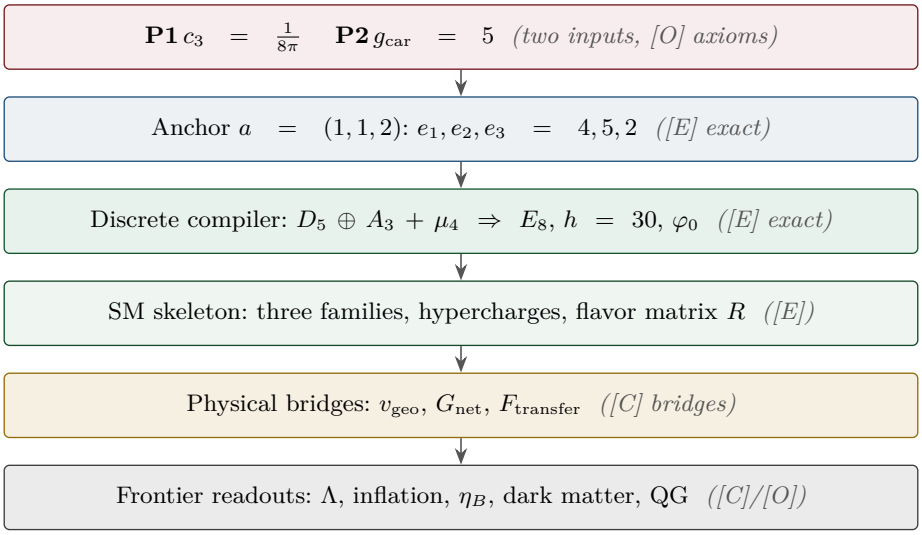
Beside that compiler rest sits the strict-physical-TOE rest: the ten-component $\text{Rest}_{\text{TOE}}$ / the eight gates of `TFPT.TOE.COMPLETE.01` [O] ($\text{TOE}_{\text{Complete}} :=$ AND(T1..T8); not a closure).

**The classical covariant field equation carries no free dimensionless Newton coupling under the named QFT/entanglement premises [C].** The classical metric-sector field equation is no longer only an $R{+}R^2$ readout: the fixed-volume entanglement equilibrium (Jacobson; Faulkner et al.), run with TFPT's atoms, gives the *full* covariant Einstein equation $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ with *both* coefficients fixed — $c_3^{-1} = 8\pi$ (no free dimensionless Newton dial; the thermodynamic origin $2\pi/\eta$ *coincides* with the geometric one $|\mathbb{Z}_2|2\pi\chi$ via $|\mu_4| = |\mathbb{Z}_2|\chi(S^2) = 4$, so $c_3$ is triply over-determined — anchor, geometry, thermodynamics, `v358`) and $\Lambda$ from $\alpha$ ($\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$, `v60`); the Einstein tensor (not Ricci) is forced by Lovelock, so matter conservation is an output (`v359`). The residual here is only the equation-of-state status (an external candidate action — Bianconi's entropic action, arXiv:2408.14391 — is quantified in `v473` without changing the typing) and the one unit $v_{\text{geo}}$; the global measure (C7 = `QG.AMB.01`) is now a [C] redundancy (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann.

The seam-net structural theorem $G_{\text{net}}$ (= `SEAM.EQUIV.01`: the raw seam *is* the holomorphic $(E_8)_1$ net) now carries two named routes (route split 2026-07-22, no status change in substance): the MMST route `SEAM.EQUIV.MMST.01` is *closed modulo cited theorems*; the twistor route `SEAM.EQUIV.TWISTOR.01` (Costello–Li, prepared by `CELEST.SEAM.01`) and the unconditional parent stay [O]. The lattice and index witnesses and the cited scaling and OS theorems establish the stated conditional route; they do not leave every old prerequisite untouched. The later charged MMST manuscript proves its uniform Green, edge graph, crossed-product and relaxed TEL-B estimates and retains the named microscopic local-algebra exhaustion condition for the full source. Its P1 corollary separately requires identification of the determinant-line response with the additive transitive four-channel response. The declared operational seed already supplies state and time data. The physical question concerns applicability of these theorems to the same source. The unconditional seam parent and T1–T8 keep their ledger statuses. Under the conformal-net hypotheses, rotation invariance makes the geometric deck a consequence (`v335`). The nonperturbative quantum-gravity measure `QG.AMB.01` is no longer carried as an open item: it is a [C] *redundancy* (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann (and TFPT is moreover rigorously gap-decoupled from the general Euclidean-QG conformal-factor problem, margin $1.648 > 0$, `v76`/`v330`). So the honest residual is the one unit $v_{\text{geo}}$ plus the typed $F_{\text{transfer}}$ interfaces, above the cited-theorem ceiling on the seam and with `QG.AMB.01` a [C] redundancy. Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels ($U_{\text{wall}}, G_{\text{metric}}, F_{\text{frontier}}$) are retained only for ledger continuity — the live residual is $v_{\text{geo}} \oplus F_{\text{transfer}}$, with the seam $G_{\text{net}}$ closed modulo cited theorems on its MMST route (parent [O]) and `QG.AMB.01` a [C] redundancy.

*Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.*

[E] exact / machine-proven [C] conditional [O] open / axiom [X] kill test




# Contents

**I Architecture and the derivation map** **4**

**1 How to read this document** **5**

**2 Shared origin of the seam normalization and carrier data** **7**

**3 Headline: the Pascal compiler on five carrier slots** **20**
3.1 $g_{\text{car}} = 5$ as the Pascal closure . . . 21
3.2 One master moment generates 40, 41, 240, $\delta_2$ . . . 22
3.3 Compact $E_8$ compression and the role of $\nu^c$ . . . 22

**4 The $D_5 \times A_3$ compiler: $E_8$ as a closure, not an input** **22**
4.1 Carrier $= D_5 = \mathfrak{so}_{10}$ (proved) . . . 24
4.2 Family $= A_3 = \mathfrak{su}_4$ from $\mathbb{P}^1 \setminus \mu_4$ (proved, with the metric settled) . . . 24
4.3 The $\mu_4$ glue and the $5/4 + 3/4 = 2$ discriminant theorem (proved) . . . 25
4.4 Glue-norm arithmetic, and three readings of 41, 60, 120 . . . 26
4.5 What the compiler outputs for the Standard Model and our reality . . . 27

**5 Grammar I — the carrier grammar $(3+2)$** **29**
5.1 Dual-root dictionary (everything from two charges) . . . 29
5.2 Carrier arithmetic (proved identities) . . . 30

**6 Grammar II — the seed grammar (one decoder $u$)** **30**

**7 Grammar III — the action grammar (exponentials of $\alpha_\star^{-1}$)** **31**
7.1 The EM closure: $\alpha_\star^{-1}$ as an explicit fixed point (made explicit here) . . . 31
7.2 The decuple bridge (new, exact headline) . . . 36
7.3 The full power tower: $x^1, x^5, x^{10}$ on the carrier graph $K_5$ . . . 36
7.4 Hubble is the square root of $\Lambda$ (corollary, exact) . . . 36
7.5 The $\sim 123$ orders of the cosmological constant . . . 37
7.6 The ladder is binomial; $2g_{\text{car}} = \binom{g_{\text{car}}}{2}$ singles out $g_{\text{car}} = 5$ . . . 37
7.7 Discriminant action theorem: the ladder reads the hypercharge discriminant . . . 37
7.8 A two-dimensional action lattice . . . 38

**8 The Einstein normaliser as a corrected tree value** **38**

**9 Λ-anchored metrology closure, and two anchors** **38**

**10 The landscape: $g_{\text{car}} = 5$ is overdetermined** **39**

**11 The $E_8$ carrier atlas — stated unambiguously** **40**
11.1 Level 1 — original $E_8$ data of the series ([E]) . . . 40
11.2 Level 2 — new exact counting identities ([E]) . . . 40
11.3 Level 2.5 — the even-flip transition atlas (a falsifiable program) . . . 40
11.4 Level 2.7 — the carrier Coxeter theorem (the strongest new bridge) . . . 41
11.5 Level 2.8 — positive-root moment: $|R^+(E_8)| = \mathrm{Tr}_{S^+} X^2 = 5!$ . . . 42
11.6 Level 2.9 — the 2·3·5 Coxeter cyclotomic compiler (meta-theorem) . . . 42
11.7 Level 3 — conjectural geometric readings ([C]) . . . 42

**12 $\mathbb{Z}_6$ monodromy, flavor and CP readouts** **43**
12.1 CP as triple seed variance; a closed observable quadrilateral . . . 43

**13 Mass hierarchy as a hexagon with one extra turn** **44**
13.1 Lepton flavor decuple: the charged-lepton determinant carries $u^{10}$ . . . 44
13.2 Why the top Yukawa is ≈ 1: the $L = 0$ anchor (Froggatt–Nielsen ladder) . . . 45

**14 Inflation interface** **45**

**15 Further simple bridges** **46**

**16 Closures that the architecture rests on** **46**

**17 Status summary and open audit points** **47**

**18 Consolidated picture** **48**

**II The Coxeter–cyclotomic compiler and the explicit $E_8$ construction** **50**

**Theorem 1** **50**

**Theorem 2** **51**

**Theorem 3** **52**

**Theorem 4** **52**

**Theorem 5** **52**

**Theorem 6** **53**

**Theorem 7** **54**

**Theorem 8 — anchor microcode** **58**

**Master statement** **59**

**Appendix: explicit $E_8$ construction** **59**

**III Appendix: ablation of the EM fixed point** **61**

**Shared source and transfer update: 3 November 2026** **63**

**Appendix: computational verification** **71**

# Part I
# Architecture and the derivation map

**Abstract**

This document develops the *architecture layer* of TFPT: the two axioms, the derivation map, and the explicit $E_8$ construction. The thesis is that the bridge layer of TFPT is governed by *three nested grammars*: a **carrier grammar** (from $3+2$ come charges, families, hypercharge, $\Omega_{\rm adm} = 48$, $b_1 = 41/10$), a **seed grammar** (one seed $u = \varphi_0^{\rm ret}$ generates the Cabibbo angle, birefringence, the baryon fraction and the reactor angle), and an **action grammar** (the large scale ratios are exponential actions of $\alpha_\star^{-1}$ with rungs $1/g_{\rm car}$, 1, 2). Two new exact results sharpen the action grammar: the *decuple bridge* $\rho_\Lambda/\bar{M}_{\rm Pl}^4 = \frac{3}{4\pi^2} e^{2\delta_{\rm ph}} [\, v_{\rm geo}/(5\beta_{\rm rad}^2 \bar{M}_{\rm Pl})\,]^{10}$ (the cosmological constant is, in essence, the tenth power of the stripped electroweak scale, with $10 = 2g_{\rm car}$), and the Hubble corollary $H_0/\bar{M}_{\rm Pl} = e^{-\alpha_\star^{-1}}/(2\pi\sqrt{\Omega_\Lambda})$. We also give the exact corrected tree form of the Einstein normaliser $\xi = \frac{3}{4}/(1 + 9/128\pi^3)$, the dual-root dictionary that generates all hypercharges and transport cusps from $y_\pm = \{1/2, -1/3\}$, the $\Lambda$-anchored metrology closure, the overdetermination of $g_{\rm car} = 5$, and the $E_8$ carrier atlas in three clearly separated levels. Section 4 adds the strongest structural result: the carrier code is the $D_5 = \mathfrak{so}_{10}$ half-spinor (Aut $= W(D_5)$, order 1920; $R_4 =$ Clebsch graph), the family geometry $\mathbb{P}^1 \setminus \mu_4$ is an $A_3$, and $E_8$ is the explicit unimodular glue closure of $D_5 \oplus A_3$ — with the discriminant-form norms $q(D_5) = 5/4$ and $q(A_3) = 3/4$ summing to the $E_8$ root norm 2 and coinciding with the TFPT constants $\delta_2/\delta_{\rm top}^2$ and $\xi_{\rm tree}$. Status markers used throughout: [E] exact identity, [E] Lie/lattice theorem, [E] formalised, [E] numerical fixed point, [C] physical/conditional, [O] axiom/open. (All markers are explicit; the legacy `[M]`/`[C]` compatibility aliases have been removed.)

# 1 How to read this document

**Notation convention.** We write $D_5 \oplus A_3$ for the *root-lattice* direct sum (used in the glue/lattice statements: discriminant forms, $\mu_4$ glue, $E_8$ as an even unimodular lattice), and $D_5 \times A_3$ for the *group/representation* branching (used for subalgebra embeddings and $E_8$-rep decompositions). The objects are the same atoms; the symbol marks whether the statement is lattice-theoretic ($\oplus$) or branching/representation-theoretic ($\times$).

Nothing here introduces a new primitive assumption; it reorganises existing outputs and proves the cross-links. The whole bridge layer rests on two inputs,

$$\boxed{c_3 = \frac{1}{8\pi} \ \text{(axiom P1)}, \qquad g_{\text{car}} = 5 \ \text{(axiom P2)}.} \tag{1}$$

**Declared structural postulates and their conditional reductions**

The compiler uses two declared structural inputs. Their algebraic consequences and their conditional source reductions are separate statements; the axiom markers record the interface:

(P1) Boundary kernel.
A single oriented seam $\Sigma$ carries a primitive, reflection-positive boundary kernel with unit winding $[u_\Sigma] = 1$; its self-linking normalisation fixes $c_3 = 1/(8\pi)$. *Status:* primitive postulate. *Organizing identity (2026-08-06, markers unchanged):* the closed normalization $I \cdot \beta_{\text{angle}} \cdot c_3 = 4 \cdot 2\pi \cdot \frac{1}{8\pi} = 1$ holds exactly with the Jones index $I = 4 = |\mu_4|$ a $c_3$-free theorem and $\beta_{\text{angle}} = 2\pi$ independently measured — the identity *explains* the value, it does not derive it; the open bridge is the even modular-response normalization (`P1.INDEX.KMS.01` **[O]**, [`verification/v813_p1_index_kms.py`]). *Hardenable by:* the Calderón projector / determinant-character chain (the P1 hardening) and the Lean carrier-rigidity development (`TfptCarrier.*`, boundary-polarization and Calderón interfaces).

(P2) Carrier interface.
This splits into an *axiom* and a *theorem*, which must not be conflated:

- **P2**$_{\text{interface}}$ **[O] (axiom):** the seam is read out by a five-slot carrier $g_{\text{car}} = 5$ (3 colour + 2 weak) — a postulate (the readout interface).
- **P2**$_{\text{algebra}}$ **[E] (theorem):** *given* the carrier, hypercharge ($y_- = -\frac{1}{3}$, $y_+ = \frac{1}{2}$), rigidity and the family count follow from the idempotent/lattice axioms — machine-verified in Lean (`Hypercharge.lean`, `LatticeRigidityGeneral.lean`, `Rigidity.lean`).

So P2 is an *interface axiom* whose *algebraic consequences are a formalised theorem* — "axiom or theorem?" has the precise answer: interface axiom, algebra theorem.

The structured P1 kernel and P2 readout remain the declared interfaces. Existing relative derivations explain their numerical data under named geometric and source hypotheses; see Section 2. Every other number in this note — $N_{\text{fam}} = 3$, $\Omega_{\text{adm}} = 48$, $b_1 = 41/10$, $\alpha_\star^{-1}$, the flavor matrix, the $E_8$ glue, $A_s$ — is a consequence of (P1),(P2).

**Anchor-first refinement.** The two inputs are themselves the elementary symmetric polynomials of the single parabolic anchor $a = (1, 1, 2)$ (the exponents-at-infinity of the splitting type $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$): $e_1(a) = 4 = |\mu_4|$, $e_2(a) = 5 = g_{\text{car}}$, $e_3(a) = 2 = |\mathbb{Z}_2|$, with $c_3 = 1/(2e_1(a)\pi) = 1/(8\pi)$. The power sums $p_n(a) = 2 + 2^n$ then give $|R(E_8)| = p_1p_2p_3 = 240$, $\dim E_8 = 240 + (p_4 - p_3) = 248$ and the binary ladder $p_{n+1} - p_n = 2^n$. So the input set reduces to $\{a = (1, 1, 2),\ \pi\}$. [`verification/v23_anchor_generator.py`] The whole ladder is one affine map: $p_{n+1} = 2p_n - 2$ identically, with unique fixed point $2 = |\mathbb{Z}_2|$ and the compiler quintet $(4, 6, 10, 18, 34)$ as the $T$-orbit of 4 — the affine normal form of Theorem 8 ([`verification/v832_anchor_flavor_checksum.py`]).

**The parabolic self-code of the anchor ([`verification/v566_parabolic_anchor_selfcode.py`]).** The anchor is not only generated — it is *coded twice* by one operator structure that already lives in the compiler. The sheet-diamond direction operators $U = Q\,\text{diag}(1, 0, 0)$ and $V = Q\,\text{diag}(0, 1, 1)$ (v218; $\text{Spec}\, V = \{0, 1, 2\}$) generate the *full* 2|1 parabolic (line-stabiliser) algebra $\mathcal{A}$ (dimension 7, word determinant $-81$, multiplicatively closed; a perturbed $Q$ breaks it) **[E]**. The Lagrange projector onto $V$'s dominant eigenvalue gives $H = I + \frac{1}{2}V(V - I)$ with spectrum *exactly* $\{1, 1, 2\} = a$, and $\chi_H(x) = x^3 - 4x^2 + 5x - 2$ carries the double code $(4, 5, 2) = (|\mu_4|, g_{\text{car}}, |\mathbb{Z}_2|) = (\dim_{\mathbb{R}}(GL_3/P), \dim(\mathcal{A}/J), \dim J)$ — flag, Levi quotient and radical of $\mathcal{A}$ **[E]**. A uniqueness theorem closes the family: the line-stabiliser structure polynomial factorises as $(x - m)(x^2 - mx + 1)$, and all-positive-integer spectra force $m = 2$

— $a = (1, 1, 2)$ is the *only* anchor this parabolic family can code [E]. Honest typing: this is a *compression* of one operator system, not new independent evidence (no-free-pattern), and it is *not* a $P2$ reduction — the route "$g_{\text{car}} = \dim(\mathcal{A}/J)$ as a derived value" is conditional on the *No Carrier Reconstruction* audit [O]; the axioms' markers are untouched. *Stage A of that audit is since executed* ([`verification/v567_carrier_free_q_reconstruction.py`]): with the v11 carrier budgets *forbidden* (rows $(4, 5, 6)$, columns $(9, 5, 1)$, $\chi_Q$ with its 5), the carrier-free inputs — sheet involution, winding normalisation, $A_3$ exponent spectrum, monodromy budget, seam-line demand — admit exactly 28 parabolic reconstructions, and **all** 28 **code the same anchor** $\chi_{H'} = x^3 - 4x^2 + 5x - 2$: the anchor self-code is carrier-free; the single family invariant $81 = N_{\text{fam}}^4$ then pins $Q$ to a two-element residual family ($q_{32} \in \{2{=}|\mathbb{Z}_2|, 0\}$) on which every classical carrier-free certificate coincides — the one separating certificate is exactly the forbidden column budget. Verdict RECONSTRUCTS-TO-FAMILY; the audit relocates to `GATE.QGEO.01` (derive the geometric inputs themselves) and to the one binary sheet-coupling selector [O] — which is since *located and given a canonical witness* ([`verification/v568_bit_selector_ladder.py`]): the twins generate different integral orders, and the *ladder self-consistency* (the dominant mode reproduces its own eigenvalue as the geometric ladder $(1, 2, 4) = (2^0, 2^1, 2^2)$, the binary AnchorLadder) selects the compiler $Q$ uniquely on all 28 members; the split geometry ($q_{32} = 1$) is excluded by the demands already imposed; *why* the ladder self-consistency should hold physically stays open (the RP-selection slot) [O]. Three follow-ups since executed: the *kinematic RP form* on the twins ([`verification/v572_rp_bit_form.py`]) is spectrally blind (identical inertia for every declared corpus state; never PSD — no kinematic positivity selection exists) but *integrally anchor-visible*: the Smith towers of the anchor-state Gram separate the twins in one slot (18 vs 2, a factor $9 = N_{\text{fam}}^2$) — the bit shows to the RP form only through anchor-coupled states [E]; and *Stage B is narrowed* ([`verification/v574_stageb_winding.py`]): the involution *type* is forced by a rank theorem (a two-dim-positive involution cannot carry the binary trace code), and the winding normalisation $N_{\text{fam}} = 3$ is *forced* by the double self-consistency — the ladder selects one member for every $N \le 3$ (higher $N$ die on the exponent demand), and sat $= N^4$ holds only at $N = 3$ ($81 = 3^4$; $N = 1, 2$ give $32 \neq 1$, $512 \neq 16$): two of the five declared geometric inputs relocate to consistency-forced; `GATE.QGEO.01` is narrowed, not closed [E]/[C]. The involution-*existence* input is since *derived* ([`verification/v590_involution_existence.py`]): the integral $V$-commuting involutions are exactly the spectral sign group $(\mathbb{Z}/2)^3$ (rigidity, both twins), and the v574 type theorem plus two named demands (seam reversal, zero-mode fixing) single out the explicit $S_0$ with exact $\Sigma$-signature uniquely — existence is unconditional; the two demands' continuum derivation stays the named open [E]/[C]. The *interacting* follow-up ([`verification/v584_ext_source_response.py`]) then separates the twins *directionally*: rank-one Ext-source coupling on the Gibbs pencil has exact rational flip thresholds $J_c(s) = -1/(v^T K(s)^{-1} v)$ whose numerators are exactly the twins' pencil factors; in the crossed phase positive coupling $J_c \le 15/16 < 1$ heals exactly the compiler twin (through either Ext arrow), while the alt twin cannot be flipped by any positive coupling in the open unit box — the healing direction is strictly positive, consistent with the v534 toy RP selection; the response pole sits on the golden polynomial $s^2 - s - 1$ (observed). Honest: a response-level selection, not an OS/RP positivity theorem — the alignment-bit contract stays open [E]/[C]. A *geometric candidate* for the whole `GATE.QGEO` realization has since emerged ([`verification/v597_mu4_cover_model.py`]): the $\mu_3$-cover $y^3 = x^4 - 1$ of $\mathbb{P}^1$ branched at $\mu_4$ hits four compiler fingerprints at once — $H_1$ is a rank-3 module over the Eisenstein integers; the $D_4$ rotation closes projectively via the deck ($r^4 = \omega$); the cusp-class group is $(\mathbb{Z}/3)^4$ of order 81 (the saturation index, with the per-mark $\mathbb{Z}/3$ mechanism the self-code program asked for); and the unique invariant hermitian form has Lorentz signature $(1, 2)$ — so the RP-positivity route would become Riemann bilinear positivity if the identification completes. The dictionary since stands at *three* exact entries ([`verification/v598_cover_dictionary.py`]): the winding operator $U$ is a single-cusp Gram, the $\mu_4$ sheet coupling $Q_-$ is the *difference of the opposite cusp Grams* (the diamond diagonal), and the binary AnchorLadder $(1, 2, 4)$ is identity plus the diagonal sum; the grading half ($Q_+$, hence $V$ and $Q$) is absent from 7296 scanned canonical combinations — the named suspect for the missing generator is the anti-holomorphic $D_4$ reflection (the real structure); the gate does not move [E]/[C]. The suspect has since been *constructed* ([`verification/v599_real_structure.py`]: $R$ unique up to scale with $R^2 = +1$; its grading functor closes the three gaps at char level, with the anchor spectrum $(1, 1, 2)$ appearing; the invariant form has $\det J = 8$ and Lorentz signature $(1, 2)$), and the *joint embedding has landed* ([`verification/v600_joint_embedding.py`]): a $\Gamma$-even degree-2 pair $(U^*, V^*)$ is $\mathbb{Z}[\omega]$-integral, $\mathrm{GL}_3$-conjugate to the compiler pair (explicit $g$, exact), restricts to *integer* matrices on the saturated real fixed lattice, and its seven-word $\mathbb{Z}$-order has saturation index $81 = 3^4$ with the exact v566 divisor pattern — the self-code algebra lives on the cover; canonicity stays open and the gate still does not move [E]/[C]. The canonicity question has since been answered ([`verification/v601_equivariant_dual.py`]):

the J-adapted cusp projectors give a presentation that is $D_4$-*equivariant by construction* (mark choice = winding choice) and realizes the *dual* parabolic exactly (conjugate to the transposed compiler pair), and the v590 Klein four-group lives integrally on the curve [E]/[C]. *The cover program at a glance (v597–v605).* Nine rounds close into one narrative: the $\mu_3$-cover $y^3 = x^4 - 1$ carries the fingerprints (v597), the dictionary (v598), the real structure with $R^2{=}1$ and the Lorentz form $\det J = 8$ (v599), the joint pair with saturation index $81 = 3^4$ (v600, Lean-certified), the $D_4$-equivariant canonical presentation and the integral Klein group (v601), the J-adjoint duality, the verbatim anchor formula $H = 1 + V(V-1)/2$ and exact Weil positivity (v602), the *real* seam line with the geometric seam-reversal demand and the first per-mark $\mathbb{Z}/3$ witness (v603), the anti-isometry $M^\dagger J M = J^T$ with the integer form $J_{\text{fix}}$ ($\det 72$), the full four-frame order of index $27 = 3^3$ and the torsion localization (81 at the marks, 27 at infinity) (v604), and the RP translator $\theta_{\text{corpus}} = J\text{adj} \circ \text{inner}(c)$ with $14c$ integral (v605). The gate does *not* move: the residual is the conformal seam identification (`QGEO.SYM.01`), the zero-mode selection (the zero-mode line is *not* $R$-real — honest negative), and the geometric identity of $c$ [E]/[C]/[O]. The honest decomposition (spectrum ablation): freeing the exponent spectrum too, the one surviving double-selfconsistent family is exactly $\{1, 2, N\}$ with the compiler shape — the self-consistency forces the *sheet* spectrum $\{1, 2\}$ (the binary code) and the shape, and the $A_3$ exponent demand contributes exactly *one* datum, $N = 3$ [E]. And the redundancy ablation shrinks the input list further: relative to the self-code demands, the monodromy budget ($M$) and the seam-line demand ($L$) are *redundant* (both dropped, the class widens to 1274 members and the self-code still picks the compiler $Q$ uniquely), and the maximal ablation drops ($E$) too: the winding column $3(1,1,1)$ plus integrality plus the self-code already pin the compiler alone. Even the winding *form* is now located: dropped entirely, the self-code leaves a four-column family of which exactly one is uniform — the compiler; the winding demand *is* the uniformity choice (1 of 4). What remains declared: the involution *existence*, the column *uniformity*, and $N_{\text{fam}} = 3$ (independently anchored in the $E_8$ glue, $240 = 16 \times 5 \times 3$) [E]. On the bit itself, the anchor-Hamiltonian *Gibbs pencil* delivers the first dynamical distinction: the compiler twin's RP kernel degenerates at a finite critical temperature $s^* = (8 - 4\sqrt{19})/15$ (determinant constant $81^2$), the alternative twin's never does — and the critical mode is the seam-row, sheet-flow-fixed direction (the $A_2$-Ext carrier): a rank-drop marker, not yet a positivity selection, but the resonance with the alignment bit now has a shared carrier [E].

## 2 Shared origin of the seam normalization and carrier data

**What is being explained.** P1 supplies a primitive reflection-positive kernel on an oriented seam with unit winding. The number $c_3$ is its normalization. P2 declares the five-slot $3 + 2$ readout; its algebraic consequences and its conditional source reductions are already available. Calling both entries "axioms" records these interfaces and does not erase those reductions.

**One marked geometry, several typed readouts.** In the flat spherical $\mathbb{Z}_2$ branch class, each branch point has deficit $\pi$. Gauss–Bonnet gives $n\pi = 4\pi$, hence four marks ([`verification/v216_marks_gauss_bonnet.py`]). For the common divisor $D = \mu_4 \subset \mathbb{P}^1$,

$$\operatorname{rank} H_1(\mathbb{P}^1 \setminus D) = 3, \qquad h^0(\mathcal{O}(D)) = 5.$$

The functions and cycles belong to the same marked geometry ([`verification/v189_riemann_roch_carrier.py`], [`verification/v197_rr_carrier_clifford_d5.py`]). They are different spaces: meromorphic functions have four independent principal parts, while meromorphic differentials have the residue-sum constraint and dimension three. These dimensions alone do not identify a physical colour/weak $3 + 2$ split. The square position of the marks is additional to their count; in the flag-symmetric branch it is equivalent to $\tau = i$ ([`verification/v512_seam_tau_flag.py`]).

**The anchor and the two numerical readouts.** The typed parabolic splitting $a = (1, 1, 2)$ gives

$$e_1(a) = 4, \quad e_2(a) = 5, \quad e_3(a) = 2, \qquad c_3 = \frac{1}{2\pi e_1(a)} = \frac{1}{8\pi}, \qquad g_{\text{car}} = e_2(a) = 5.$$

The uniqueness of three positive integer weights summing to four is exact; the Deligne extension, Birkhoff–Grothendieck splitting and stability route explain the typing for the specified connection

([verification/v491_p2_partition_corollary.py], [verification/v499_p2_weights_deligne_bg.py]). Later analytic descent of the actual $A_0^\star$ system gives finite $S_4$ monodromy, so a positive invariant Hermitian form exists by finite-group averaging. Historical analytic-monodromy and unitarizability obligations are therefore not a current independent numerical gap. Identifying that mathematical module and its frame with the actual physical source retains its stated scope.[1]

**Two normalization arguments with different hypotheses.** Unit seam winding gives $\Delta = 2\pi$ and the relative Chern normalization $c_3 = \Delta/(16\pi^2) = 1/(8\pi)$. The one-sided normal-sphere argument gives the same value.[2] The index/KMS argument instead uses $I = 4$, $\beta_{\text{angle}} = 2\pi$ and $c_3 = 1/(I\beta_{\text{angle}})$. Its arithmetic is exact ([verification/v813_p1_index_kms.py]); the determinant-line response must also be identified with the additive transitive four-channel response. The charged MMST paper explicitly retains this additional operator identification even after its net/KMS rigidity theorem. Equal channel weights alone do not supply the variation formula.

**Which former prerequisites have already been derived.** The W-bridge anchor direction is derived by the winding/triple lock ([verification/v1014_bridge_refinements.py]). Its democratic response is derived for the finite factorized abelian collar model ([verification/v1015_axiom_core_closure.py]): $r = (1, 1, 1)$ at sizes $12, 16$, with character-dependent and deck-breaking controls. Both finite W-bridge premises are closed; its finite axiom core is zero modulo the externalized MMST identification. This does not remove the continuum determinant identification in the P1 route. The earlier MMST route retains its conditional ledger status. The later charged master theorem proves its implication under microscopic local-algebra exhaustion, which remains unproved for the frozen collar. The unconditional parent retains its ledger marker.

The 2 November R102 result also removes evenness, unimodularity, fixed rank eight and minimality as inputs to a stronger selection theorem: for the standard positive $D_{n+1} \oplus A_{n-1}$ pairing, an integral full-rank extension containing a primitive $A$ discriminant projection forces $n \mid 4$, hence $n = 4$ for $n \geq 3$, and the whole completion is $E_8$. The standard positive normalization and the actual one-sided integral primitive insertion remain its source hypotheses. Full doubled locality alone does not imply that one-sided hypothesis. The complementary metric rigidity theorem assumes unimodularity, so the two results cannot be used to prove each other's premises in a circle.

**The common explanation and its boundary.** The marked seam supplies the geometric setting; its cycles, sections, parabolic anchor and primitive normalization explain the shared $4, 3, 5, 2$ data in their specified branches. P1, its scalar normalization and P2 are three levels of this construction. The operational original seed already contains the local net, state, reflection, time flow and collar generator. Existing conditional OS and common-action reconstructions must be reused with those data. Source identification means proving the named maps for the same source; it does not mean constructing state and time anew from the scalar $c_3$. The consolidated original-source audit is `REC20261105C3`; it changes no axiom or T1–T8 ledger marker.

**The Gaussian code bridge and the quartic half (2026-08-03) — the code section closes two loops**

Two exact-arithmetic results tie the binary code layer directly to the $E_8$ lattice and its complex $\mu_4$ structure. *The Gaussian code bridge* ([verification/v689_gaussian_code_bridge.py], 26 checks, exact): the $\mu_4$ complex structure $J$ makes $E_8$ a rank-4 module over the Gaussian integers $\mathbb{Z}[i]$, and the quotient $E_8(\mathbb{Z}[i])/(1+i)$ *is* $\mathbb{F}_2^4$ — the information space of the Reed–Muller code RM$(1,3)$ (Smith normal form exact; standard model $|\det| = 256 = 2^8$, SNF $[1^4, 2^4]$, presentation-independent). The 240 roots fall $15 \times 16$ over the fifteen nontrivial classes (the zero class provably empty), each class is 4 of the 60 G31 lines ($60 = 15 \cdot 4$), the compiler clock $\sigma = c^4$ acts as the RM$(1,3)$ *family permutation* (3-cycle plus fixed anchor — the v638 info-bit action) with a residual gauge group of order 18, and the coordinate block is

[1] TFPT Gesamtsynthese, 28 September 2026, the fourfold cover section; recovered source proof in `REC20261105C3`.
[2] Boundary-kernel source, primitive seam theorem; original source paths and spans are listed in `REC20261105C3`.

the sum-of-families label $F_1{+}F_2{+}F_3$. Three must-fail controls fire (naive placement, non-$\pi J$ placement, plain $\mathbb{Z}[i]^4$) **[E]**. *The quartic half* ([verification/v690_quartic_half.py], 22 checks, no floats): the vanishing of the $W(E_8)$ invariants of degrees $\{2, 14, 18, 30\}$ under holomorphic $\mu_4$ restriction is proven symbolically but *trivial* — $W(J\alpha) = -i\,W(\alpha)$ kills every $J$-orbit contribution for all $d \not\equiv 0 \bmod 4$ (honest typing); the substance is the surviving half: the restrictions $F_8|, F_{12}|, F_{20}|, F_{24}|$ are G31-invariant (60/60 mirrors), algebraically independent (exact Jacobian), with Molien-unique degrees of product $46080 = |G31|$ — by Chevalley (cited) a system of *basic* invariants, i.e. the holomorphic restriction of the $W(E_8)$ ring generates the full G31 invariant ring; the Jacobian carries the 60 mirror lines point-verified. Controls: a non-root-preserving complex structure breaks the informative half, a root-preserving one keeps it — the statement is about the compiler-compatible conjugacy class **[E]**. No marker moves; both are structure statements of the code layer, registered as `E8.GAUSSCODE.01` and `E8.QUARTICHALF.01`. *The orbit packet lemma (2026-08-04):* the code bridge's number family closes as one lemma ([verification/v736_orbit_packet.py], 26/26, ORBIT-PACKET-EXACT; companion Lean module `OrbitPacket.lean`, 34 theorems, kernel-checked):

| | |
|---|---|
| $\sigma$ orbit census (Burnside) | $7 = (15 + 3 + 3)/3$ |
| canonical packet split | $240 = 5 \times 48$ (1 fixed + 4 moved, 12 Gaussian lines each, label-free) |
| carrier reading | packets $= 1 + 4 = 5 = g_{\mathrm{car}}$, $240 = g_{\mathrm{car}} \times 48$ |
| anchor | leader weights $(0, 1, 1, 2) \to (1, 1, 2)$, pair $e_6/e_7$ |
| $q = 16$ normal form | $240 = 16 \cdot 15, \quad 248 = 240 + 8$ |

*The circularity fence is part of the lemma:* P1 ($c_3 = 1/(8\pi)$) and P2 ($g_{\mathrm{car}} = 5$) come back from the code as *exact internal reconstruction* — the packet law reconstructs the axioms, it does *not* derive them; and the $1 + 4$ law is $\sigma$-*class*-specific (the 160 free order-3 elements of G31 give Burnside 5, not 7). The sharpening to a canonical intertwiner with the 48 regular five-cycles is *dead* on its preregistered criterion ([verification/v737_seam48_intertwiner.py], 24/24 machinery, SEAM48-DEAD: incidence distribution 0:96/1:64/2:64/3:16 instead of $\equiv 1$) — $240 = 48 \cdot 5$ remains a factorization *without* coordinatization **[E]**. *The symplectic code lemma (2026-08-04):* the quotient carries its own incidence geometry ([verification/v752_projective_hamming_incidence.py], 36/36, PROJ-HAMMING-EXACT; companion Lean module `ProjectiveHamming.lean`, 32 theorems): $h = H/4$ is $\mathbb{Z}[i]$-valued with hermitian Gram determinant 1 — the *unimodular* hermitian E8 lattice over $\mathbb{Z}[i]$ — and the reduction $\bar{h}$ makes $V = L/(1 + i)L$ a symplectic $\mathbb{F}_2^4$ (doubly-even $\Rightarrow$ alternating; explicit Darboux basis); on the 15 nonzero classes the hyperplane incidence matrix satisfies $BB^{\mathsf{T}} = 4I + 3J$ *exactly*, with design parameters 2/3/4/7/15/16 and eigenvalues $7^1(+2)^9(-2)^5$ (the multiplicities $(1, 9, 5)$ are the symplectic signature), the channel $K = B/7$ obeys $K^2 = (4/49)I + (45/49)\Pi_0$, and the NS/R parity is the point $y_0 = \mathrm{FSUM}$ of the geometry ($\chi_{\mathrm{NSR}} = \bar{h}(\cdot, y_0)$; $112 + 128 = 240$, $248 = 120 + 128$). The lemma also *resolves* the seam-48 kill: the distribution above is derived exactly (16× the $3 \times 5$ label table), and the $5 \times 48$ coordinatization exists via exactly one $\bar{u}$-invariant spread — the kill is explained (wrong blocks), not revised **[E]**. *The Arf/spinor compiler (2026-08-05):* the code layer now carries its canonical quadratic refinement ([verification/v774_arf_spinor_compiler.py], 46/46, ARF-SPINOR-EXACT; companion Lean module `ArfSpinorCompiler.lean`, 23 kernel theorems): of the exactly 16 refinements of $\bar{h}$ the frozen selector ($\sigma$-invariance, $q(A){=}1$, $q(F_\Sigma){=}0$) picks a *unique* $q^*$, which *is* the parity-lift weight form under $\iota\colon V \cong C_{\mathrm{even}}(5)$ (crosslink $\beta = \bar{h}$ exact in all 256 cells); $\mathrm{Stab}(q^*) \cong S_5$ makes $16 = 1 + \bar{5} + 10$ an *Arf geometry* ($Sp(4, 2) \cong S_6$; $K_6$ duads; $B = I + A_{KG(6,2)}$ with exact ovoid eigenvectors), and the hypercharge table $X = (-2, -2, -2, 3, 3)$ reproduces the `v310/v14` dictionary with the master-moment chain 120/40/41/240/248; the same $\sigma$ acts as the family 3-cycle $A_3^{\mathrm{Fam}}$ on message bits and as the 3+2 slot split $A_3^{\mathrm{PS}}$ on carrier legs — naming discipline, not a physics claim **[E]**. *The fired matter fence (2026-08-05):* the preregistered kill test of the code→matter reading *fired* ([verification/v775_gaussian_class_d5_purity.py], 18/18, ROOTCLASS-MIXED): no $D_5 \oplus A_3/SU(5)$ convention makes even one full set of 15 classes pure ($\geq 7$ classes are adjoint-side for *every* embedding, by the 128-spinor-root counting bound); the code→matter table — including the "empty mailbox = right-handed neutrino" reading — must not enter any main claim **[E]**. *Boundary code uniqueness (2026-08-05):* the non-circular selection chain exists ([verification/v776_boundary_hamming_uniqueness.py], 31/31, BOUNDARY-CODE-UNIQUE): among all doubly-even self-dual binary codes of length 8 (direct census 30, no Hamming/$E_8$ input) the boundary data — four $J$-paired marks, $\mathbb{Z}_3$ deck pair-coherence, and *one* deck-invariant $\mathbb{Z}_2$ (the product of the four side orientations) — select exactly one code; Construction A of it *is* $E_8$, and the deck packets output $g_{\mathrm{car}} = 1 + 4 = 5$. The circularity fence

above is thereby *relocated and shrunk*, not removed: the residuals named then were the doubly-even self-dual *universe premise* (R1), the semantic deck placement (R2) and the anchor naming (R3); all three moved in round 21 (see the Type-II and metaplectic entries below) — R1 is replaced by three physical axioms, R2 is placed, R3 is named; no P2-removal claim — the sharpened contract form is `SEAM.CODE.TYPEII.01` [C]. *The $d = 4$ forcing census (2026-08-05):* the glue pattern itself forces the four marks ([`verification/v781_one_object_clock_census.py`], $21/21 + 23/23$, ONE-OBJECT-PARTIAL+CLOCK-FIBER-DEAD): the TFPT glue conditions (cyclic isotropic glue, evenness, unimodularity $|H|^2 = 4d$) admit an even unimodular *diagonal* glue of $A_{d-1} \oplus D_{d+1}$ iff $d = 4$ (exact census $d = 2..12$; at $d = 4$ exactly the two `v92` Lagrangians with the explicit 240-root $E_8$ certificate; $d = 9$ dies at isotropy; both relaxation controls fire) — so $g_{\mathrm{car}} = d{+}1 = 5$ and $N_{\mathrm{fam}} = d{-}1 = 3$ are the two functor readouts of *one* four-point boundary object. *Honesty gate:* $c_3 = 1/(2\pi d)|_{d=4} = 1/(8\pi)$ is a corpus-legitimate *rewriting* of the anchor-first refinement ($e_1(a) = 4$), not a new derivation of P1 — P1 stays a primitive postulate. The companion clock census is an honest kill: no $\mathbb{Z}_{12} \times_{\mathbb{Z}_6} \mathbb{Z}_{30}$ realization can act on the 60 Gaussian lines (full line-compatible group order 23040, maximal element order 12) [E]. *The transition-bus torsor (2026-08-05):* the $240 = 16 \times 15$ reading is settled at its sharp level ([`verification/v782_e8_transition_bus.py`], 35/35, TRANSITION-BUS-TORSOR): the roots are 15 coordinate 8-frames; each class fiber is a canonical *torsor* under $v^{\perp}$ on $L/2L$ (no distinguished origin), the deck acts as translation by $v$, and the obstruction census is proof-grade — *no* deck-equivariant labeling $R(E_8) = V \times (V \setminus 0)$ exists; the frozen symplectic predicate reproduces all 57600 inner products, and the $A_{15}$ comparison quantifies what a strict transition bus would need (a canonical start state, which $E_8$ provably lacks) [E]. *The two-qubit Clifford identification (2026-08-05):* under one predeclared chart the 60 Gaussian lines *are* the 60 pure two-qubit stabilizer rays, entry-exactly over $\mathbb{Z}[i]$ ([`verification/v783_two_qubit_clifford.py`], 32/32, CLIFFORD-PARTIAL(H4)): $UG_{31}U^{-1}$ is an index-2 subgroup of the Clifford matrix group, $46080 = \mu_4 \times \mathrm{Pauli}_{16} \times Sp(4,2)$, classes = Pauli contexts and spreads = MUB pentads; the index bit is the Galois $\sqrt{2}$/Hadamard bit (not $J \leftrightarrow -J$, the honest H4 partial; identified in round 21 as the $\tau = i$ metaplectic bit — see below). Carrier mathematics only; the fired matter fence above is unaffected [E]. *The curve/code 2-torsion decider (2026-08-05):* the seam-curve realization of the code geometry is honestly *partial* ([`verification/v784_curve_code_2torsion.py`], 33/33, CURVE-CODE-PARTIAL): $(A[2], e_2) \cong (V, \bar{h})$ as bare symplectic spaces, but the correspondence dies at two exact layers — the deck acts syntheme-side ($[3,3]$-free) while the code acts duad-side, and the curve's canonical theta refinement is even (Arf 0, 1+6+9) while $q^*$ is odd (Arf 1, 1+5+10); joint census 0 with positive control 6 [E]. *The canonical outer bridge (2026-08-06):* the two kill layers above are the two faces of one outer twist, and that twist is canonical ([`verification/v796_duad_syntheme_bridge.py`], 24/24, BRIDGE-CANONICAL): the Sylvester duality $\beta$ (code odd forms $\to$ curve isotropic spreads) with $\beta\sigma = \tau\beta$ and $\beta(q^*) = S^*$ — the unique $\langle\tau, \rho\rangle$-fixed spread — exists and is unique up to the order-6 code joint stabilizer (census $18 + 18 \to 6 + 6$, one orbit including the $\tau/\tau^2$ swap); it resolves the Arf mismatch canonically ($q_\Theta$ pulls back constantly to $q^*_{\mathrm{even}} = q^* + \bar{h}(\cdot, A)$ — the anchor shift), transports $\bar{5}/10$ and 3+2 exactly onto the $S^*$-line geometry, sends the anchor transvection $t_A$ to the curve rotation $\rho = J$ on every census bridge, and types $\chi_{\mathrm{NSR}}$ as an honest exact two-variant torsor. The inner identification stays dead; structure labels only [E]. *Type II from the seam axioms (2026-08-06):* residual R1 of the boundary-code chain is replaced by three physical axioms ([`verification/v799_seam_code_typeii.py`], 19/19, TYPEII-FORCED): exhaustively over all 308,993 subspaces of $\mathbb{F}_2^8$ ($\dim \le 4$), mutual locality of the Construction-A boundary vertex words $\iff$ self-orthogonal, integer conformal spin $\iff$ doubly even, and holomorphy/index one $\iff$ self-dual — the survivor set is exactly the 30 $\hat{e}_8$ copies, and no boundary-legal non-Type-II code exists; the universe premise reduces to A1 (bose statistics / trivial monodromy), A2 (modular $T$-alignment), A3 (holomorphic index one), each a standard physical demand; imported lattice-VOA/theta ingredients cited and machine-verified in every instance. No P2 marker move [E] + [C] (the seam satisfies A1–A3). *The metaplectic $S$-lift (2026-08-06):* the missing Clifford coset is identified ([`verification/v798_seam_clifford_modular_s.py`], 27/27, MODULAR-S-PARTIAL): the per-binary-factor Hadamard *is* the metaplectic/Weil lift of modular $S$ at its fixed point $\tau = i$ and lands exactly in $\zeta_8 G_{31}$ ($S^2 = I$, $T^2 =$ Pauli parity, $(TK)^3 = \zeta_8 I$); the total Fourier $H \otimes H$ lies inside $G_{31}$ with metaplectic anomaly $(T_3 K_3)^3 = iI = J$ — *the deck is the anomaly of the total modular lift* (R2 placed), and $\zeta_8 = (1{+}J)/\sqrt{2}$ is the real half-deck swapping the seam pair $\{L, \zeta_8 L\}$; the strict uniqueness census over all 46080 coset elements is *empty* (no $\Theta$-real metaplectic involution fixes $q^*$ — the Arf defect is unavoidable), one $G_{31}$-conjugacy class survives retyped, and the $\sigma$-orbit Arf defects XOR to the anchor $A$ itself (R3 named, not fixed) [E]. *The torsor Fourier modes (2026-08-06):* the origin-free fibers are read at character level ([`verification/v800_e8_torsor_fourier.py`], 30/30, TORSOR-FOURIER-PARTIAL): the 120 projective Fourier rays are canonical (an origin shift is a global

sign), orthogonal, deck/Pauli-diagonal and $G_{31}$-monomial through $Sp(4,2)$; the carrier block — the 8 R-fibers' 64 rays — factorizes as $(1{+}5{+}10)\boxtimes(1{+}3)$ under $W(D_5)\times W(A_3)$ with $\sigma$ the family 3-cycle plus fixed anchor — the carrier Pascal at *character* level, provably impossible at point level; the honest cap: the weight and character bases are mutually unbiased (1/8 exactly), so the fired matter fence stands unmoved; the 45 deck-even modes are the 45 context Pauli observables under the certified chart **[E]**. *The jet-code normal form (2026-08-06):* the no-origin torsor becomes a ring-theoretic theorem ([`verification/v803_e8_jetcode_affine_nsr.py`], 18/18+15/15, JETCODE-EXACT+AFFINE-NSR-EXACT): because 2 ramifies in $\mathbb{Z}[i]$, $\mathbb{Z}[i]/(2)\cong\mathbb{F}_2[\epsilon]/(\epsilon^2)$ (unique iso) and $W=L/2L$ is *free* of rank 4 over the dual numbers — the mod-2 root system is a first-order jet space, the deck is the jet unit $J=1{+}\epsilon$, and among all 65536 sections of $0\to\epsilon W\to W\to V\to 0$ there are *zero* $\mu_4$-equivariant splittings: non-splitness *is* the no-canonical-origin obstruction; $240=15\times8\times2$ (address × error jet × orientation). The affine companion: $\mathrm{Stab}(\chi_{\mathrm{NSR}})=\mathrm{AGL}(3,2)$ of order $1344=|\mathrm{Aut}(H_8)|$ in the same natural 8-point action, and affine functions on the NS/R torsor reconstruct $\mathrm{RM}(1,3)=H_8$ — a typed internal reconstruction, not a P2 removal **[E]**. *The Petersen carrier machine (2026-08-06):* the $K_5$ edge geometry is certified ([`verification/v808_petersen_sixthroot.py`], $19/19+16/16$, PETERSEN-RADIAL-EXACT+SIXTHROOT-SPECTRAL-ONLY): edge charges $(-4,+1,+6)$ = the certified $SU(5)$ decuple, the edge graph is Petersen $\mathrm{SRG}(10,3,0,1)$, the flavor hexagon *is* its distance-2 shell edge-by-edge, and $\mathrm{spec}(P^6)=\{1,64/729,1/729\}$ *is* the deployed transport spectrum exactly — a graph-geometric origin for the $\{1,(2/3)^6,(1/3)^6\}$ triple; the sixth root is spectral-only (the corpus freezes the spectrum, not the basis) **[E]**. *[Dated correction, 2026-08-06 round 23: the sixth-root census had missed the deployed* `v221` *evaluation — $T_{v221}$, the exact spectral sum of* `v221`*'s transfer $(1-w)^6$, equals $B^6$ bit-exactly, so the verdict upgrades to* BITEXACT-VS-V221 *([`verification/v814_k5_sixstep_transport.py`]); the frozen* SPECTRAL-ONLY *stands as that probe's adjudication.] The doily incidence identity (2026-08-06):* the outer bridge sharpens ([`verification/v809_curve_code_doily.py`], 21/21, DOILY-EXACT): $NN^T=B+2I$ on the $15\times15$ duad–syntheme doily, $N/3$ with singular values $\{1,2/3,0\}$ — the recovery base rate 2/3 as the unique nontrivial singular value; every census bridge factors exactly $N_{\mathrm{code}}=P_\Delta^T N_{\mathrm{curve}}^T P_\Gamma$ (a transpose duality, not an isomorphism); the $(2/3)^6$ identification stays typed pending the six-step composition **[E]** *[decided 2026-08-06 round 23: clock-only — see the six-step entry below]. The cut code (2026-08-06):* the Arf split is the cut classification of $K_5$ ([`verification/v810_cutcode_moment.py`], $17/17+16/16$, CUTCODE-EXACT+MOMENT-INCIDENCE-PARTIAL): the cut map is injective-isometric on all 256 cells, $16=1+\bar{5}+10$ *is* $(0|5)+(1|4)+(2|3)$, Clebsch $40=5+20+15$ over the Arf shells with within-10 = Petersen; the $210=\binom{10}{4}$ map is decorative per the frozen rule (the equivariant bijection killed exhaustively); Arf/cut grading, not matter semantics **[E]**. *The six-step composition and the $T_{10}$ proposal (2026-08-06):* the $(2/3)^6$ question of the doily entry is decided ([`verification/v814_k5_sixstep_transport.py`], 23/23, SIXSTEP-CLOCK-ONLY+TRANSPORT10-PROPOSED): the rate arises canonically on *three* spatial routes (doily alternation, hexagon compression, hexagon circulation) but each carries a typed obstruction (no unit mode; the bipartite sign $-2/3$; the wrong space), and sixth-power blindness ($C_6$ vs $2K_3$) makes per-step data mandatory — the deployed six is the *clock* exponent (`v124`/`v486`, fixed eigenbasis); the canonical proposal $T_{10}=P_{10}^6$ in the $\Lambda^2$ basis passes all gates with the equitable `v774` quotient $=(Q_{\mathrm{Pet}}/3)^6$ bit-exact — a proposal, no deploy claim **[E]**. *The $(1{+}i)$ ramification ladder (2026-08-07):* the four documented roles of the ramified Gaussian prime $\pi_2=1{+}i$ are *one* machine-checked object with four rungs, each re-certified from scratch ([`verification/v833_gaussian_ramification_ladder.py`], 33/33, RAMIFICATION-LADDER-EXACT). *Rung 1, norm doubling:* $N(\pi_2)=2$, $(\pi_2^2)=(2)$; on the rebuilt Gaussian lattice $(1{+}J)^{\mathsf{T}}(1{+}J)=2I_8$ exactly, $\min L=4$ by census, so the zero class of $L/(1{+}i)L$ carries *no* root (0/240 directly), $240=15\times16$, each class 4 of the 60 Gaussian lines, elementary divisors $(1^4,2^4)$, index $16=N(\pi_2)^4$. *Rung 2:* $L/(1{+}i)L\cong\mathbb{F}_2^4$ with the deck acting *trivially* — the 4-bit address. *Rung 3, the non-split jet:* $\mathbb{Z}[i]/(2)\cong\mathbb{F}_2[\epsilon]/(\epsilon^2)$ is the unique ring iso among all 24 bijections, $W=L/2L$ is free of rank 4, the deck is the jet unit $1{+}\epsilon$, and exactly 0 of the 65536 sections are deck-equivariant (the `v803` obstruction re-proved); the 120 root pairs are exactly the $q=1$ shell of the plus-type form (136/120, Arf 0). *Rung 4, the metaplectic lift:* $\zeta_8=(1{+}i)/\sqrt{2}$, $(1{+}J)^2=2J$, $(SH)^3=\zeta_8 I$ per binary factor and $(T_3K_3)^3=iI$ = the deck (`v798`); the exact $\mathbb{Z}[\zeta_8]$ BFS census gives $|C_2/\mu_8|=11520=16\times720$ with *zero* Galois-mixed classes, hence $|C_2|=92160$ and $C_2=K\sqcup\zeta_8K$ (the `v783` chart theorem cited), with witness parities $\chi(H\otimes I)=1$ (the RM–CSS $\zeta_8$ phase bit) and $\chi(H\otimes H)=\chi(CZ)=0$. Controls: the split ring $\mathbb{F}_2\times\mathbb{F}_2$ admits 0 ring isos (2 *must* ramify) and the split self-extension admits 65536/65536 equivariant sections (non-splitness is contentful). One ladder — the same ramified prime read at $(\pi_2)$, $(\pi_2^2)$ and $\pi_2/\sqrt{2}$; a compression of `v689`/`v803`/`v783`/`v798`, no marker moves **[E]**. *The message ladder (2026-08-07):* the budget lines of this paper are *one formula* ([`verification/v844_message_doily_rank.py`], 23/23, MESSAGE-LADDER-EXACT): $M_n=15\cdot\Delta p_n=15\cdot2^n=(15,30,60,120,240)$ — the anchor's binary shift

(v832) times the 15 nonzero classes of $V = L/(1+i)L$ (v833) — with each rung realized by a *named structure of the same rebuilt object*: $M_0 = 15$ the address space; $M_1 = 30 = h(E_8)$, derived here *from the rebuilt root system* (exactly 8 simple roots, connected simply-laced Cartan matrix with det = 1 and the $E_8$ degree sequence, exact nonnegative integer simple-root coordinates, the highest root unique at height 29 with marks $\{2,2,3,3,4,4,5,6\}$, $h = 1+29 = 30 = |R|/8$); $M_2 = 60$ the Gaussian root lines ($\mu_4$ orbits, 4 per class); $M_3 = 120 = |R^+|$ (the sign pairs); $M_4 = 240 = |R|$ with the census $240 = 15 \times 16$ (zero class empty); and the coda $248 = 15 \cdot 16 + 8 = M_4 +$ rank against the actual census and the actual HNF rank. The structural doubling $60 \to 120 \to 240$ (each line exactly two sign pairs, each pair two roots) and the halving $60 \to 30 = |\mu_4|$ vs. rank are typed. Controls fire: the wrong anchor $(1,2,2)$ gives $(45, 105, 225, 465)$ — all four targets missed — and the wrong multipliers 14 (the weight-4 codeword count) and 16 (the class size) miss all four targets and break the coda **[E]**. *The Doily–Pascal rank theorem (2026-08-07):* the doily incidence identity above sharpens into a *rank theorem* ([verification/v844_message_doily_rank.py], 16/16, DOILY-PASCAL-RANK-EXACT): under the $K_6$ duad model of the 16 quadratic refinements (Arf census $6+10$, rebuilt by brute force over $2^{16}$), the duad–syntheme incidence $N$ of the Cremona–Richmond doily carries $NN^{\mathsf{T}} = B + 2I$ *entrywise* (all 225 cells) — duad collinearity *is* symplectic orthogonality, and the spectrum shift $7/2/-2 \to 9/4/0$ identifies the ovoid $(-2)$-eigenspace of v774 with the kernel: $\mathrm{charpoly}(NN^{\mathsf{T}}) = (x-9)(x-4)^9x^5$ exactly, so $\operatorname{rank} N = 10 = A_\Lambda = \binom{g_{\mathrm{car}}}{2}$ (the decuple sector), $\dim\ker N = 5 = g_{\mathrm{car}}$ (the five-slot carrier; the six ovoid indicators $3 \cdot \mathbf{1}_{O_q} - 1$ *span* it), multiplicity $9 = N_{\mathrm{fam}}^2$, top singular value $3 = N_{\mathrm{fam}}$ — *the P2 integers are the singular-value data of the Cremona–Richmond incidence.* The recovery value: $\mathrm{sing}(N/3) = \{1,\ 2/3^{\times 9},\ 0^{\times 5}\}$ exactly, $2/3 = (N_{\mathrm{fam}}-1)/N_{\mathrm{fam}}$; the six-step exponentiation identity gives exactly $(2/3)^6 = 64/729$ as the deployed $\lambda_2$, and the typed gap is re-affirmed with no upgrade: $1/729$ is *not* a root of the six-step charpoly — the deployed $(1/3)^6$ stays on the clock side (SIXSTEP-CLOCK-ONLY, cited). Both wrong-pairing controls (circulant, frozen-seed LCG; same row sums 3) fire with rank $13 \neq 10$ **[E]**. *The Arf–vacuum syndrome theorem and the code pencil (2026-08-07):* the flat counts and orbit splits of this code section compress into *one two-bit syndrome table* ([verification/v852_arf_vacuum_ovoid.py], 24/24, ARF-VACUUM-SYNDROME-EXACT). The Arf bit is a *position-independent direction bit*: for every affine flat $t+U$ of $(\mathbb{F}_2^4, \hbar)$ the sum $a(t+U) = \sum_{x \in t+U} q^*(x) \bmod 2$ equals $\hbar(u,v)$ of the underlying plane, for *all* 140 flats; the pair $(s_{\mathrm{vac}}, s_{\mathrm{Arf}}) =$ (origin bit, direction bit) splits the 140 flats exactly into 15/20/45/60 — the doily lines, the non-isotropic lines, and the two off-origin plane classes, i.e. the v834 counts as rows of *one* table. The code side is a *pencil*: $H = [15, 11, 3]$ (weight-3 words = the 35 lines, weight-4 = the 105 off-origin flats, set equalities) $\supset \{D = [15,10,4],\ C_{\mathrm{iso}},\ C_{\mathrm{non}} = [15,10,3]\} \supset K = [15,9,4]$, all pairwise intersections $K$, all pairwise sums $H$, syndrome fibers $4 \times 512$; the *canonicity lemma*: all 16 refinements differ from $q^*$ by simplex $(= H^\perp)$ words, hence induce *one* Arf functional on $H$; and puncturing/shortening $\mathrm{RM}(2,4) = [16,11,4]$ at the vacuum coordinate gives exactly $H/D$ — *the vacuum bit is the parity bit.* Controls fire (dot form: no refinements, ill-defined direction bit; Arf-0: table invariant but labeling broken, zero set $10 \neq 6$; frozen-LCG column permutation: parameters preserved, line labeling destroyed) **[E]**. *The bent phase code and the two CSS codes (2026-08-07):* the canonical $q^*$ is *bent* ([verification/v853_bent_css_mangoldt.py], 22/22, ARF-BENT-CSS-EXACT): its phase vector satisfies $\hat{s}_q = -4\, s_q$ on all 16 characters ($W^2 = 16I$; perfect autocorrelation $16\,\delta_{a,b}$ on all 256 pairs), the zero set $\{0\} \cup \overline{5}$ is a $(16,6,2)$ Hadamard difference set (every nonzero shift meets it in exactly 2 points — the arithmetic face of the ovoid), and $A_q = C_{\mathrm{iso}}^\perp = [15,5,6]$ carries the 31-word structural census (15 weight-8 = the characters, 10 weight-6 = the Arf-0, 6 weight-10 = the Arf-1 refinements). Two CSS codes follow: $A_q \subset C_{\mathrm{iso}}$ self-orthogonal gives $[[15,5,3]]$ (distance words = the doily lines), and $A_{16} = \mathrm{RM}(1,4) + \langle q^* \rangle = [16,6,6]$ inside $C_{16} = A_{16}^\perp = [16,10,4]$ gives $[[16,4,4]]$ whose 60 minimal words are *exactly* the 60 isotropic affine planes — the $s_{\mathrm{Arf}} = 0$ rows of the syndrome table; the *vacuum transformation* $[[15,5,3]] \to [[16,4,4]]$ is typed (one physical slot added, one logical qubit paid, distance $3 \to 4$ bought: shortening $A_{16}$ at the vacuum slot $= A_q$, puncturing $C_{16} = C_{\mathrm{iso}}$). The bent Gauss sum and Walsh eigenvalue are both $-4$; the 16 bent translates form a second Gram-$16I$ frame *mutually unbiased* against the 16 characters (all 256 cross products $\pm 4 = \sqrt{16}$) — the unbiased partner of the v800 torsor Fourier rays, reported with no upgrade. Controls fire (Arf-0 flips the Walsh sign to $+4$ and kills the $(16,6,2)$ set; the dot form has no bent seed and its $[16,6,6]$ extension dies at dimension $5 \neq 6$) **[E]**. *The ovoid decoder (2026-08-07):* the doily rank theorem above gains its *integral* layer ([verification/v852_arf_vacuum_ovoid.py], 23/23, OVOID-DECODER-EXACT): $N^{\mathsf{T}} v_a = 0$ for all six ovoid vectors $v_a = 3 \cdot \mathbf{1}_{O_a} - 1$ (each syntheme covers each letter once — kernel facts from perfect matchings; any five of the six independent, $\sum_a v_a = 0$); the Gram is $V^{\mathsf{T}}V = 36I - 6J$ entrywise; the Smith normal form is $\mathrm{SNF}(N) = \mathrm{diag}(1^{10}, 0^5)$ exactly over $\mathbb{Z}$ — rank 10, kernel 5, *no torsion*, with a det $= \pm 1$ minor exhibited; the Moore–Penrose inverse has the closed form $N^+ = N^{\mathsf{T}}/4 - J/36$ (all

four Penrose axioms in exact $\mathbb{Q}$), the carrier projector is $P_5 = I/2 - B/4 + J/12$ with $\operatorname{tr} P_5 = 5 = g_{\mathrm{car}}$ and $NN^+ = I - P_5$, and the closed-form decoder $\hat{m} = (I - P_5)m$ is exact on the predeclared test set (kernel modes invisible, $\operatorname{im} N$ modes fully visible) — the recovery channel now has an explicit integral inverse, not just a spectrum. The kernel *is* the $S_6$ standard representation ($\operatorname{tr}(P_5\Pi(\pi)) = \operatorname{fix}(\pi) - 1$ on all 11 conjugacy classes), and the mod-2 shadow $v_a \bmod 2 = q_a$ identifies $\ker_{\mathbb{F}_2}(N^{\mathsf{T}}) = A_q = [15, 5, 6]$ — *three rings, one kernel*: the $\mathbb{R}$ ovoid span, the torsion-free $\mathbb{Z}$ kernel, and the $\mathbb{F}_2$ code of the bent layer. The wrong-pairing control fires (torsion appears, the $\mathbb{F}_2$ kernel collapses to 4) **[E]**. *The simplex-Fourier character theorem (2026-08-08):* the census $240 = 15 \times 16$ (zero class empty) *is* a Fourier law ([`verification/v857_simplex_fourier_winding.py`], 26/26, SIMPLEX-FOURIER-EXACT + INTERTWINER-IDENTIFIED): the Walsh transform of the root multiplicity is $\hat{r} = (240, -16^{15})$, i.e. the uniform-nonzero channel $P = (J-I)/15$ with spectrum $\{1, (-1/15)^{15}\}$ and two-step contraction $1/225$ (exact in Fractions). Ramification (the zero class is empty at *every* odd theta level — one-line proof) and equidistribution (one value over the 15 nonzero classes at every level $\leq 16$) force the character theorem. Typed honestly: the identification is *spectral* (the packet grading $(52, 60, 128)$ and the code grading $15 \times 16$ do not refine each other); controls fire, and the law is invariant under the 24 explicit isometries and the other invariant placement $C^{**}$ **[E]**. *The G31 clock alphabet (2026-08-08):* the compiler clock alphabet is a *normal form* of $G_{31}$ ([`verification/v858_g31_clock_alphabet.py`], 19/19, CLOCK-ALPHABET-SELECTIVE): $\operatorname{Deg}(G_{31})/4 = \{2, 3, 5, 6\} = \{|Z_2|, N_{\mathrm{fam}}, g_{\mathrm{car}}, |R^+(A_3)|\}$ with $\gcd = 4 = |\mu_4|$, $\operatorname{lcm} = 120 = |R^+(E_8)|$, $\operatorname{lcm}/\gcd = 30 = h(E_8)$, and the audit identity $(\sum d/4)(\sum(d-1)/4) = 16 \times 15 = 240$ *realized* on the roots (the 60 reflections are the 60 Gaussian lines, 4 per class — the same $15 \times 16$ census as the code side); $(8, 12, 20, 24)$ is the *unique* integer quadruple with product 46080 and sum 64; the kill scan over all 607 rank-4 Shephard–Todd groups leaves $G_{31}$ the sole full-battery passer (weak-triple impostors and raw-sum-64 coincidences typed by name; honesty line: degree-only audits cannot see the reducible torus), and the $W(D_5) \times W(A_3)$ order coincidence is re-killed by computed centers, $|Z(G_{31})| = 4 = \langle J \rangle$ vs 1 (the structural refutation cited) **[E]**. *The global Gaussian shell theorem (2026-08-08, round 36):* the $-1/15$ census law above is now a *theorem for all shells* ([`verification/v875_shell_global_lean.py`], 15/15, SHELL-GLOBAL-THEOREM): on $V = L/(1+i)L$, for every $n \geq 1$, $\Theta_v(n) = (\Theta_L(n) - \Theta_0(n))/15$ for all $v \neq 0$ (the division exact), $\hat{\Theta}_u(n) = 16\,(16 \cdot \mathbf{1}_{2|n}\,\sigma_3(n/2) - \sigma_3(n))$, and at every *odd* $n$: $\hat{\Theta}_u(n)/\Theta_L(n) = -1/15$ *exactly* — no census cap. The three inputs, each warded: the $G_{31}$ class action is the 60 symplectic transvections generating $\mathrm{Sp}(4, 2)$ (order 720), *transitive* on the 15 nonzero classes and identical to the shell-module action; norm doubling by $(1+i)$ ($\Theta_0(2m) = \Theta_L(m)$, $\Theta_0(\mathrm{odd}) = 0$, exact from $2 = -i(1+i)^2$); and $\Theta_L = 240\,\sigma_3$ (classical Eisenstein $E_4$; the $n \leq 16500$ census retained as the exact ward). A non-transitive control subgroup admits an invariant *non*-uniform census (the law breaks — transitivity is load-bearing), and the even-shell formula differs correctly (7/135 at $n = 2$). The consequence chain is kernel-checked from the named hypothesis bundle in `TfptCarrier/GaussianShells.lean` (7 theorems, exact integer identities, standard axioms, no `sorry`) **[E]**. *The winding decoder of the clock alphabet (2026-08-08, round 36):* the Paper-2 winding quadratic $q_{\mathrm{wind}}(t) = t^2 - g_{\mathrm{car}}t + |Z_2|$ *decodes* the clock alphabet ([`verification/v874_feedback_decoder_lean.py`], 11/11, WINDING-DECODER-EXACT): $q(\{2, 3, 5, 6\}) = (-|\mu_4|, -|\mu_4|, |Z_2|, h(D_5)) = (-4, -4, 2, 8)$, with $q(2) = q(3)$ *forced* by the palindrome axis $g_{\mathrm{car}}/2$ and $2 + 3 = 5 = g_{\mathrm{car}}$; the decoder polynomial $D(y) = (y+4)^2(y-2)(y-8) = y^4 - 2y^3 - 48y^2 - 32y + 256$ carries the budget coefficients ($|Z_2|$, $\Omega_{\mathrm{adm}} = 48$, $2^{g_{\mathrm{car}}} = 32$, $2^{\mathrm{rank}(E_8)} = 256$); and the Fourier bridge lands: $|\mu_4|\,q(|Z_2|)/|R(E_8)| = 4 \times (-4)/240 = -1/15$ is *exactly* the nonzero-character Walsh eigenvalue of the census above ($-16 = |\mu_4|\,q(|Z_2|) = \hat{r}(u \neq 0)$). Sibling deformation directions collapse the bridge and alternative alphabets change the decoder; honest lines typed: the $-32$ coefficient alone is not selective (the selective object is the full vector), and the 6-vs-7 tension carries over verbatim. Type: exact *audit* theorem, not a physical functor — an explicit intertwiner would be needed to make it causal **[E]**.

**The finite compiler normal form $C_{\mathrm{fin}} = (V, \hbar, q^*, \sigma, \iota)$ (2026-08-07) — one object, the kill built in [E] ([`verification/v845_cfin_normal_form.py`])**

The four-bit symplectic compiler that this paper assembles piecewise — quotient census, Arf refinements, selector, parity lift, $S_5$ geometry, Pascal counting — is now *one* machine-checked object, read off the Gaussian quotient and not postulated (28/28, NORMAL-FORM-ASSEMBLED, SPEC v2 declared): $V = L/(1+i)L$ (census $240 = 15 \times 16$, zero class empty); $\hbar$ the descended symplectic form with a *found* family/anchor basis $(F_1, F_2, F_3, A)$ of Gram $J - I$, equal to the bit form on all 256 label pairs; $\sigma$ the descended order-3 deck ($\sigma^3 = \mathrm{id}$, symplectic, exactly three fixed nonzero classes $\{A, F_\Sigma, F_\Sigma{+}A\}$ and four free 3-orbits — the family cycle); $q^*$ the *unique* selector-picked quadratic refinement (exactly 16 refinements by brute force over $2^{16}$, Arf census $6 + 10$; $\sigma$-invariant: $4 \to q(A){=}1$: $2 \to q(F_\Sigma){=}0$: 1;

Arf type 1, $\sigma$-preserved); $\iota$ the parity lift, bijective onto the even five-slot code $C_{\text{even}}(5)$ with $\beta = \hbar$ on all 256 cells, $q_{\text{wt}} = q^*$, and $\sigma$ acting on the five carrier slots as the 3+2 split — message register and carrier register intertwined. *Derived from the one object:* $|Sp(4,2)| = 720$ by full census, $\text{Stab}(q^*) = 120$ acting *faithfully* as $S_5$ on the five, orbits $[1,5,10]$; $16 = 1+5+10$ and $15 = 5+10$; the Pascal reading (the $\iota$-weight census $(1,10,5) = (\binom{5}{0}, \binom{5}{2}, \binom{5}{4})$); the projective reading (the $\bar{5}$ is an *ovoid* of the doily in $PG(3,2)$: it meets every isotropic line exactly once); and the budget against the actual lattice objects: $240 = 16 \cdot 15$, 60 lines $= 4 \cdot 15$, $30 = 2 \cdot 15 = h(E_8)$ (rung 1 of the message ladder), rank 8, $248 = 16 \cdot 15 + 8$. *The kill built in* (mandatory negative, `v775` re-anchored): the exact counting bound — every $D_5 \oplus A_3 \subset E_8$ has 128 spinor-side roots, so at most $\lfloor 128/16 \rfloor = 8$ all-spinor classes and at least 7 of the 15 classes *always* contain adjoint-side roots; the concrete doubled-model census *saturates* both bounds ($8+7$, zero mixed classes — the $J$-quotient respects the $D_8$ grading), and the weight-level kill at the canonical split shows each spinor-pure class restricting to 8 distinct half-spinor weight pairs, never one — the 15 mailboxes are *not* 15 matter multiplets; ROOTCLASS-MIXED re-verified, not overturned. *What the normal form does not remove:* all 30 placements are Type II $[8,4]$ codes (the `v799` echo), and $C_{\text{fin}}$ does *not* derive the three seam axioms, the OPE-linearity scope note, or the `v776` selection of one copy — P2 ($g_{\text{car}} = 5$, entering as the $\iota$ slot count) is *narrowed* to the `v799` residual R1′ ([C] class), not eliminated. Controls fire (the non-alternating dot form admits zero refinements; the trivial deck leaves four selector candidates; the wrong lift breaks $\beta = \hbar$). A normal form, not a new claim; no marker moves [E].

**The uniqueness of $C_{\text{fin}}$ and $\text{Aut}(C_{\text{fin}}) \cong C_6$ (2026-08-07) — one orbit, six automorphisms [E]** ( [`verification/v849_cfin_unique_cofinal_lean.py`] )

The normal form above is not merely assembled — it is *unique* (22/22, CFIN-UNIQUE, exhaustive finite verification over $\mathbb{F}_2^4$). The WLOG ladder is exact: the 28 nondegenerate alternating forms are one $GL(4,2)$ orbit; $Sp(4,2)$ (order 720 by full enumeration) is transitive on the 6 Arf-1 refinements; per refinement there are exactly 20 admissible $\sigma$ (one orbit under the order-120 stabilizer) and per $(q^*, \sigma)$ exactly 120 admissible equivariant $\iota$ (one orbit). *The uniqueness theorem*: the full admissible tuple census at $\omega = \hbar$ has 14400 members and the $Sp(4,2) \times S_5$ action has *exactly one* orbit (orbit–stabilizer verified: $14400 \times 6 = 86400 = |G|$) — every admissible compiler object is isomorphic to $C_{\text{fin}}$. *The automorphisms*: $|\text{Aut}(C_{\text{fin}})| = 6$ with element orders $[1,2,3,3,6,6]$, abelian and cyclic — $\text{Aut}(C_{\text{fin}}) \cong C_6$, the projection to $Sp(4,2)$ faithful and the $S_5$ slot permutation *determined* by the symplectic part. The Gaussian-reduction diagram is typed: the parent lattice fixes the carrier triple $(V, \hbar, \sigma)$ up to exactly the 18-element centralizer torsor, the selector pins $q^*$ uniquely, the 120 $\iota$-completions form a single orbit — and the *strict* initial/terminal reading fails honestly (120 completions against 6 automorphisms: unique up to *non*-unique isomorphism, a one-object groupoid, not a terminal object). Controls fire: dropping alternation kills the object (0 refinements); the wrong Arf class kills $\iota$ (the $C_{\text{even}}(5)$ weight census $(1,10,5)$ *forces* Arf 1); dropping $\sigma$ kills *canonicity* (the completion census becomes $16 \to 8 \to 4 \neq 1$) while the bare $(q, \iota)$ orbit count survives — $\sigma$ is load-bearing for the selector, typed either way. P2 sharpens accordingly: the interface object is unique up to isomorphism with $C_6$ automorphisms; what remains of P2 is unchanged — the `v799` residual R1′ (why *this* normal form at all). No marker moves [E].

**The four finite anchor theorems (2026-08-09) — the census-free $q^*$, the exclusive anchor classification, the doily as a multiplication table, and the $g = 5$ exhaustion [E]** ( [`verification/v880_finite_anchor_theorems.py`] )

Four theorems close the finite-anchor layer of this section at identity grade (26/26, FINITE-ANCHOR-CLOSED; the discovery probe embedded byte-exact). **(1) The census-free normal form.** The selector-picked refinement has the closed weight-only form

$$q^*(x) \;=\; \sum_{i<j} x_i x_j + \sum_i x_i \;=\; \binom{|x|+1}{2} \bmod 2,$$

value table $(0,1,1,0,0)$ on weights 0–4 — and the `v845` selector *forces* the linear part $(1,1,1,1)$: the $2^{16}$ brute-force search of the normal-form keybox above is replaced by a theorem. **(2) The exclusive anchor classification.** Every line of $PG(3,2)$ is classified by its $q^*$-weight multiset, and the classification is *exclusive*:

| lines | count | weight multiset | anchor |
|---|---|---|---|
| doily (isotropic) | 15 | $\{2,1,1\}$ | $a=(1,1,2)$ |
| secants | 10 | $\{2,2,1\}$ | the *wrong* anchor $(1,2,2)$ |
| external | 10 | $\{1,1,1\}$ | — |

All 15 doily lines carry the anchor $a=(1,1,2)$ with per-line characteristic polynomial $(t-2)(t-1)^2 = t^3-4t^2+5t-2$ and Vieta data $(e_1,e_2,e_3)=(4,5,2)$ — $e_2=g_{\text{car}}$ and $e_3=|Z_2|$ read off one line; the 20 non-isotropic lines split *exactly* as 10 secants carrying the wrong anchor $(1,2,2)$ (Vieta $(5,8,4)$ — the `v844` control multiset) plus 10 external lines $\{1,1,1\}$. The incidence budget lands as counting: $15=M_0$ and $30=M_1=h(E_8)$ (counting only, no new structure claimed). The anchor–Vieta report closes the loop: $p_2(a)=6$, $\chi_L-\chi_R=-p_2(a)\,(t^2-e_2t+e_3)$, and $M_n=15\,(1,e_3,e_1,e_1e_3,e_1^2)$. **(3) The doily is a multiplication table.** The doily *is* the multiplication table $\Lambda^2E\times\Lambda^2E\to\Lambda^4E$ of the five-slot carrier: each doily line has support partition $\{i\}\cup\{j,k\}\cup\{l,m\}$, the wedge $b_{jk}\wedge b_{lm}=\pm f_i$ holds exactly, the assignment is a bijection onto all $5\times3=15$ such configurations, and the wedge is nonzero *exactly* on the doily lines. **(4) The $g=5$ exhaustion.** The quadratic one-step exhaustion $\Lambda^2\wedge\Lambda^2=\Lambda^{g-1}$ *forces* $g_{\text{car}}=5$ uniquely among odd $g\in\{3,\dots,11\}$. The honest split is explicit and load-bearing: the *theorem* (exhaustion $\Rightarrow g=5$) is **[E]**; the *physical premise* (why the carrier must satisfy the quadratic one-step exhaustion — the Calderón reading) stays **[O]** — P2 is *narrowed*, not eliminated; no marker moves.

**The finite Pfaffian/$K_6$ signature (2026-08-09, rounds 44–49) — the $Gr(2,5)$ Plücker reading, the canonical sign character, the self-hosting counting theorem, the flavor hexagon as the automorphism orbit, and the vacuum selection [E]** ( [`verification/v888_finite_pfaffian_signature.py`] )

Rounds 44–49 close the finite Pfaffian/$K_6$ layer of the compiler at identity grade $(27/27+19/19+42/42)$; verdicts S6PLUCKER-CLOSED, C6FLAVOR-MEASURED + INTERTWINED, K6PFAFFIAN-CLOSED; three discovery probes embedded byte-exact; exact integer / Fraction / sympy arithmetic in every decision — no floats, no RNG, no fits). **(1) The $Gr(2,5)$ reading.** The 15 doily lines *are* the 15 monomials of the five Plücker quadrics of $Gr(2,5)$: the bijection is 15/15 with *perfect* sign match — every quadric reads $(+,-,+)$, the `v880` Hodge table (converse warded: secants/externals produce no monomial). The $S_6$ duad representation decomposes as $15=1+5+9$ (Specht $S^{(6)}+S^{(5,1)}+S^{(4,2)}$, characters from scratch by Murnaghan–Nakayama) and the normalized doily operator acts on the isotypics by the exact scalars $(1,0,2/3)$, with the anchor identities $2/3=e_3/(e_2-e_3)$ and $15=1+e_2+(e_2-e_3)^2$ at $(e_1,e_2,e_3)=(4,5,2)$; and the $q^*$ zero set is a $(16,6,2)$ Hadamard difference set with the anchor signature $(v,k,\lambda,v-k)=(2^{e_1},p_2(a),e_3,\binom{e_2}{2})=(16,6,2,10)$. **(2) The $K_6$ motif theorem.** Under the duad labeling the 35 lines of $PG(3,2)$ are *exactly* the 15 perfect matchings of $K_6$ (the doily) plus the 10 triangles through the vacuum vertex (the secants — the wrong-anchor class) plus the 10 triangles avoiding it (the external lines, 20 triangles exhausting $\binom{6}{3}$), and the edge law is $q^*(e)=0$ iff $e$ touches the vacuum vertex — $q^*$ *is* vacuum incidence. **(3) The canonical sign character.** The fermionic Pfaffian signs of the symbolic $6\times6$ expansion (15 monomials, coefficients $\pm1$) are gauge-matched to the Hodge table via the canonical global character $\chi(i)=(-1)^{i+1}$ on the five vacuum edges — triple-identified as the vacuum-row Laplace sign of $\mathrm{Pf}(A)=\sum_i(-1)^{i+1}a_{0i}\,\mathrm{Pf}(A_{\hat{0}\hat{i}})$, the $\Lambda^4$-vs-$\Lambda^5$ volume-form conversion (verified on all 15), and the $C_6$ crossing character ($\mathrm{Pf}(PAP^{\mathsf{T}})=\det(P)\,\mathrm{Pf}(A)$ symbolically) — a global gauge, not a line-dependent fudge (the flipped-$\chi$ control produces exactly the 3 expected mismatches); each $4\times4$ minor Pfaffian *equals* the corresponding $Gr(2,5)$ Plücker quadric. **(4) The self-hosting counting theorem.** The pairing count $\binom{g+1}{2}=g!!$ has exactly the solutions $\{1,5\}$ over odd $g$ ($g=1$ trivial, stated; growth separation $g!!>\binom{g+1}{2}$ for all odd $g\geq7$ by symbolic induction), so $g=5$ is the *unique* nontrivial fixed point, with $(g-2)!!=3=N_{\text{fam}}$. The honest split is explicit and load-bearing: the counting *theorem* is **[E]**; the *physical premise* (the boundary grammar is a self-hosting Wick pair compiler) stays **[O]** — P2 is narrowed to *one* premise, not eliminated; no marker moves. **(5) The flavor hexagon is the automorphism orbit.** $\mathrm{Aut}(C_{\text{fin}})\cong C_6$ is rebuilt exactly; the pinned generator $g$ satisfies $g^2=\sigma$ (the deployed family 3-cycle) *exactly* and $g^3=$ the $W$-edge sheet swap; its action on the ten pair-messages has cycle type $1+3+6$ with orbits equal to the charge shells, the $g$-orbit of $\{f_1,w_1\}$ traverses the deployed flavor hexagon edge by edge ($D_6$-equivalent to the transport cycle), and the recovery chain reads $(|Z_2|/N_{\text{fam}})^{|\mathrm{Aut}(C_{\text{fin}})|}=(2/3)^6=64/729$ — the frozen deployed transfer eigenvalue as an exact Fraction (typed honestly as a consistent *new reading*: the corpus derivation of the exponent via $p_2=|R_+(A_3)|$ stands; 0 of 600 symplectic non-automorphisms factor, scrambled transport fails).

**(6) The vacuum-selection theorem.** $Sp(4,2) \cong S_6$ acts transitively on the six Arf-1 refinements; $\mathrm{Stab}(q^*)$ has order 120 and acts faithfully *and* surjectively as $S_5$ on the remaining five, with orbits on the 16 messages exactly the $\iota$-weight classes $1+5+10 = (\Lambda^0, \Lambda^4, \Lambda^2)$ — the five-slot carrier is the residual symmetry after choosing one of six equivalent vacua. Two side reports: the anchor power-sum clock $p_n = 2+2^n$ has mod-30 orbit $4 \to 6 \to 10 \to 18$ of period exactly $4 = |\mu_4|$, typed CLOCK-DISTINGUISHED (the purely period-4 moduli are exactly $\{5, 10, 15, 30\}$, all dividing 30); the operative-Coxeter-30 reading is separately dead (sandbox, both sides). All controls fired (Arf-0 refinements, secant substitution, wrong scalar, flipped sign character, $g = 3/7$ counterexamples). No marker moves **[E]**.

**The Wick functor arc (2026-08-09, rounds 50–55) — the six compiler roles in the deployed seam, the proved scalar obstruction, the block resolution, the seam-diagonal demand, and the constructed channel-mixing candidate [E]/[O] ( [`verification/v896_wick_block_functor.py`], [`verification/v898_kms_schur_mixing.py`])**

Rounds 50–53 take the one remaining P2 premise of the keybox above — the physical Wick-compiler reading, [O] — and close its *structure* level (27/27+33/33+26/26; verdicts WICKFUNCTOR-PARTIAL, WICKMIX-MEASURED (WICKMIX-OBSTRUCTED), WICKBLOCK-MEASURED (BLOCKFUNCTOR-CONSTRUCTED / SEAM-DIAGONAL); three discovery probes embedded byte-exact; exact integer / rational / symbolic checks in every decision typed PROVED or SYMBOLIC). **(1) The six roles exist.** The compiler roles are present in the deployed seam one-particle space — five $D_5$ carrier slot pairs plus the $A_3$ boundary block (dims $2,2,2,2,2,6$ from the `v155/v156` split $16 = 10+6$); the $\mathfrak{so}(16)$ sector law holds (current-sector dims 12 on vacuum edges / 4 on pair edges = the $q^*$ edge grading, ratio $3 = N_{\mathrm{fam}}$); and the $C_6$ intertwining is *exact*: the $\mathrm{Aut}(C_{\mathrm{fin}})$ generator $g$ ($g^2 = \sigma$) induces the channel permutation of cycle type (1)(2)(3) fixing channel 0. But the deployed *vacuum kernel* is channel-diagonal, so all 15 duad two-point functions vanish (DUADMAP-DEGENERATE-ON-VACUUM): the structure level of the functor exists, the value level needs a channel-mixing covariance — WICKFUNCTOR-PARTIAL, the honest typing (the candidate is constructed in round 55, item (5) below). **(2) The scalar obstruction is proved.** On the 6-dimensional scalar channel space *every* $C_6$-covariant antisymmetric covariance vanishes on the transposed duad — the $C_6$ 2-cycle $\{4,5\}$ forces a duad zero, so the scalar route reaches at most 14/15 nonzero duads, never 15/15: a *theorem of the deployed action*, exhaustively verified over the isotypic decomposition (the multiplicity list $[3,0,1,1,1,0]$ re-derived by hand and by exact root-of-unity sums; the frozen expected list carried a transcription error, caught fail-first); no equivariant scalarization rescues the scalar space, and both relaxation witnesses are constructed. **(3) The block resolution.** The full block-valued functor on the 16-dim one-particle space is BLOCKFUNCTOR-CONSTRUCTED — block commutant dimension 33; all 15/15 duad cross-blocks nonzero; all 15/15 block Wick monomials obey the canonical sign law $\chi(i) = (-1)^{i+1}$ (the round-49 gauge of the keybox above); the interleaving character $+1$ *symbolically*; the review's five conditions hold at block level. **(4) The seam-diagonal demand.** The decisive fact: the *deployed* seam covariance (the `v113` family rebuilt exactly) has *zero* cross-blocks everywhere (SEAM-DIAGONAL) — the value level of the reality bridge needs a *physical channel-mixing mechanism* (a boundary/transport-twisted state, not the vacuum); with it, the $K_6$/Pfaffian grammar of `v888` would be the measured correlation grammar of the seam. **(5) The constructed mixing candidate (round 55, [`verification/v898_kms_schur_mixing.py`], 30/30; verdict kmsmix-measured [kmsmix-found, schurmix-generates, rp-theta-open]).** The demanded state now *exists as a candidate*, by two routes. *The KMS route:* the Majorana-compatible $C_6$-invariant Hamiltonians $K = ih$ are parametrized exactly (the hermitian $O_{16}$-commutant has dimension $112 = 62 + 50$; the antisymmetric slice splits as 33 cross — exactly the block space of (3) — plus 17 diagonal, with the two forced zeros of the transposed $\{4,5\}$ duad), and already the *first* point of the frozen grid, $h = -(A_{16}^{\mathrm{dep}} + \frac{1}{8}A_{\mathrm{int}})$ at $\beta = 1$, passes *all* gates: strict CAR positivity, exact $C_6$ block covariance, all 15 duad cross-blocks at the frozen floor, the $q^*$/sheet grading carried exactly (5/5 vacuum + 10/10 carrier), and all 15 block Wick monomials nonzero with the *canonical* $\mathrm{Pf}_4$ sign pattern of (3); robust at $\beta = 2$. *The Schur route (exact, sympy):* eliminating the boundary of the coupled diagonal state $A_{\mathrm{full}} = [[\kappa A_{CC}, tV], [-tV^{\mathsf{T}}, mJ_3]]$ gives $A_{\mathrm{eff}} = \kappa A_{CC} + \frac{t^2 m}{1-m^2} V J_3 V^{\mathsf{T}}$ *exactly*, and $S = V J_3 V^{\mathsf{T}}$ deposits nonzero correlation on *all ten* carrier duads (uniform $3J$ block, canonical $\mathrm{Pf}_4$ signs) from the bare diagonal state — the same Schur/Feshbach elimination that creates the RH port creates the carrier mixing: on both fronts the real correlation lives in the *eliminated boundary*. The frozen 15-signature table of the winner is the registered prediction any physical seam state must reproduce; the OS-positivity under a *spatial* reflection is the named open check (RP-THETA-OPEN — no spatial $\theta$ is deployed on the 16-dim space; the Majorana real structure satisfies the `v426`/`v440` dictionary automatically but is particle-hole, not spatial). The honest split: the

obstruction theorem, the block construction and the Schur generation identity [E]; the constructed states are *candidates* — whether the *actual* seam realizes the signature pattern is untouched, so the physical Wick-compiler premise (`SEAM.CFIN.WICKFUNCTOR.01`, `SEAM.CFIN.KMSMIX.01`) stays [O], unmoved — the physical realization is the named demand. *Dated addendum (2026-08-10, rounds 57–59):* the open check RP-THETA-OPEN is now *measured* at family level and the answer is an exclusion — strict reflection positivity under both deployable spatial reflections forces $t = 0$ (OS positivity $\iff$ the mixing floor falls: mutually exclusive on this family), the complete $C_6$-twisted census is exclusionary (0/6), and the parent-dilation route yields only a marginal cone-boundary witness (with the exact floor identity $t^2 \cdot 3m/(1-m^2) = 1/200$) while the strict collar route is obstructed by an exact linear $2t$ law ([`verification/v903_seam_rp_exclusion.py`]; a measurement on the complete finite candidate set, not a universal no-go); the [O] premise stays unmoved. *Dated addendum (2026-08-10/11, rounds 60–63):* the exclusion is now *anatomized* — strict 2-cycle RP is impossible on the *whole* $C_6$-covariant class (the seat law reduces exactly to $\mathrm{diag}(a_J, -a_J)$; the only escape is covariance-breaking at exact linear price $2\varepsilon$, with the floor exchange $\mathrm{Pf}_4(\varepsilon) = (\varepsilon - \frac{1}{200})(\varepsilon + \frac{1}{200})$), while the strict-collar obstruction is a two-seat linear law whose *equilibrium* kernel witnesses carry the full 1/200 mixing at zero entropy production — no non-equilibrium state is needed, and whether the deployed PURE-I wiring is compiler-forced became the registered successor contract ([`verification/v908_seam_equilibrium_wiring.py`], `SEAM.STATE.WIRING.SELECTOR.01`); the [O] premise stays unmoved. *Dated addendum (2026-08-12, rounds 69–71):* the successor contract is *closed*, and the answer is a *freedom theorem* ([`verification/v911_wiring_freedom.py`]): the exact Gröbner census of the orbit/edge rules leaves exactly *one* admissible three-dimensional wiring component modulo the rule gauge (the rotation cell); $Z/X$ wirings are excluded gauge-invariantly by orientation propagation; the deployed PURE-I point is an *interior* point of that cell, connected to the pure-$J$ ray by an integer rule-gauge element — PURE-I IS A DEPLOYMENT REPRESENTATIVE, NOT A COMPILER THEOREM. No compiler-side demand distinguishes the $\theta_S$ sheet-swap frame within the rule-gauge algebra (five demand classes, all silent on the angle), the strict-collar RP Grams transform *exactly covariantly* under the rule gauge, and the same deployed parent that fails strict collar RP at $\theta_S$ *passes it exactly* in the integer-conjugated frame: the rounds-58/59 obstruction is frame-relative (a statement at the fixed $\theta_S$ frame, not a wiring no-go), and a canonical wiring selection would require an additionally *derived* physical frame. The [O] premise of the Wick-compiler demand stays unmoved. No marker moves.

**The P2 reduction as recorded on 2026-07-22, with later closures below**

This paragraph records the July branch, not the final residual inventory. The later analytic monodromy, invariant Hermitian form and finite W-bridge closures are collected in Section 2. The axiom P2 stays *declared* (`AX.P2.01`); what has changed is how much of it is now carried by theorems. The honest ledger state: P1 ($c_3 = 1/(8\pi)$) and the *four marks* are [E]-anchored (Gauss–Bonnet $n = 2\chi = 4$, [`verification/v216_marks_gauss_bonnet.py`]); given the four marks, $g_{\mathrm{car}} = 5$ follows from the partition $4 = 2{+}1{+}1$ plus the Deligne/Birkhoff–Grothendieck/stability typing ([`verification/v491_p2_partition_corollary.py`], [`verification/v499_p2_weights_deligne_bg.py`]; residual R1, the module identification, [C]); and the clock's *order*, the $\mathbb{Z}_8$ tower, the deck's *class* (a half-period translation) and the *position* half of the alignment bit are theorems ([`verification/v506_seam_clock_rigidity.py`], [`verification/v507_seam_bit_origin.py`], [`verification/v510_seam_bit_freedom.py`]). What remains as carrier input is *one discrete symmetry-lift bit*: flag transitivity of the four marks (marks *and* their two sides indistinguishable, $V_4 \to D_4$) $\iff$ $\tau = i$ (equivalently: the clock / the CM-fixed half-period / the non-split extension class on the free seam circle) — on the free seam circle bare mark-transitivity is automatic for *every* configuration (the pair-exchanging $V_4$) and no local jet sees the side bit, so the flag form is the sharp discrete statement of the datum and its negation is concretely measurable (odd-doublet split $(2/\pi)\ln\cot(\delta/2) \neq 0$) ([`verification/v512_seam_tau_flag.py`], `SEAM.TAU.FLAG.01`). The bit has since survived *ten* side-blind derivation attacks on that scoreboard — free RP/Θ existence (the eighth, v521), the exhausted Wick-computable free-plus-twist state class (the ninth, v525) and the first genuinely interacting seam toy (the tenth, v529) — and is now physically *defined* as the *twist-class choice*: facets #14/#15 extend the web to 15 exact equivalences (both directions each), the flip-axis fraction is a gauge-robust order parameter ($O = \frac{1}{2}$ at $m = 4$, 0 elsewhere), and its $\eta$ holonomy is in principle interferometrically readable (v528, `SEAM.BIT.TWISTCLASS.01`; the bit remains formal input — a definition replaces no derivation, no marker moves); and it now carries the first *positive* selection datum: reflection positivity, imposed on the interacting mark family, keeps exactly *one* member alive — $\delta = \pi/2$ with positive coupling, the carrier of that order parameter ([`verification/v534_seam_straddle_cone.py`], `SEAM.STRADDLE.CONE.01`; toy-level evidence for a dynamical origin, not a derivation); plus the [C] identifications R1/R2 of v499.

In one sentence: **the compiler's dimensionless inputs reduce to four marks (derived from P1-side topology), one discrete symmetry-lift bit — the twist-class choice (flag transitivity $\iff \tau = i$) — and $\pi$.** This is a reduction *state*, not an abolition of the axiom: `AX.P2.01` remains the declared interface input, and its dynamical realisation (`QGEO.REALIZE.01`) stays [C]/[O]. *[2026-08-07 addendum:* the finite normal form of the interface is now assembled as *one* object — $C_{\text{fin}} = (V, \hbar, q^*, \sigma, \iota)$, the keybox above ( [`verification/v845_cfin_normal_form.py`]) — with the code-to-matter kill built in as a checked negative; its honesty section types exactly what the assembly does *not* remove: P2's remaining content is the `v799` residual R1′ (the three seam axioms, the OPE-linearity scope, the `v776` uniqueness selection). The assembly narrows; it does not eliminate. Markers unchanged.]

**Seed normal form, the factorial spine, and the degree-2 pattern [E]/[C] ( [`verification/v106_review_validation.py`])**

Three review-validated sharpenings of the anchor picture. **(1) Seed normal form [E].** The seed formula is exactly

$$\boxed{\varphi_0^{\text{ret}} = \frac{|\mu_4|}{N_{\text{fam}}}\, c_3 \;+\; \Omega_{\text{adm}}\, c_3^{|\mu_4|}} \qquad \left(\tfrac{4}{3} = \tfrac{|\mu_4|}{N_{\text{fam}}},\;\; 48 = \Omega_{\text{adm}} = N_{\text{fam}} \dim S^+\right),$$

a *linear seam term* plus a *topological fourth-order correction* over all admissible fermion states — a structured anchor readout, not a fitted expression. **(2) Factorial spine [E].** With $X = 6Y$ on one 16-state generation, $\operatorname{Tr} X = 0$ and $\operatorname{Tr} X^2 = 120 = 5!$ (computed from the explicit charge table); the *same* factorial moment is $|R^+(E_8)| = \sum(E_8 \text{ exponents}) = 120$, doubles to $|R(E_8)| = 240$, and scales to $|W(D_5)| = 2^4 \cdot 5! = 1920$ (the Hawking-power fingerprint) — one moment, four projections (hypercharge, roots, Weyl group, horizon). **(3) The degree-2 inventory.** The compiler repeatedly selects *degree two*: the Pascal closure $2^{g-1} = \sum_{k\le 2}\binom{g}{k}$ truncates at $K{=}2$ with the *unique* solution $g = 5$ (scan 1..40); the glue norms add to the root norm, $\frac{5}{4} + \frac{3}{4} = 2$; $c_3 = \frac{1}{2} \times \frac{1}{4\pi}$ (variational half × two-dimensional-slice Gauss–Bonnet); the pair sector $\binom{5}{2} = 10 = \mathcal{A}_\Lambda$; the hypercharge moment is a *second* moment; the seam is codimension *two*. *Named hypothesis [C] (recorded, not claimed):* **Quadratic Boundary Locality** — the seam supports only bilinear data, so the truncation $K = 2$ is *forced* rather than chosen; if proven, the Pascal *selection* (red-team Target B residual, typed [O]/[C]) upgrades to [E] and the chain locality $\Rightarrow K{=}2 \Rightarrow g{=}5 \Rightarrow 16 \Rightarrow D_5 \Rightarrow E_8$ closes the carrier choice. *Audit:* the even-code transition reading $240 = 16 \times 15$ is exact arithmetic but non-unique ($240 = 16 \times 5 \times 3$ is the established glue factorisation) and stays recorded, not promoted.

**The Sheet-Pairing Lemma ( [`verification/v109_sheet_pairing.py`]) — the first exact anchor of the hypothesis.** Character-exact weight-multiset combinatorics of the $D_5$ spinor weights $(\pm\frac{1}{2})^5$ gives: *(i)* **no scalar within a sheet** — the zero-weight multiplicity of $S^+ \otimes S^+$ is 0, because the odd slot count $g = 5$ flips the sign parity of $-w$ (a parity theorem of the five-slot code); *(ii)* **the scalar lives across the sheets**:

$$\boxed{S^+ \otimes S^- = \Lambda^0 \oplus \Lambda^2 \oplus \Lambda^4} \qquad (256 = 1 + 45 + 210),$$

exact as weight multisets, with zero-mode grading $(1, 5, 10)$ = the carrier code; *(iii)* **sheet-diagonal = odd forms**: $(S^+ \otimes S^+) \cup (S^- \otimes S^-) = 2\Lambda^1 + 2\Lambda^3 + \Lambda^5$ exactly — within-sheet pairings are odd-form-valued, never scalar; *(iv)* the scalar-bearing tower **tops at** $\Lambda^{2K}$ with $K = 2$ — bilinear seam data reach exactly the Pascal-ladder degree (the source-layer transport theorem; document 3). *Consequence [C]:* the seam's single scalar two-point kernel is necessarily a **sheet pairing** (chirality-off-diagonal — the same $\mathbb{Z}_2$ as the glue chiralities and the branch-kernel selection); the remaining QBL step is to show the Calderón kernel supplies exactly this pairing and nothing beyond.

**The Calderón-sheet selection theorem ( [`verification/v110_calderon_sheet.py`]) — which involutions certify a scalar kernel.** On $H = S^+ \oplus S^-$ the certified bilinear data of a Calderón involution $\varepsilon$ ($\varepsilon^2 = 1$) are the matrix elements $\langle f, \varepsilon g\rangle$. The lemma's singlet counts per sheet block, $(0, 0, 1, 1)$, give the exact selection statement [E]:

$$\boxed{\text{a scalar two-point datum exists} \iff \varepsilon \text{ is sheet-odd}}$$

— a sheet-odd involution (swapping $S^+ \leftrightarrow S^-$) certifies exactly $1 + 1 = 2 = |\mathbb{Z}_2|$ invariant scalar kernels (one per orientation), a sheet-even involution certifies *none*. The kernel count $2 = |\mathbb{Z}_2|$ matches exactly the *two Lagrangian glues* (swapped by single spinor conjugation): the Calderón scalar channel carries precisely the glue ambiguity, resolved by the same sheet choice. *Anti-overclaim control [E]:*

the parity theorem is generic over the odd ladder ($g = 3, 5, 7$: within-sheet count 0, cross count $2^{g-1}$) — sheet-oddness forces the *half-spinor relation* $K = (g-1)/2$, **not** the value $g = 5$; the $g$-selection stays with the independent rank-8/integrality closure. *Consequence:* the QBL residue splits into two named parts [C] (recorded, not claimed): **(a) interface** — "the seam Calderón involution is sheet-odd" (structurally natural: the one-sided collar/double-cover deck is the *same* $|\mathbb{Z}_2|$ that halves $c_3 = \frac{1}{2}(4\pi)^{-1}$, and exchanging cover sides is sheet-odd by construction); **(b) analytic core** — "the two-point kernel certifies only slot-degrees $\leq 2$". Proving both closes QBL $\Rightarrow K{=}2 \Leftrightarrow g{=}5$.

**The Quadratic-Transport Theorem (`[verification/v111_quadratic_transport.py]`) — the transport half of the analytic core *is* a theorem.** Grade seam transport by Clifford degree in the ten field operators on the five-slot Fock space ($S = \Lambda^\bullet(\mathbb{C}^5)$, $S^+$ the 16-state code; integer Jordan–Wigner model, spanning claims by full-rank certificates mod $p$, valid over $\mathbb{Q}$). *Parity* [E]: all linear words are sheet-odd, all 45 quadratic words sheet-even — code-preserving transport has *even* Clifford degree (the same $\mathbb{Z}_2$ again). *Minimality* [E]: sheet-even words of degree $\leq 1$ are the scalars alone. *Generation and completeness* [E]: the 45 quadratics ($= \mathfrak{so}(10)$, exactly the $\Lambda^2$ term of the certified tower $1 + 45 + 210$) generate the whole code from the vacuum *and* span the full operator algebra $\mathrm{End}(S^+)$ ($\dim = 256$) at word length two — every code operation whatsoever is a word in pair transport, and all higher even-degree elements (e.g. the 210 quartics) are redundant. Hence

| the transport degree is exactly 2 |
|---|

— degree $\leq 1$ generates nothing, degree 2 generates everything: a theorem, not a hypothesis. *Anti-overclaim control [E]:* the $g = 3$ world behaves identically (15 quadratics generate $\mathrm{End}(S^+) = 16$) — the theorem selects the *degree*, not the rank; the $g$-selection stays with Pascal closure + rank-8. *Residue after v109–v111* [C]: QBL's remaining content is two *interface* statements — (a) the seam Calderón involution is sheet-odd; (b′) the certified *state* inventory is the $\leq 2$-slot tower. Nothing about transport remains hypothetical.

**The Self-Counting Channel (`[verification/v112_selfcounting_channel.py]`) — the Pascal closure is an *identity* of the certified channel.** For odd $g$, negation $w \mapsto -w$ maps the $S^+$ weights bijectively into the opposite sheet, so the Cartan-neutral kernels of the certified channel are exactly the pairs $(w, -w)$ — *one per code state*: $\#\{\text{neutral kernels}\} = \dim S^+ = 2^{g-1}$ exactly [E]. Grading the *same* neutral set by pair-degree $m$ gives sizes $\binom{g}{m}$, $m = 0..K$ — for $g = 5$ the triple $(1, 5, 10)$ [E]. Two countings of one set:

$$2^{g-1} = \sum_{m \leq K} \binom{g}{m} \quad \text{is the channel counting itself}$$

(symbolically verified for all odd $g = 3..13$) — the Pascal *closure* (`v2`/`v108`) is not an extra requirement but an identity of the certified channel. *Hodge fold* [E]: the code's minus-sign grading $\binom{g}{2j}$ equals the folded binomial $\binom{g}{\min(2j,\, g-2j)}$ with $\min(2j, g-2j) \leq K$ — every code sector appears at pair-degree $\leq K$. *Anti-overclaim:* the identity holds for every odd $g$ and carries no $g$-selection; the selection stays with the rank-8 closure $g + N_{\mathrm{fam}} = 8$ and ladder integrality. *Residue re-type [C] (recorded, not claimed):* clause (b′) loses its closure burden — given the sheet-odd kernel, the channel *automatically* contains one neutral kernel per code state at pair-degree $\leq K$; what remains physical is the single input "the seam certifies through one scalar two-point kernel" (for the established free $c = 8$ seam net, Wick extends the one kernel to the entire even tower by determinants). QBL is thereby the structural consistency frame of the carrier choice; the *selection* is carried by rank-8 + integrality.

**One kernel is the whole net (`[verification/v113_quasifree_kernel.py]`) — the QBL input merges with the holomorphy premise.** The remaining input is now anchored in the established Gate-A objects. *Bookkeeping* [E]: $c(D_5)_1 = \frac{45}{9} = 5 = g_{\mathrm{car}}$, $c(A_3)_1 = \frac{15}{5} = 3 = N_{\mathrm{fam}}$, and the carrier net $(D_5)_1 \times (A_3)_1 = SO(10)_1 \times SO(6)_1$ is $10 + 6 = 16$ *free Majorana fermions* ($c = \frac{16}{2} = 8$); the whole extension tower carrier $(\mu{=}16) \to SO(16)_1(\mu{=}4) \to (E_8)_1(\mu{=}1)$ carries the *same* sixteen fermions — only the certified glue grows (index $2 \times 2 = 4 = |\mu_4|$). *Wick/Pfaffian* [E]: in the exact ten-Majorana model *all* 210 vacuum four-point and all 210 six-point functions equal the Pfaffian of the single two-point kernel — one kernel determines all correlations. *The kernel is a Calderón involution of rank g* [E]: $M = \mathbf{1} + iA$ with $A^2 = -\mathbf{1}$, so $P = M/2$ is a projection with $\mathrm{rank}\, P = 5 = g_{\mathrm{car}}$ and $\varepsilon = iA$ an involution; at seam level (sixteen Majoranas) the rank is $8 = \mathrm{rank}\, E_8$:

| the central charge is the rank of the one kernel |
|---|

($c = 5$ carrier block, $c = 8$ seam hull); and the polarization fixes the vacuum *uniquely* (joint annihilator kernel one-dimensional). *Premise merge [C] (recorded, not claimed):* "single scalar two-point kernel" = "the seam state is quasi-free" = the defining property of the free-fermion net the holomorphy gate already posits — QBL adds *no* assumption beyond Gate A; when the gate closes, the carrier choice closes **[E]** along free net ⇒ one kernel ⇒ sheet-odd ⇒ self-counting tower ⇒ complete transport ⇒ carrier. Gate typing **[C]**/**[O]** unchanged.

**Hardening (P1): the Gauss–Bonnet origin of $c_3 = \frac{1}{8\pi}$**

$c_3 = 1/(8\pi)$ is not an arbitrary number: it is the *one-sided Gauss–Bonnet normalisation of the seam.* The seam construction compactifies the oriented two-dimensional normal slice $N_\Sigma$ to a sphere $S^2$, so Gauss–Bonnet gives the seam curvature integral

$$\oint_{S^2} K\, dA = 2\pi\, \chi(S^2) = 4\pi \qquad (\chi(S^2) = 2),$$

and the boundary datum is *one-sided* (the Calderón/double-cover construction uses a single collar), contributing a factor $1/|\mathbb{Z}_2| = \frac{1}{2}$. Hence

$$\boxed{c_3 = \frac{1}{|\mathbb{Z}_2|\ \oint_{S^2} K\, dA} = \frac{1}{2 \cdot 4\pi} = \frac{1}{8\pi}}$$

equivalently $c_3 = \frac{1}{2}(4\pi)^{-1}$, the one-sided 2D heat-kernel coefficient. This locates both factors precisely:

- the *discrete* $8 = |\mathbb{Z}_2| \cdot 2 \cdot \chi(S^2) = 2 \cdot 2 \cdot 2$ *coincides with* $h(D_5) = \operatorname{rank} E_8 = \varphi(30)$ — the geometric and the bootstrap readings of the "8" agree;
- the *continuous* $\pi$ is the irreducible Gauss–Bonnet / 2D-Gaussian primitive ($\int e^{-x^2} = \sqrt{\pi}$), the single primitive the self-consistency loop also isolates.

*Status:* this *hardens* (P1) — it gives $c_3$ a named geometric origin and reconciles the "8" with $E_8$ — but does *not* eliminate $\pi$; $\pi$ remains the one continuous primitive. **[C]**

**The one-paragraph version.** There is a boundary (the seam). It yields $c_3 = 1/(8\pi)$ and a five-slot carrier $g_{\text{car}} = 5$, split 3 colour + 2 weak. Two base charges $y_- = -1/3$ and $y_+ = 1/2$ generate the whole fermion family; the 16 even-occupation states are one generation, three families give $3 \times 16 = 48$. One seed $u = \varphi_0^{\text{ret}}$ drives four low-energy observables. Finally $\alpha_\star^{-1} \approx 137$ is the exponential engine: the electroweak scale takes one fifth of it ($e^{-\alpha_\star^{-1}/5}$), the cosmological constant takes it twice ($e^{-2\alpha_\star^{-1}}$), and the Hubble scale is the square root of $\Lambda$. A small grammar generates many sectors.

## 3 Headline: the Pascal compiler on five carrier slots

The strongest new statement is that three of the theory's separate structures are the *same* truncated row of Pascal's triangle on the five-slot carrier.

**Family = action ladder = transition count (one Pascal row)**

The even exterior algebra of $E$ ($\dim E = g_{\text{car}} = 5$), the action ladder, and the $E_8$ root count are one Pascal row:

$$S^+ = \Lambda^0 E \oplus \Lambda^2 E \oplus \Lambda^4 E, \qquad (\dim \Lambda^0, \dim \Lambda^2, \dim \Lambda^4) = (1, 10, 5), \qquad \dim S^+ = 1 + 10 + 5 = 16,$$

$$A_{\text{EW}} : \mathcal{A}_H : \mathcal{A}_\Lambda = \binom{5}{0} : \binom{5}{1} : \binom{5}{2} = 1 : 5 : 10 = (\dim \Lambda^0 E,\ \dim \Lambda^4 E,\ \dim \Lambda^2 E),$$

$$|R(E_8)| = \dim S^+ (\dim S^+ - 1) = 16 \cdot 15 = 240.$$

Family (exterior degree), cosmology (action charge) and $E_8$ (root transitions) read the same $\{1, 5, 10\}$. **[E]**

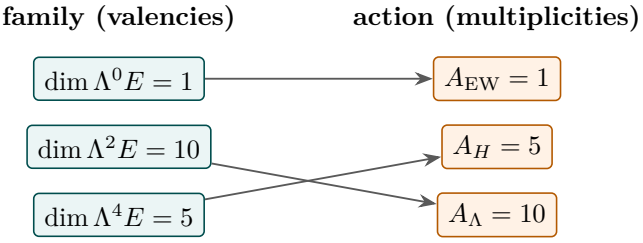



Figure 1: Exterior–Action duality. The Standard-Model family is the valency side of the even Hamming scheme on the five-slot code; the cosmological action ladder is its spectral (multiplicity) side. The Λ (pair) sector $\Lambda^2 E$ has dimension $\binom{5}{2} = 10$, the origin of the decuple exponent.

Natural sector assignment:

$$EW \leftrightarrow \Lambda^0 E \text{ (singlet)}, \quad H \leftrightarrow \Lambda^4 E \cong E^\vee \text{ (dim 5)}, \quad \Lambda \leftrightarrow \Lambda^2 E \text{ (dim 10)}.$$

The cosmological constant lives on the *pair* sector $\Lambda^2 E$, dim $= \binom{5}{2} = 10$; this is why the decuple bridge has exponent 10 — Λ is a product over all $\binom{5}{2}$ carrier pairs ($K_5$ edges), not over the full 16-state family.

### 3.1 $g_{\text{car}} = 5$ as the Pascal closure

A family is exactly the complete Pascal truncation to degree two:

$$\dim S^+ = 2^{g_{\text{car}}-1} = \binom{g_{\text{car}}}{0} + \binom{g_{\text{car}}}{1} + \binom{g_{\text{car}}}{2} \iff g_{\text{car}} = 5 \qquad (16 = 1+5+10), \quad \text{[E]}, \quad (2)$$

the unique positive-integer solution of $f(g) := 2^{g-1} - 1 - \frac{g(g+1)}{2} = 0$. *Proof of uniqueness.* Direct evaluation gives $f(1){=}-1$, $f(2){=}-2$, $f(3){=}-3$, $f(4){=}-3$, $f(5){=}0$, $f(6){=}+10$; so $g{=}5$ is the only root in $\{1, \dots, 6\}$. For $g \geq 6$ the forward difference is $f(g{+}1){-}f(g) = 2^{g-1}{-}(g{+}1) \geq 2^5{-}7 = 25 > 0$ and increasing, so $f$ is strictly increasing on $[6, \infty)$ from $f(6){=}10 > 0$; hence $f(g) > 0$ for all $g \geq 6$ and there is *no* further root. Thus $g_{\text{car}} = 5$ is the unique solution (no finite-range caveat). [verification/v2_carrier_pascal.py] This is sharper than $2g_{\text{car}} = \binom{g_{\text{car}}}{2}$: the family code (even states, $2^{g-1}$) equals the action code (sum of the first three Pascal entries) only at $g_{\text{car}} = 5$.

**Carrier uniqueness theorem: $g_{\text{car}} = 5$ and the 3+2 split are forced [E]**

The carrier rank and its colour/weak split are not chosen — they are the unique solutions of two integer conditions tied to the $E_8$ closure and the SM gauge algebra:

$$g_{\text{car}} = 5 \text{ is the unique } g \text{ with } N_{\text{fam}} = \frac{2^{g-1}-1}{g} \in \mathbb{Z}^+ \text{ and } \operatorname{rank} E_8 = g + N_{\text{fam}} = 8 \qquad (3)$$

$$(b, s) = (3, 2) \text{ is the unique split of } \; b + s = g_{\text{car}} = 5, \;\; b^2 + s^2 = 13 = |R(A_3)|{+}1 \qquad (4)$$

The integer-family carriers are $g \in \{3, 5, 7, \dots\}$ (rank $4, 8, 16$), so rank $E_8 = 8$ picks $g_{\text{car}} = 5$ uniquely; and $b^2 + s^2 = 13$ with $b{+}s = 5$ forces $(3, 2)$, reproducing dim $\mathfrak{g}_{\text{SM}} = (b^2{-}1) + (s^2{-}1) + 1 = 12 = |R(A_3)|$. The induced hypercharge is anomaly-free ($\operatorname{Tr} Y = \operatorname{Tr} Y^3 = 0$ over the 16 states). This hardens P2: the *rank* and the *split* are forced; only the seam→carrier interface itself remains a declared input. [verification/v14_carrier_uniqueness.py]

**The Pascal grammar: one $\Lambda^\bullet$ mechanism in four places (anti-numerology) [E].** The recurring small integers are *not* coincidences — they are the *same* exterior-algebra (Pascal)

mechanism read in four spaces, which is why the binomials of rows 4 and 5 keep reappearing:

| space | Pascal reading |
|---|---|
| carrier (one generation) | $\dim S^+ = \Lambda^{\text{even}}(5) = \binom{5}{0}+\binom{5}{2}+\binom{5}{4} = 1+10+5 = 16$ |
| scale ladder (cosmology) | $A_{\text{EW}} : \mathcal{A}_H : \mathcal{A}_\Lambda = \binom{5}{0} : \binom{5}{1} : \binom{5}{2} = 1 : 5 : 10$ |
| flavor $u/d$ readout | $\|\,\text{Pl}(K)\|_1 = \binom{4}{0}+\binom{4}{1}+\binom{4}{2} = 1+4+6 = 11 = 16-g_{\text{car}}$ |
| mass volume | $\det M_{\text{SM}} \sim (\varphi_0^{\text{ret}})^{6+9+10} = (\varphi_0^{\text{ret}})^{25} = (\varphi_0^{\text{ret}})^{g_{\text{car}}^2}$ |

So the carrier is the even row 5, the scale ladder is the lower row 5, the $u/d$ leg is the cumulative row 4 up to $\deg u^c = 2$, and the mass volume closes on $g_{\text{car}}^2$. *One* grammar ($\Lambda^k$ of the 5-slot carrier and the $4 = |\mu_4|$ family fundamental), not four lucky hits. [verification/v44_carrier_exterior.py] [verification/v45_family_exterior.py] [verification/v46_grand_mass_volume.py]

### 3.2 One master moment generates 40, 41, 240, $\delta_2$

With $X = 6Y$, the quadratic hypercharge trace on one family is $\text{Tr}_{S^+} X^2 = 120 = 5!$, and it generates the whole integer chain:

$$\sum_{f,j} L_{f,j} = \frac{\text{Tr}_{S^+} X^2}{3} = 40, \qquad 10b_1 = \frac{\text{Tr}_{S^+} X^2}{3} + 1 = 41,$$
$$|R(E_8)| = 2\,\text{Tr}_{S^+} X^2 = 240, \qquad \delta_2 = 4!\ \text{Tr}_{S^+} X^2\, c_3^8. \tag{5}$$

The flavor budget, the $U(1)$ budget, the $E_8$ count and the second seam defect all hang on the second hypercharge moment 5!. **[E]** *Round-22 sharpening:* the moment itself is one symbolic identity, $\text{Tr}_{S^+} X^2 = 4|X|^2 = 4h(E_8) = 120$ — the Coxeter number carries the whole chain ([verification/v810_cutcode_moment.py]) **[E]**.

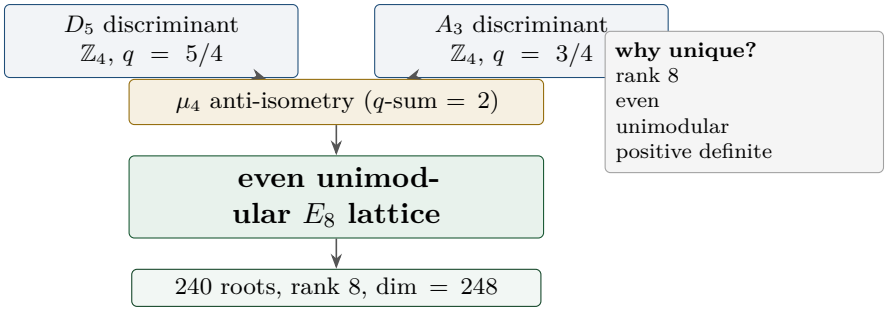
### 3.3 Compact $E_8$ compression and the role of $\nu^c$

$$\boxed{\begin{aligned} |R(E_8)| &= \dim S^+ \cdot g_{\text{car}} \cdot N_{\text{fam}} = 16 \cdot 5 \cdot 3 = 240, \\ \dim E_8 &= \dim S^+(\mathcal{A}_H + \mathcal{A}_\Lambda) + (g_{\text{car}} + N_{\text{fam}}) = 240 + 8 = 248 \end{aligned}} \tag{6}$$

with $N_{\text{fam}} = (2^{g_{\text{car}}-1} - 1)/g_{\text{car}} = 15/5 = 3$ and $\text{rank}\, E_8 = g_{\text{car}} + N_{\text{fam}} = 8$. **[E]**. The right-handed neutrino $\nu^c$ is the Pascal singleton $\Lambda^0 E = (1,1)_0$ that closes $16 = 1 + 5 + 10$; removing it breaks the truncation and shifts $\alpha^{-1}$ by $2.16 \times 10^{-3}$.

## 4 The $D_5 \times A_3$ compiler: $E_8$ as a closure, not an input

This section upgrades the $E_8$ atlas (Section 11) from a counting coincidence to an explicit lattice construction. The whole content is consolidated and verified in the standalone companion note *"The Even Carrier Code and the $\mathbb{Z}_{30}$ Coxeter Compiler."* The thesis: the carrier code is the $D_5$ half-spinor, the family geometry is an $A_3$, and $E_8$ is the standard unimodular glue closure of $D_5 \oplus A_3$ — with the four punctures $\mu_4$ as the glue code.




*$D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ — the unique even unimodular rank-8 glue [E] exact (lattice).*

**The compiler in one line:**

> five carrier slots $\Rightarrow$ $D_5$, four family corners $\mu_4$ $\Rightarrow$ $A_3$, $(D_5 \oplus A_3) + \mu_4$-glue $\Rightarrow$ $E_8$.

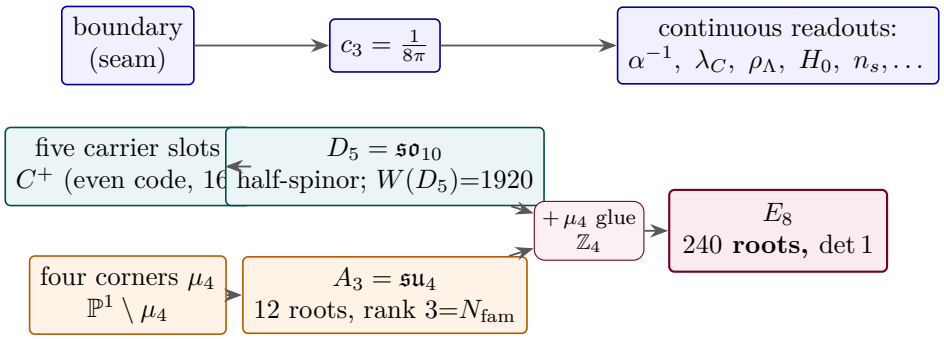
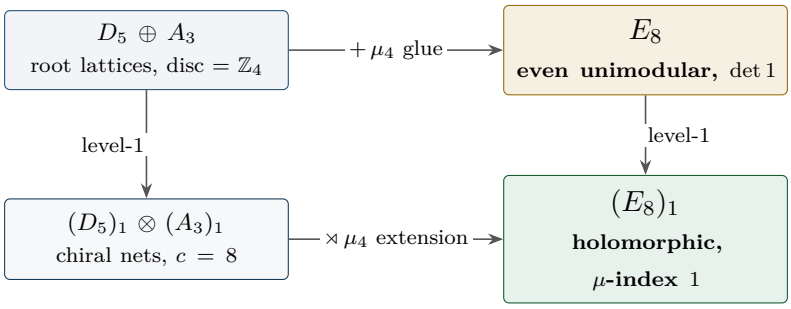



Figure 2: The explanation hierarchy. The boundary fixes the seam seed $c_3$; the five-slot even carrier code is the $D_5$ half-spinor; the four punctures $\mu_4$ give $A_3$; their common $\mathbb{Z}_4$ discriminant ($\mu_4$) glues $D_5 \oplus A_3$ into the unimodular $E_8$. $E_8$ is the *closure*, not the input; $c_3$ dresses the continuous observables.

**The glue is *one* operation in two categories.** The same order-$|\mu_4|$=4 discriminant glue closes $D_5 \oplus A_3$ into $E_8$ *twice over*: once on *root lattices* (the unimodular pushout $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$, v1) and once on *chiral conformal nets* (the simple-current / anyon-condensation extension $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 \cong (E_8)_1$, v92/v281). These are not two coincidences but the two functorial faces of a single commuting square (v234/v277/v260); the boundary realisation on which the strict-TOE residual turns (`SEAM.EQUIV.01`) is exactly the statement that the raw seam realises this square — now *closed modulo cited theorems on its MMST route* `SEAM.EQUIV.MMST.01` (the lattice/$S_3$ stack pins it at every computable level, Lean `FORM.SEAM.MMST.01`; only the cited MMST/OS continuum-existence theorems stay [O], v367/v368/v376–v379/v336; the twistor route `SEAM.EQUIV.TWISTOR.01` and the unconditional parent stay [O]). The extension leg of that citation package is itself re-founded at net level ([`verification/v469_seam_crossedproduct_route.py`]): the 128-spinor step is the *local* $\mathbb{Z}_2$ simple-current crossed product ($h_s(SO(16)_1) = 16/16 = 1 \in \mathbb{Z}$, the Longo–Rehren locality criterion; KLM $\mu = 4/2^2 = 1$), on 1995–2001 peer-reviewed subfactor theory, with the lattice-VOA preprint route demoted to an independent second witness, and the realisation input reduced to invariant level (R1′). Premise (i) of that package ($c_-$=8) now also carries a *ground-state* witness ([`verification/v489_seam_modular_commutator.py`]): the Kim–Shehab–Kim modular commutator $J(A,B,C) = (\pi/3)c_-$, evaluated on the exact BdG ground state of the collar model, gives $c_- = 8 = g_{\text{car}} + N_{\text{fam}}$ for the 16-layer carrier (per layer 0.500000 at $L$=48; trivial and orientation-flip controls pass) — independent of both the Chern route (v367) and the KLM count (v378). `SEAM.EQUIV.01` stays [O]. *Follow-up (2026-08-03, the $G_{\text{net}}$ witnesses):* the index-4 statement now carries two *exact* arithmetic/index witnesses. The NS/R grading *is* the parity character of $E_8(\mathbb{Z}[i])/(1{+}i) = \mathbb{F}_2^4$ — $(1{+}i)L$ lies in the even sublattice (proved on a $\mathbb{Z}$-basis), all $15 \times 16$ root classes are parity-pure (7 NS + 8 R), so the Ramond projection is $(1{+}i)$-*adic* at the one ramified edge of the $\mathbb{Z}[i]$-E8 Hecke tower ([`verification/v722_phys_ramified_ns_r.py`], 9/9, RAMIFIED-SEES-NSR); and on the CAR ladder the $\mu_4$-average expectation has optimal Pimsner–Popa constant exactly 1/4 and Watatani index exactly 4 via an explicit quasi-basis with *forced* weights, with $\mu_4$ *derived* from the clock and the Ramond sector healed state-preservingly by the Klein zero-mode dressing ([`verification/v726_phys_car_pp_index.py`], 57/57, T2-SLICE-GO). The remaining half is the *identification* of the ladder Q-system with the constructed $\mathbb{C}[\mathbb{Z}_4]$ object (`GNET.RAMIFIED.01`, kills registered); `GATE.METRIC.08/.10` typing unchanged. *Follow-up (2026-08-05, the continuum strand):* the finite-level precursors of the GNS limit are measured *coherent* ([`verification/v779_gnet_gns_arf.py`], 27/27 + 27/27): the Haar scaling tower intertwines exactly, the index-4 expectation squares commute exactly (the finite half of the Q-system identification), the state converges geometrically, and the exact dim law $d_c(\ell) = 4^{\ell-1} + 2^{\ell-1}\cos(\pi c/2)$ types the

$\mathbb{C}[\mathbb{Z}_4]$ obstruction at exactly $2^{-\ell}$ (explaining the $\ell = 1$ three-sector anomaly); the honest negative: the $16 = 1{+}5{+}10$ counting echo with the Arf geometry is exact but *non-geometric*, and the two fours (index 4 vs channel 4/49) stay in their registers. The round-21 martingale probe measures the limit-existence hypotheses of the continuum strand as met ([verification/v802_gnet_martingale.py], 15/15, MARTINGALE-HYPOTHESES-MET; the analytic remainder stays typed). Typing unchanged.




*One operation, two categories: the order-$|\mu_4|$=4 glue is the lattice pushout $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and the simple-current / anyon-condensation extension $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 \cong (E_8)_1$; the square commutes. [E] exact.*

## 4.1 Carrier $= D_5 = \mathfrak{so}_{10}$ (proved)

The 16 even-occupation states $C^+ = \{x \in \mathbb{F}_2^{g_{\rm car}} : |x| \text{ even}\}$ are exactly the weights of the $D_5$ half-spinor $\frac{1}{2}(\pm1, \ldots, \pm1)$ (even number of minus signs). The flip-weight-4 relation is the **Clebsch graph** SRG$(16, 5, 0, 2)$ and the flip-weight-2 relation is its complement SRG$(16, 10, 6, 6)$; the *affine* automorphism group of the scheme (translations $\rtimes$ coordinate permutations) is the $D_5$ Weyl group,

$$\boxed{\mathrm{Aut}_{\rm aff}(C^+) = C^+ \rtimes S_5, \quad |\,\mathrm{Aut}_{\rm aff}(C^+)| = 2^{g_{\rm car}-1} g_{\rm car}! = 1920 = |W(D_5)| = r\,|R(E_8)|} \quad \text{[E]}, \tag{7}$$

(the linear code automorphism group alone is only $S_5$; the order-1920 group is the affine/scheme one) and the Clebsch edge count is the flavor budget $|E(R_4)| = \frac{16\cdot5}{2} = 40 = |R(D_5)| = \sum_{f,j} L_{f,j}$. *Round-22 grading (2026-08-06):* the Arf split $16 = 1 + \bar{5} + 10$ *is* the cut classification $(0|5) + (1|4) + (2|3)$ of $K_5$ (the cut map injective-isometric on all 256 cells), and the Clebsch edges grade as $40 = 5{+}20{+}15$ over the Arf shells with the within-10 block equal to the Petersen graph — an Arf/cut geometry, exact but carrying no matter semantics ([verification/v810_cutcode_moment.py]) [E].

## 4.2 Family $= A_3 = \mathfrak{su}_4$ from $\mathbb{P}^1 \setminus \mu_4$ (proved, with the metric settled)

The carrier geometry fixes $X_f^\circ \cong \mathbb{P}^1 \setminus \mu_4$, $H_1 = \mathbb{Z}^3$, rigid $D_4$ generated by $z \mapsto iz$, $z \mapsto 1/z$. The twelve puncture-difference cycles $\gamma_i - \gamma_j$ carry the residue pairing $\mathrm{Res}(\omega_i, \gamma_j) = \delta_{ij}$ and form the $A_3$ root system:

$$\boxed{\#\{\gamma_i - \gamma_j\} = 12 = |R(A_3)| = \dim \mathfrak{g}_{\rm SM}, \quad \mathrm{rank} = 3 = N_{\rm fam}, \quad \mathrm{Gram} = \mathrm{Cartan}(A_3),\ \det = 4} \quad \text{[E]}. \tag{8}$$

The corner rotation $z \mapsto iz$ acts on $H^1 = \mathbb{C}^3$ as the $A_3$ Coxeter element (order $4 = h(A_3) = |\mu_4|$, spectrum $\{i, -1, -i\}$, $\chi = \Phi_4\Phi_2$), an isometry of the Cartan form — the exact analogue of the $E_8$ Coxeter element reading $\Phi_{30}$.

**The $\mu_4$ clock is the order-4 character of the $E_8$ cycle, and $(2, 3, 5)$ is the icosahedron ([verification/v223_coxeter_totative_clock.py], [verification/v219_icosahedral_mckay.py]).** The two "Coxeter" clocks are one object read at two scales. The $E_8$ exponents are the totatives $(\mathbb{Z}/30)^\times = \{1, 7, 11, 13, 17, 19, 23, 29\}$, and the multiplicative order-4 element 7 ($7^2$=19, $7^4$=1 mod 30) generates a literal $\mu_4$ clock $\langle 7\rangle = \{1, 7, 13, 19\}$ that permutes the four conjugate Coxeter 2-planes — the same $\mu_4$ as the seam corner rotation. And the period $30 = 2{\cdot}3{\cdot}5$ itself is not a free factorisation: by the McKay correspondence $E_8$ is the exceptional *top* of the tower of finite $SU(2)$ subgroups ($2T \to \hat{E}_6$, $2O \to \hat{E}_7$, $2I \to \hat{E}_8$), so its branch data is the *icosahedron* and the three atoms $(|\mathbb{Z}_2|, N_{\rm fam}, g_{\rm car}) = (2, 3, 5)$

are its rotation-axis orders. The binary icosahedral group $2I$ (order $120 = |R^+(E_8)|$) has nine irreducible-representation degrees $\{1,2,2,3,3,4,4,5,6\}$ equal to the affine-$E_8$ Kac marks, with $\sum d_i = 30 = h(E_8)$ and $\sum d_i^2 = 120$. This is a *backward certificate* of the closed $E_8$ [E] (it explains the $(2,3,5)$ choice [C]; it does *not* replace the P2 input).

**The metric, settled (former [C]).** The transcendental geometric (Green) metric $G_{ij} = -\log|p_i - p_j|$ on $\mu_4$ is only $D_4$-symmetric, with a regularisation-independent anisotropy $-\log 2$ on the root space, so it is *not* proportional to the $A_3$ Cartan form. But it does not need to be: the $E_8$-relevant $A_3$ is the canonical *residue* pairing (integral, $S_4$-symmetric, $=$ Cartan$(A_3)$), and the geometric metric is a $D_4$-equivariant deformation preserving the root system, Weyl group and Coxeter element. So the lattice structure is exact [E]; the analytic metric is a characterised deformation, not an open gap.

### 4.3 The $\mu_4$ glue and the $5/4 + 3/4 = 2$ discriminant theorem (proved)

The two factors have equal discriminant groups and $E_8$ is unimodular, so the glue index is exactly $|\mu_4|$:

$$\mathrm{disc}(D_5) = \mathrm{disc}(A_3) = \mathbb{Z}_4, \qquad [E_8 : D_5 \oplus A_3] = \sqrt{\frac{\det D_5 \det A_3}{\det E_8}} = \sqrt{\frac{4\cdot 4}{1}} = 4 = |\mu_4|. \tag{9}$$

The discriminant-form norms sum to the $E_8$ root norm, and — strikingly — they are pre-existing TFPT constants:

$$\boxed{\begin{aligned} &q(D_5) = \tfrac{5}{4} = \frac{\delta_2}{\delta_{\mathrm{top}}^2}, \qquad q(A_3) = \tfrac{3}{4} = \xi_{\mathrm{tree}}, \\ &q(D_5) + q(A_3) = 2 = |E_8 \text{ root}|^2 \end{aligned}} \qquad \text{[E]} \ [\texttt{verification/v1_e8_glue.py}] \tag{10}$$

This is literally the $E_8$ spinor-root split $\frac{1}{2}(\pm)^8 = \frac{1}{2}(\pm)^5 \oplus \frac{1}{2}(\pm)^3$, $5\cdot\frac{1}{4} + 3\cdot\frac{1}{4} = 2$.

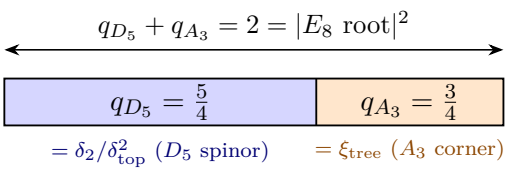



Figure 3: The discriminant-norm glue. The $E_8$ spinor root $\frac{1}{2}(\pm)^8$ (norm 2) splits across $D_5 \times A_3$ as $5\cdot\frac{1}{4} + 3\cdot\frac{1}{4}$; the two halves are the pre-existing TFPT constants $\delta_2/\delta_{\mathrm{top}}^2 = \frac{5}{4}$ and $\xi_{\mathrm{tree}} = \frac{3}{4}$.

The standard branching is realised explicitly:

$$\mathfrak{e}_8 = (45,1)\oplus(1,15)\oplus(10,6)\oplus(16,4)\oplus(\overline{16},\overline{4}), \qquad 248 = 45{+}15{+}60{+}64{+}64, \quad 240 = 40{+}12{+}60{+}64{+}64, \tag{11}$$

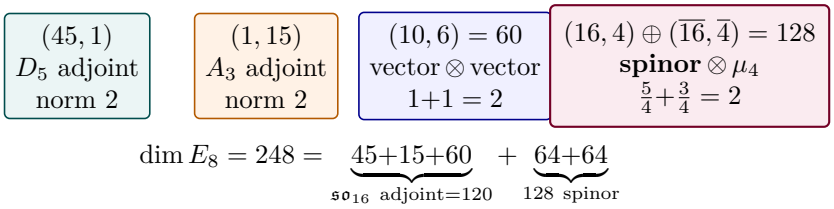



Figure 4: The $E_8 \supset D_5 \times A_3$ branching as norm-2 blocks. The adjoint half is the hypercharge moment $\mathrm{Tr}_{S^+} X^2 = 120 = \dim \mathfrak{so}_{16}$; the spinor half is two sheets over the four corners $\mu_4$.

and assembling $D_5 \oplus D_3 \subset D_8$ with the even spinor coset produces all 240 roots (norm 2, Cartan $\det = 1$, every root an integer combination of one simple-root basis): *the* 240*-metrisation is no longer blind — $E_8$ is built.* [E]

### 4.4 Glue-norm arithmetic, and three readings of 41, 60, 120

The two glue norms generate the carrier integers through sum, difference and product. With $\dim S^+ = 16$:

$$\boxed{\begin{aligned} &16(q_D{+}q_A) = 32 = 2^{g_{\rm car}} = h(D_5)h(A_3), \qquad 16(q_D{-}q_A) = 8 = r(E_8) = h(D_5), \\ &16\,q_Dq_A = 15 = \dim S^+{-}1 \end{aligned}} \quad \textbf{[E]}, \quad (12)$$

so the glue already contains the full carrier exhaustion $2^{g_{\rm car}}$, the $E_8$ rank, and the non-trivial transition count $\mathcal{A}_H{+}\mathcal{A}_\Lambda = 15$; and $|\mu_4| = (q_D-1)^{-1} = (1-q_A)^{-1} = 4$ is the inverse displacement of the two norms from the unit norm.

**Spine Quotient Lemma: the recurring factors are adjacent quotients of $(2,3,4,5)$ [E]** ( [verification/v74_compiler_micro_lemmas.py] )

The integers of the spine are $(|\mathbb{Z}_2|, N_{\rm fam}, |\mu_4|, g_{\rm car}) = (2,3,4,5)$. Every "mysterious" $O(1)$ factor in TFPT is an *adjacent quotient* of this chain, not a separate special number:

$$\tfrac{|\mathbb{Z}_2|}{N_{\rm fam}} = \tfrac{2}{3}, \quad \tfrac{N_{\rm fam}}{|\mu_4|} = \tfrac{3}{4} = q(A_3), \quad \tfrac{|\mu_4|}{g_{\rm car}} = \tfrac{4}{5}, \quad \tfrac{g_{\rm car}}{|\mu_4|} = \tfrac{5}{4} = q(D_5), \quad \tfrac{|\mu_4|}{N_{\rm fam}} = \tfrac{4}{3}.$$

These are exactly the factors that recur across the theory: $\lambda_2 = (\frac{2}{3})^6$ (the transport gap = Page recovery rate, and the Koide target $Q_\star = \frac{2}{3}$); $\varepsilon = \frac{3}{4}\varphi_0 = q(A_3)\varphi_0$ (the solar misalignment); $\varphi_0 = \frac{4}{3}c_3 + 48c_3^4$ with leading $\frac{4}{3}c_3 = \frac{1}{6\pi}$; and $q(D_5){+}q(A_3) = \frac{5}{4}{+}\frac{3}{4} = 2$ (the $E_8$ root norm). So $(\frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{4}, \frac{4}{3})$ are five projections of one discrete chain — not scattered constants. The spine is moreover *one* finite object: as a tetrahedron on $T = \{e_3(a), p_0(a), e_1(a), e_2(a)\}$ its edge and face products and its volume $120 = |R^+(E_8)| = 5!$ reproduce the central integer grammar, with the honest negative control that the graph reading $240 = |\mu_4| \cdot |E(K_4)| \cdot |E(K_5)|$ breaks at $K_6$ ( [verification/v91_spine_tetrahedron.py] ).

The carrier adjoint $\dim D_5 = |R(D_5)| + g_{\rm car} = 45$ then gives a second reading of the $U(1)$ integer,

$$\boxed{10b_1 = \dim D_5 - |\mu_4| = 45 - 4 = 41 = \underbrace{40}_{\sum L} + \underbrace{(g_{\rm car} - |\mu_4|)}_{=N_\Phi=1}} \quad \textbf{[E]}, \quad (13)$$

i.e. the single Higgs doublet is the carrier rank minus the glue corners, $5 - 4 = 1$.

*A fourth, metric reading of* 41 *( [verification/v222_cm_norm_duality.py] ).* On the square seam ($j$=1728, complex multiplication by the Gaussian integers $\mathbb{Z}[i]$) the $U(1)$ index is the Gaussian *norm* of the carrier-glue vector,

$$10b_1 = N_{\mathbb{Z}[i]}(g_{\rm car}{+}i|\mu_4|) = g_{\rm car}^2 + |\mu_4|^2 = 5^2 + 4^2 = 41,$$

and the hexagonal partner ($j$=0, Eisenstein $\mathbb{Z}[\omega]$) reads the same $(3,2)$ carrier split as the scalaron exponent $N_{\mathbb{Z}[\omega]}(3{+}2\omega) = 3^2{-}3{\cdot}2{+}2^2 = 7$. So 41 (EM) and 7 (scalaron) are the two exceptional CM norms of the one carrier, square vs. hexagonal — not two unrelated $O(1)$ numbers **[E]**. The start defect 60 and the hypercharge moment 120 are "bilingual":

$$\begin{aligned} D_{\rm start} = \dim D_5 + \dim A_3 = 45 + 15 = 60 = \mathcal{A}_\Lambda \cdot |R^+(A_3)| = 10 \cdot 6, \\ \mathrm{Tr}_{S^+} X^2 = 120 = \underbrace{60}_{\text{adjoint}} + \underbrace{60}_{\text{mixed }(10,6)}, \end{aligned} \quad (14)$$

and $\dim E_8 = 248 = \mathrm{Tr}_{S^+} X^2 + 2\dim S^+|\mu_4| = 120 + 2 \cdot 16 \cdot 4$. Finally the hypercharge norm and the EW exponent are $A_3$-rational,

$$\boxed{\gamma_{\rm car} = \frac{\mathcal{A}_\Lambda}{|R(A_3)|} = \frac{10}{12} = \frac{5}{6}, \qquad I_1^{\rm EW} = \frac{|E(K_5)|}{|R(A_3)|} + \frac{1}{|\mu_4||R(A_3)|} = \frac{10}{12} + \frac{1}{48} = \frac{41}{48}} \quad \textbf{[E]}. \quad (15)$$

*Transport reduction.* The flavor matrix $L$ has its mod-6 content fixed by a single combinatorial datum: $\{0,1,3\}$ is the *unique* (up to $D_6$) 3-subset of $C_6 = |R^+(A_3)|$ with all distinct cyclic distances

$\{1,2,3\}$, and the three sectors are its $D_6$ orbit ($u\equiv\{0,1,3\}$, $d\equiv 2-\{0,1,3\}$, $e\equiv 2+\{0,1,3\}$). The total is fixed by the carrier, $\sum L = 40 = |R(D_5)|$, with winding unit $6 = |R^+(A_3)|$. Given the residue *set* of a sector, the full ordered packet is reproduced by two universal rules — the lightest generation carries the single $+6$ winding, and the entries are sorted in decreasing length (monotone hierarchy). The *only* remaining datum is then, per sector, which residue is the first-generation anchor that receives the winding:

$$\boxed{\text{anchors } (a_u,a_d,a_e)=(1,1,2) \;\Rightarrow\; L_u=(7,3,0),\; L_d=(7,5,2),\; L_e=(8,5,3)} \qquad \text{[E] (given anchors).} \tag{16}$$

So the open step of a full "Clebsch walk" theorem is sharply isolated: it is exactly these three per-sector anchors. They are *not* free: the three transport cusps are exactly the $D_5$ dual-root combinations, and each sector is tied to its cusp by the right-handed hypercharge,

$$\boxed{\{1,\tfrac{2}{3},\tfrac{1}{3}\}=\{2y_+,\,-2y_-,\,2(y_++y_-)\},\qquad \text{lepton}\leftrightarrow 2y_+,\;\text{up}\leftrightarrow -2y_-,\;\text{down}\leftrightarrow 2(y_++y_-)} \qquad \text{[E].} \tag{17}$$

The canonical-transport-kernel theorem (document 3, Part II) derives the residue packets — and hence the anchors — as the *word-lengths of the family-connection holonomy* $\mathrm{Hol}_{\mathcal{G}_{D_4}}(\nabla_F^\star)$ along the $D_4$-invariant geodesic spine of $\mathbb{P}^1\setminus\mu_4$ with these cusps as boundary data. The anchors are therefore a genuine $D_5\times A_3$ readout (cusps from the $D_5$ carrier, spine holonomy from the $A_3$ four-punctured sphere), not an extra parameter. A compact candidate closed form is the nearest-integer cusp map

$$\boxed{a_f=\mathrm{round}(2\,\mathrm{cusp}_f):\quad (a_u,a_d,a_e)=\mathrm{round}(\tfrac{4}{3},\tfrac{2}{3},2)=(1,1,2)} \qquad \text{[C]}, \tag{18}$$

which reproduces the matrix exactly. The remaining open step is now precise and purely geometric: an *independent* closed-form evaluation of the $D_4$-spine holonomy word-lengths from the lattice data alone (the same $\mathbb{P}^1\setminus\mu_4$ whose residue pairing is Cartan$(A_3)$ and whose corner rotation is the $A_3$ Coxeter element), rather than from the family-connection theorem (document 3).

**Riemann–Hilbert check, and what it does (and does not) fix.** Setting up the rigid $\mathbb{Z}_4$-symmetric rank-3 $SU(3)$ local system explicitly — four puncture monodromies $M_k=R^kCR^{-k}$ with the order-3 cusp class $C\sim\mathrm{diag}(1,\omega,\omega^2)$ ($\omega=e^{2\pi i/3}$, the three cusps as eigenvalue-exponents) and the rotation $R$ ($R^4=\mathbb{1}$) — the closure $M_0M_1M_2M_3=\mathbb{1}$ has a solution to machine precision, and the unique $R$ has eigenvalues $\{i,-1,-i\}$: the $A_3$ Coxeter element. This *independently* confirms the rigidity of the family connection and the corner-rotation = Coxeter identity. However, the flat monodromy *alone* does not fix the integer word-lengths $\{0,1,3\}$: those are *parabolic degrees* (the cusp exponents as parabolic weights, together with the $\mathbb{Z}_2$ sheet parity), i.e. Hodge/metric data on the spine, not topological monodromy. *Status update (the computation has since been carried out):* the exact residue normal form $A_0^\star$ with spec $=\{0,\frac{1}{3},\frac{2}{3}\}$ is pinned algebraically ([verification/v115_anchor_residue.py], [verification/v116_resonance_uniqueness.py]), the unitarised holonomy is the exact 24-element $W(A_3)=S_4$ ([verification/v117_monodromy_weyl_a3.py]), and the parabolic weights are the $\mu_4$-character degrees of $H^1(\mathbb{P}^1\setminus\mu_4)$ — one spectrum in four readouts ([verification/v137_qplus_cohomology.py]); the deck itself is *derived* — the integer model selects the sheet-twisted class, no discrete freedom left ([verification/v141_deck_selection.py]), and the full Möbius $D_4$ of the curve matches the integer model parity by parity ([verification/v146_moebius_d4.py]). What remains of the old residual is the much smaller selector establishment R4′ (research contracts): derive the *one* lift map of the torsion normal — the pairing values are atom identities ([verification/v142_frame_integrality.py]/[verification/v145_pairing_atoms.py]). [E]/[C]

## 4.5 What the compiler outputs for the Standard Model and our reality

Every entry below is an *output* of $(c_3, g_{\mathrm{car}})$ through the $D_5\times A_3$ compiler; none is an independent input. "Reads" names the lattice object the observable comes from.

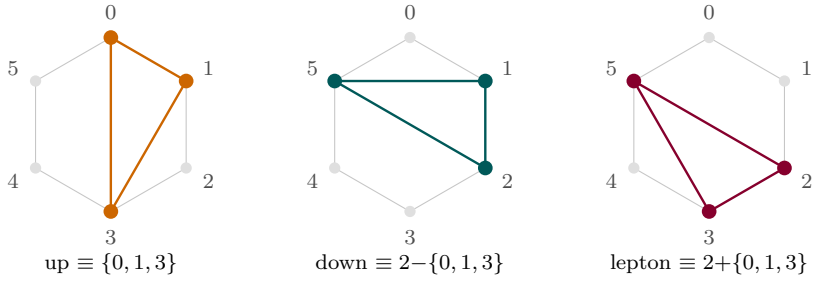



Figure 5: Flavor transport on the hexagon $C_6 = Z_6 = |R^+(A_3)|$. The unique distinct-distance triple $\{0,1,3\}$ (cyclic distances $\{1,2,3\}$) and its $D_6$ orbit give the three sectors' residues mod 6; the total $\sum L = 40 = |R(D_5)|$ and the first-generation +6 winding come from the carrier.

| **SM / cosmology observable** | **Compiler output** | **Reads** | **St.** |
|---|---|---|---|
| Number of generations | $N_{\text{fam}} = 3 = \text{rank}\, A_3 = \dim H^1(\mathbb{P}^1 \setminus \mu_4)$ | $A_3$ | [E] |
| Gauge algebra dimension | $\dim \mathfrak{g}_{\text{SM}} = 12 = \lvert R(A_3)\rvert = \gcd(6r, 2h)$ | $A_3$ | [E] |
| One generation (16, incl. $\nu^c$) | $S^+ = D_5$ half-spinor $= \Lambda^{0,2,4}E$ | $D_5$ | [E] |
| Hypercharge spectrum | all $Y$ from $y_\pm = \{\frac{1}{2}, -\frac{1}{3}\}$; $\text{Tr}\, X^2 = 120$ | $D_5$ root data | [E] |
| Abelian index $b_1$ | $10b_1 = 1 + \lvert\mu_4\rvert\lvert E(K_5)\rvert = 1 + 4 \cdot 10 = 41$ | $\mu_4 \times K_5$ | [E] |
| Fermion family occupancy | $\Omega_{\text{adm}} = N_{\text{fam}} \dim S^+ = (N_{\text{fam}}+N_\Phi)\lvert R(A_3)\rvert = 48$ | $A_3$ | [E] |
| Fine structure $\alpha_\star^{-1} = 137.0359992168$ | root of $F_{U(1)}(\alpha) = 0$ (explicit, unique) | EM closure | [E] |
| Cabibbo angle | $\lambda_C = \sqrt{u(1-u)}$, $u = \frac{4}{3}c_3 + 48c_3^4$ | seed | [E] |
| CKM CP phase | $\delta_{\text{CKM}} = \frac{2\pi}{\lvert R^+(A_3)\rvert} + N_{\text{fam}} u(1-u) = \frac{\pi}{3} + 3\lambda_C^2$ | $A_3^+$, seed | [E]/[C] |
| PMNS angles/phase | $\sin^2\theta_{12} = \frac{1}{3} - \frac{u}{2}$, $\sin^2\theta_{13} = e^{-5/6}u$, $\delta^\nu = \frac{4\pi}{3}$ | $A_3^+$, seed | [E]/[C] |
| Charged-lepton hierarchy | $\det M_\ell \propto u^{\,h/N_{\text{fam}}} = u^{10}$; $(5,3,2)$ | decuple | [E]/[C] |
| Mass-hierarchy transport | $\sum L = 40 = \lvert R(D_5)\rvert$; entries on $C_6 = \lvert R^+(A_3)\rvert$ | $D_5 \times A_3$ | [E] |
| Strong CP | $\theta_{\text{eff}} = 0$ (determinant erasure, given RP; document 3, Part III) | — | [E]/[C] |
| Weak CP | holonomy: $\delta_{\text{CKM}} - \frac{\pi}{3} = N_{\text{fam}}\, u(1-u)$ | $A_3^+$ | [C] |
| Gravity tree normaliser | $\xi_{\text{tree}} = \frac{\dim \mathfrak{g}_{\text{SM}}}{\dim S^+} = \frac{3}{4} = q(A_3)$ | $A_3$ glue | [E] |
| Cosmological constant | $\rho_\Lambda/\bar{M}_{\text{Pl}}^4 = \frac{3}{4\pi^2} e^{2\delta_{\text{ph}}} [v_{\text{geo}}/(5\beta_{\text{rad}}^2 \bar{M}_{\text{Pl}})]^{10}$ | decuple | [E] |
| Hubble anchor | $H_0\sqrt{\Omega_\Lambda} = \bar{M}_{\text{Pl}} e^{-\alpha_\star^{-1}}/(2\pi)$ | action | [E] |
| Baryon density | $\omega_b = \lambda_C/\mathcal{A}_\Lambda = \lambda_C/10$ | decuple | [E]/[C] |
| Inflation (scalaron) | $M_{\text{scal}}^2/\bar{M}_{\text{Pl}}^2 = c_3^{\Omega_{\text{adm}} - 10b_1} = c_3^7$ | deficit $r-1$ | [E] |
| Inflation amplitude | $A_s = N_\star^2 c_3^7/24\pi^2 \approx 2.2 \times 10^{-9}$ | deficit | [C]/[O] |

*Marker key:* ***[E]*** *exact / machine-proven ·* ***[C]*** *conditional (holds under named hypotheses) ·* ***[O]*** *open / axiom ·* ***[X]*** *falsifiable kill test. Per-claim fine type in* `verification/status_ledger.csv`.

*Reading.* The discrete sector — generations, gauge dimension, family content, hypercharge, $b_1$, occupancy — is now *group-theoretic output* of $D_5 \times A_3$. The continuous sector — $\alpha^{-1}$, $\lambda_C$, the mixings, $\rho_\Lambda$, $H_0$, inflation — is the seam seed $c_3$ and the action exponentials read on the *same* carrier graphs ($K_5$ pairs, the $C_6$ hexagon $= |R^+(A_3)|$, the decuple $\mathcal{A}_\Lambda = \binom{g_{\text{car}}}{2} = 10$). What used to be a list of separate hits is, structurally, one machine: $D_5$ carries the family and the flavor budget, $A_3$ carries the generations and the gauge dimension, $\mu_4$ glues them (and reappears as the 41 chain, the discriminant norms, and the CP phase step $2\pi/6$), and $c_3$ dresses everything.

# 5 Grammar I — the carrier grammar $(3+2)$

*Machine checks for this section:* [verification/v2_carrier_pascal.py]

From $g_{\rm car}=5$ and the split $(b,s)=(3,2)$ the carrier data follow:

$$\begin{aligned}\gamma_{\rm car}&=\frac{g_{\rm car}}{g_{\rm car}+1}=\frac{5}{6},\quad N_{\rm fam}=\frac{g_{\rm car}+1}{2}=3,\quad \dim S^+=2^{g_{\rm car}-1}=16,\\ \Omega_{\rm adm}&=N_{\rm fam}\,2^{g_{\rm car}-1}=48,\qquad b_1=\frac{g_{\rm car}\,2^{g_{\rm car}-2}+1}{10}=\frac{41}{10}.\end{aligned}\tag{19}$$

## 5.1 Dual-root dictionary (everything from two charges)

**Two carrier roots generate the family and the transport cusps [E].** With $y_+=\frac12$ and $y_-=-\frac13$ (the determinant-normalised carrier eigenvalues), the full one-family hypercharge spectrum and the flavor-transport cusps are sums of the two roots:

$$\begin{aligned}&Y(Q_L)=y_++y_-=\tfrac16,\quad Y(u^c)=2y_-=-\tfrac23,\quad Y(d^c)=-y_-=\tfrac13,\\ &Y(L_L)=-y_+=-\tfrac12,\qquad Y(e^c)=2y_+=1,\qquad Y(\nu^c)=0,\end{aligned}$$

$$\{1,\tfrac23,\tfrac13\}=\{\,2y_+,\ -2y_-,\ 2(y_++y_-)\,\},\qquad \gamma_{\rm car}=\mathrm{Tr}_E\,Y^2=3y_-^2+2y_+^2=\tfrac56,$$

$$\varphi_{\rm base}=\frac{1-\gamma_{\rm car}}{\pi}=\frac{1}{6\pi}.$$

All residuals are exactly zero. The Standard-Model charge table is not assumed; it is built by addition from $\{1/2,-1/3\}$.

**The $X=6Y$ quadratic and the hypercharge Pascal-sign duality [E]**

Clearing denominators with $X=6Y$ turns the two base charges into the integer roots $x_+=3$, $x_-=-2$ of

$$\boxed{x^2-x-6=0}:\qquad x_++x_-=1=N_\Phi,\quad x_+-x_-=5=g_{\rm car},$$

$$-x_+x_-=6=|R^+(A_3)|,\qquad \sqrt{\Delta}=5,$$

so the Higgs singleton, the carrier rank and the $A_3$ hexagon all fall out of one quadratic. Counting the signs of $X$ over a generation $(X=1^{\times6},2^{\times3},6^{\times1},(-4)^{\times3},(-3)^{\times2},0^{\times1})$ gives the *Pascal triple* again:

$$\boxed{(\#X{=}0,\ \#X{<}0,\ \#X{>}0)=(1,5,10)=\tbinom{5}{0},\tbinom{5}{1},\tbinom{5}{2}},$$

$$10{-}5{=}g_{\rm car},\quad 5{-}1{=}|\mu_4|,\quad 10{-}1{=}\,\mathrm{tr}\,R,$$

with $\nu^c\leftrightarrow\binom{5}{0}$ the neutral singleton — the right-handed neutrino is the Pascal zero with a job. The moment ladder adds $\mathrm{Tr}\,X^2=120=5!$ and $\mathrm{Tr}\,X^4/\mathrm{Tr}\,X^2=2280/120=19$ (an $E_8$ exponent; audit). [E]

### 5.2 Carrier arithmetic (proved identities)

The discrete coefficients are not isolated numbers; they are exact carrier traces:

$$10\,b_1 = 41 = g_{\rm car}^2 + (g_{\rm car}-1)^2 = \Omega_{\rm adm}\,\gamma_{\rm car} + 1 = 1 + h_{E_8}\frac{\dim S^+}{\dim \mathfrak{g}_{\rm SM}} = 1 + 30\cdot\tfrac{16}{12} = 40+1, \quad \textbf{[E]} \tag{20}$$

$$I_1^{\rm EW} = \frac{41}{48} = \mathrm{Var}_{S^+}(Y)\cdot b_1 = \frac{5}{24}\cdot\frac{41}{10} = \gamma_{\rm car} + \frac{1}{\Omega_{\rm adm}}, \quad \textbf{[E]} \tag{21}$$

$$\mathrm{Tr}_{S^+}(X^2) = 120 = 5! \quad (X = 6Y), \qquad \delta_2 = \tfrac{5}{4}\delta_{\rm top}^2 = 2880\,c_3^8 = 4!\cdot 5!\,c_3^8 = 4!\;\mathrm{Tr}_{S^+}(X^2)\,c_3^8, \quad \textbf{[E]} \tag{22}$$

$$\frac{5}{4} = \underbrace{\tfrac{3}{2}\cdot\tfrac{5}{6}}_{B\gamma} = \underbrace{\frac{D_{\rm start}}{\Omega_{\rm adm}}}_{60/48} = \underbrace{\frac{g_{\rm car}}{g_{\rm car}-1}}_{5/4} = \frac{\delta_2}{\delta_{\rm top}^2}. \quad \textbf{[E]} \tag{23}$$

So: $b_1$ is a Pythagorean carrier identity ($5^2+4^2$); equivalently the anchor-product form $10b_1 = p_1p_3 + (g_{\rm car}-p_1) = 4\cdot 10 + (5-4) = 41$ reads "all four glue corners times all ten carrier pairs, plus the one leftover Higgs slot"; the electroweak exponent $41/48$ is the hypercharge variance times the $U(1)$ budget; the second seam defect $2880\,c_3^8$ is a product of factorial traces $4!\cdot 5!$; and the universal compression $5/4 = g_{\rm car}/(g_{\rm car}-1)$ recurs whenever the five-slot carrier projects onto four effective directions. As a minor curiosity, the SU(5)-tree weak mixing angle rewrites as $\sin^2\theta_W^{\rm tree} = \frac{3}{8} = N_{\rm fam}/(g_{\rm car}+N_{\rm fam}) = N_{\rm fam}/\,\mathrm{rank}(E_8)$ (the value is the standard one; the rewrite is a coincidence to note, not a claim). **[C]**

## 6 Grammar II — the seed grammar (one decoder $u$)

*Machine checks for this section:* [`verification/v106_review_validation.py`]

The seed is, exactly, a pure function of $c_3$:

$$\boxed{u = \varphi_0^{\rm ret} = \frac{4}{3}\,c_3 + \Omega_{\rm adm}\,c_3^4 = \underbrace{\frac{1}{6\pi}}_{\varphi_{\rm base}} + \underbrace{\frac{3}{256\pi^4}}_{\delta_{\rm top}} = 0.0531719521768} \quad \textbf{[E]}. \tag{24}$$

Its two coefficients are themselves carrier data: $\frac{4}{3} = \dim S^+/\dim\mathfrak{g}_{\rm SM} = 16/12 = 8/(g_{\rm car}+1)$ (so $\varphi_{\rm base} = 1/((g_{\rm car}+1)\pi)$) and $48 = \Omega_{\rm adm}$. The seed is thus a pure function of $c_3$ and the carrier counts.

**One seed, four readouts (and their inverses) [E].**

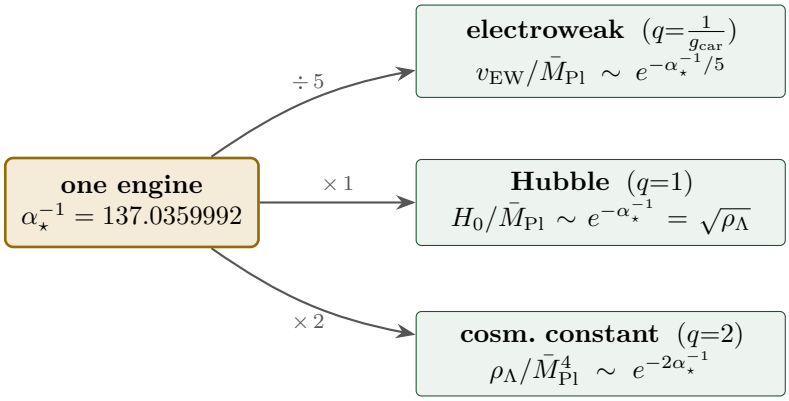
$$\lambda_C^2 = u(1-u), \quad \beta_{\rm rad} = \frac{u}{4\pi}, \quad \Omega_b = \Big(1-\frac{1}{4\pi}\Big)u, \quad \sin^2\theta_{13} = e^{-5/6}\,u,$$

$$u = \frac{1-\sqrt{1-4\lambda_C^2}}{2} = 4\pi\,\beta_{\rm rad} = \frac{\Omega_b}{1-1/(4\pi)} = e^{5/6}\sin^2\theta_{13}$$
$$= \sum_{n\ge 0} C_n\,\lambda_C^{2n+2} = \lambda_C^2 + \lambda_C^4 + 2\lambda_C^6 + 5\lambda_C^8 + \cdots$$

One knob $u$ turns four dials simultaneously: Cabibbo mixing (the Bernoulli width $\sqrt{u(1-u)}$), cosmic birefringence, the baryon fraction, and the reactor angle. Turning one forces the others; there is no free per-observable tuning. The exponent $5/6$ is the carrier trace $\gamma_{\rm car} = \mathrm{Tr}_E\,Y^2$, *not* the Euler–Mascheroni constant $0.5772\ldots$ (audit **[O]**).

# 7 Grammar III — the action grammar (exponentials of $\alpha_\star^{-1}$)

**The relation in plain words.** The fine-structure constant is not just "a number near 137": it is the *one exponential engine* behind three wildly separated scales. Take the same $\alpha_\star^{-1} = 137.036$ and divide the carrier into it: $\frac{1}{5}$ of it sets the *electroweak* scale, $1\times$ it sets the *Hubble* scale, $2\times$ it sets the *cosmological constant*. The three action charges are the Pascal row $1:5:10$ of the five-slot carrier, and the Hubble rung is just the square root of the vacuum energy. So one constant fixes the electroweak hierarchy, dark energy and the expansion rate at once.




action charges $1 : g_{\rm car} : 2g_{\rm car} = 1:5:10 = \binom{5}{0} : \binom{5}{1} : \binom{5}{2}$ (Pascal row of $K_5$)

Read every dimensionless ratio through its *action* and ask for the action charge:

$$\boxed{x = R_x\, e^{-A_x}, \qquad A_x = q_x\, \alpha_\star^{-1} + \Delta_x, \qquad q_x \in \Big\{\tfrac{1}{g_{\rm car}},\, 1,\, 2\Big\}.} \tag{25}$$

The robust ladder and its residues are

| sector | action $A_x$ | charge $q_x$ | residue $R_x$ | value of $-\ln x$ |
|---|---|---|---|---|
| electroweak $v_{\rm geo}/\bar{M}_{\rm Pl}$ | $(\alpha_\star^{-1} + \delta_{\rm ph})/g_{\rm car}$ | $1/g_{\rm car}$ | $g_{\rm car}\beta_{\rm rad}^2$ | 36.85 |
| Hubble $H_0/\bar{M}_{\rm Pl}$ | $\alpha_\star^{-1}$ | 1 | $1/(2\pi\sqrt{\Omega_\Lambda})$ | 138.68 |
| cosm. const. $\rho_\Lambda/\bar{M}_{\rm Pl}^4$ | $2\alpha_\star^{-1}$ | 2 | $3/(4\pi^2)$ | 276.65 |

$$A_{\rm EW} = \frac{\alpha_\star^{-1} + \delta_{\rm ph}}{g_{\rm car}} = 27.5259, \qquad \mathcal{A}_\Lambda = 2\alpha_\star^{-1} = 274.072, \qquad \boxed{A_{\rm EW}{:}\mathcal{A}_H{:}\mathcal{A}_\Lambda = 1{:}5{:}10}. \tag{26}$$

The leading ratio is exact: $\mathcal{A}_\Lambda/(\alpha_\star^{-1}/g_{\rm car}) = 2g_{\rm car} = 10$ (with the transport correction $\mathcal{A}_\Lambda/A_{\rm EW} = 9.957$). The claim is that the *action charges*, not the raw logarithms, form the ladder.

## 7.1 The EM closure: $\alpha_\star^{-1}$ as an explicit fixed point (made explicit here)

The engine $\alpha_\star^{-1}$ is not an input — it is the unique root of an explicit closure equation (the exact electromagnetic closure theorem). With $c_3 = 1/(8\pi)$, $\varphi_{\rm base} = 1/(6\pi)$, $q(\alpha) = \delta_{\rm top} e^{-2\alpha}$ and the seam response $\varphi_{\rm seam}(\alpha) = \varphi_{\rm base} + q(\alpha)(1-q(\alpha))^{-5/4}$,

$$\boxed{F_{U(1)}(\alpha) = \alpha^3 - 2c_3^3\alpha^2 - \tfrac{4}{5}c_3^6\Big(\textstyle\sum_{f,j} L_{f,j} + N_\Phi\Big)\log\frac{1}{\varphi_{\rm seam}(\alpha)} = 0} \qquad \text{[E]/[C]}, \tag{27}$$

where the coefficient is $\frac{4}{5}(\sum L + N_\Phi)c_3^6 = \frac{4}{5}\cdot 41\, c_3^6 = 8b_1 c_3^6$, so the fine-structure constant reads the *flavor transport budget* $\sum L = 40$ and the $U(1)$ index $b_1 = 41/10$.

**Lemma 1** (Existence and uniqueness of the EM fixed point)**.** $F_{U(1)}$ has *exactly one* positive root, and it is simple.

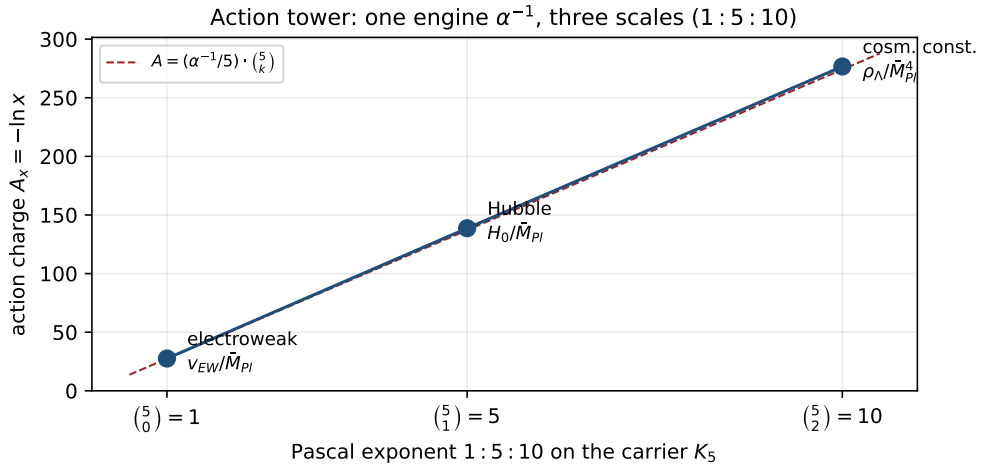



Figure 6: Action tower (data plot, `verification/make_figures.py`). The three action charges $A_x = -\ln x$ for the electroweak, Hubble and cosmological-constant ratios fall on the line $A = (\alpha_\star^{-1}/5)\binom{5}{k}$: one engine $\alpha_\star^{-1}$ generates all three scales with the Pascal exponents $1 : 5 : 10$.

*Proof.* Write $F_{U(1)}(\alpha) = h(\alpha) - C\,g(\alpha)$ with $h(\alpha) = \alpha^3 - 2c_3^3\alpha^2$, $C = \frac{4}{5}c_3^6 M > 0$ ($M = 41$), and $g(\alpha) = -\log\varphi_{\rm seam}(\alpha)$. Since $q(\alpha) = \delta_{\rm top}e^{-2\alpha} \in (0,1)$ is strictly decreasing, $\varphi_{\rm seam} = \varphi_{\rm base} + q(1-q)^{-5/4}$ is $C^1$ and decreasing, so $g$ is $C^1$, bounded, and slowly increasing with $g'(\alpha) = O(q)$ tiny; numerically $g \in [2.9342, 2.9343]$ on the admissible window. The cubic $h$ has its interior critical point at $\alpha_c = \frac{4}{3}c_3^3 = 8.399\times10^{-5}$, with $h(0) = 0$ and $h < 0$ on $(0,\alpha_c)$. *Monotonicity (corrected endpoint).* Since $F'_{U(1)}(\alpha) = 3\alpha^2 - 4c_3^3\alpha - C\,g'(\alpha)$ and at $\alpha_c$ the first two terms *cancel* ($3\alpha_c = 4c_3^3$), one has $F'_{U(1)}(\alpha_c) = -C\,g'(\alpha_c) \approx -5.89\times10^{-10} < 0$: the derivative is *not* yet positive at $\alpha_c$. Its first zero lies slightly above, at $\alpha_0 = 8.6264\times10^{-5}$ (where $3\alpha_0^2 - 4c_3^3\alpha_0 = C\,g'(\alpha_0)$), with $F'_{U(1)} < 0$ on $(\alpha_c,\alpha_0)$ and $F'_{U(1)} > 0$ on $(\alpha_0,\infty)$. Now $F_{U(1)}(0) = -C\,g(0) < 0$, and $F_{U(1)}$ *decreases* on $(0,\alpha_0]$ (both $h$ and $-Cg$ decrease on $(0,\alpha_c]$, then $F'_{U(1)} < 0$ on $[\alpha_c,\alpha_0]$), so $F_{U(1)}(\alpha_0) \approx -3.82\times10^{-7} < 0$; thereafter $F_{U(1)}$ is strictly increasing to $+\infty$. Hence $F_{U(1)} < 0$ on $(0,\alpha_0]$ and strictly increasing on $[\alpha_0,\infty)$, so it crosses zero *exactly once*. The crossing has $F'_{U(1)}(\alpha_\star) = 1.58\times10^{-4} > 0$, so the root is simple. A sign scan on $(0, 0.05)$ confirms a single change. *Rigorous version:* an **interval-arithmetic** enclosure (`mpmath.iv`) brackets the root in $[0.00729735256220970,\ 0.00729735256221000]$ with $F_{U(1)} < 0$ on the lower and $> 0$ on the upper endpoint (definite signs), and a 200-cell interval partition of $(\alpha_c, 0.05)$ contains exactly *one* indeterminate (root-bearing) band — a verified uniqueness, not merely a sampled one. [`verification/v3_em_alpha.py`] □

Solving:

$$\boxed{\alpha_\star = 0.00729735256220985\ldots,\quad \alpha_\star^{-1} = 137.0359992168407\ldots} \quad \textbf{[E]} \tag{28}$$

[`verification/v3_em_alpha.py`] versus CODATA-2022 $\alpha_\star^{-1} = 137.035999177(21)$ — a relative deviation $2.9\times10^{-10}$ ($\approx 1.9\sigma$ on the tiny experimental error; the older 2018 value 137.035999084 gave $9.7\times10^{-10}$, so the agreement *improved* in relative terms). The exponent $-5/4$ in $\varphi_{\rm seam}$ is the carrier discriminant norm $q(D_5)$, and $\delta_{\rm top} = 48c_3^4$, so the closure is built only from $c_3$, the carrier, and the flavor budget. This discharges the former open audit point: the EM closure is a stated, existence-/uniqueness-proven, numerically reproduced equation, not a fit.

**Why *this* function: the boundary $U(1)$ Ward reading (Theorem C) [E]/[C].** $F_{U(1)}$ is not a chosen ansatz but the reduced Ward identity of the boundary $U(1)$ partition function $Z_\partial = Z_{\rm Maxwell}\,Z_{\rm Calderón}\,Z_{\rm transport}$, with three terms whose coefficients are compiler atoms, not fits:

$$\underbrace{\alpha^3}_{\text{Maxwell moment}} \quad \underbrace{-\,2c_3^3\alpha^2}_{\text{Calderón sheet } (2=|\mathbb{Z}_2|)} \quad \underbrace{-\,8b_1c_3^6\log\tfrac{1}{\varphi_{\rm seam}}}_{C_6 \text{ transport det.}} = 0, \qquad 8b_1 = \tfrac{4}{5}M = \tfrac{4}{5}(\textstyle\sum L + N_\Phi) = \tfrac{164}{5}.$$

The Calderón factor $2 = |\mathbb{Z}_2|$ (sheet), the transport power $c_3^6$ is the $C_6$ hexagon, and the seam exponent $-\frac{5}{4} = -q(D_5)$ is the carrier discriminant norm. **Typing:** the three-term decomposition and the coefficient identities ($8b_1 = \frac{4}{5}M$, $-\frac{5}{4} = q(D_5)$) are exact [E]; the *physical origin* as a boundary Ward closure (the lemmas $\partial_\tau \log Z_{\text{Maxwell}} = \alpha^3$, $\partial_\tau \log Z_{\text{Calderón}} = -2c_3^3\alpha^2$, $\partial_\tau \log \det_\zeta(1 - \mathcal{T}_{C_6,U(1)}) = -8b_1c_3^6 \log \frac{1}{\varphi_{\text{seam}}}$) is a conjectured [C] interpretation, not yet machine-proven. [verification/v48_em_ward.py]

**The determinant-line (Quillen) form.** ([verification/v341_alpha_quillen.py]) The three terms assemble into a single *stationarity* statement for the boundary $U(1)$ determinant line,

$$\underbrace{[\alpha^3 - 2c_3^3\alpha^2]}_{\delta \log\det \Delta_{U(1)}} = \underbrace{[8b_1c_3^6]}_{\text{ctr.}}\ \underbrace{[\log \tfrac{1}{\varphi_{\text{seam}}}]}_{\text{seam resp.}},$$

so $F_{U(1)} = 0$ is $\delta \log\det \Delta_{U(1)}$ + counterterm + seam response = 0. Every factor is a *named* object, not a knob: the counterterm coefficient is $8b_1c_3^6 = (\text{rank } E_8)\, b_1\, c_3^6$ (index × heat-kernel), $c_3 = 1/(8\pi)$ is the Gauss–Bonnet boundary coefficient, the exponent $-\frac{5}{4} = -q(D_5)$ is one carrier discriminant norm and the seam base $\varphi_{\text{base}} = 1/(6\pi) = \frac{4}{3}c_3$ carries the *other*, $c_3/\varphi_{\text{base}} = \frac{3}{4} = q(A_3)$ — both glue forms appear, and the split is verified at the root. **Typing (and: no new gate).** The determinant-line assembly and all coefficient identities are exact [E]; $\alpha_\star^{-1}$ itself stays closed [E] (unique root + ablation). This adds *no* open item — it *sharpens* the pre-existing [O]/physical-origin obligation `EM.WARD.01` (the "why *this* function" residual, already conjectural) by naming every coefficient. The one still-open part *is* that residual, now *named* as its own tracked target `ALPHA.QUILLEN.EXACT.01` ([verification/v382_alpha_quillen_exact.py]): the from-first-principles AQFT/$\zeta$ proof that $\delta_\tau(\log\det_\zeta \Delta_{U(1)} + 8b_1c_3^6 \log\varphi_{\text{seam}}) = 0$ — i.e. that $F_{U(1)}$ is the *exact* Quillen functional of the seam $U(1)$, the variational principle *derived*, not assembled. It is a *face* of `SEAM.EQUIV.01` (v336) — the seam $U(1)$ determinant line lives on the same holomorphic net, so once the net is $(E_8)_1$ its $U(1)$ sub-determinant is fixed — hence no second independent gate, just the sharpest remaining EM-Ward obligation, now citable on its own. *Honest progress towards it ([verification/v391_alpha_quillen_progress.py]), without closing it:* a solvable model $\det_\zeta(-\partial_\tau^2+m^2)$ on a circle $= (2\sinh(\beta m/2))^2$ has the closed $m$-variation $\beta\coth(\beta m/2) = 2/m + (\beta^2/6)m - (\beta^4/360)m^3 + \ldots$, a heat-kernel/Laurent series — so a $\zeta$-determinant's variation is *structured* heat-kernel data, and the counterterm $8b_1c_3^6$ (a Gauss–Bonnet heat-kernel power) is the right *kind* of object. This *reduces* `ALPHA.QUILLEN.EXACT.01` to the named sub-target "the seam $U(1)$ Seeley–DeWitt coefficient $= 8b_1c_3^6$", but the target itself stays [O] (an external math proof, v384); the value $\alpha_\star^{-1}$ stays [E]. *A second honest step ([verification/v433_alpha_quillen_heatkernel.py]) bridges this reduction to the heat-kernel origin below, without closing it:* v391's circle reached only the low orders $a_0, a_1$, whereas a solvable *4D* model — the flat $\Delta = -\partial^2+m^2$ on $T^4$, $\text{Tr}\, e^{-t\Delta} = V/(4\pi)^2\, t^{-2}e^{-tm^2}$ — has $a_4 = Vm^4/(2(4\pi)^2) \neq 0$, *reaching* the $t^0$ (conformal-anomaly) order the circle could not, the very order in which the quadratic term is grounded below (v342, Gilkey $a_4$). So a $\zeta$-determinant's variation is heat-kernel data *at the $a_4$ order*, and the measure itself is $c_3$-built, $(4\pi)^{-2} = (2c_3)^2 = 4c_3^2$; the *actual* seam $a_4 = 8b_1c_3^6$ and the cubic $\alpha^3$ Chern moment stay [O] (v384), the seam $F$-normalisation [C] (retyped by the fifth step below, v470, as the embedding index $k_Y$=5/3).

**The heat-kernel origin of the terms ([verification/v342_em_ward_heatkernel.py]).** The determinant-line form is more than bookkeeping: its terms are textbook heat-kernel coefficients. [E] $c_3 = 1/(8\pi)$ *is* the boundary Gauss–Bonnet coefficient ($\oint_{S^2} K = 2\pi\chi = 4\pi$, $c_3 = 1/(|\mathbb{Z}_2| \oint K)$, the same $\chi$=2 as the marks, v216). [E] the $\alpha^2$ (Calderón) term *is* the Gilkey gauge-curvature coefficient: the $U(1)$ Laplacian's Seeley–DeWitt $a_4$ contains $30\,\Omega_{ij}\Omega^{ij}/360 = \frac{1}{12}\Omega^2$ (Gilkey, *Invariance Theory*, Thm 4.8.16), and with $\Omega = i\alpha F$ this forces an $-(\alpha^2/12)F^2$ contribution to $\log\det \Delta_{U(1)}$ — so the Calderón $-2c_3^3\alpha^2$ is the gauge-curvature (Quillen-curvature) piece, not an ansatz. [E] the $c_3$-orders $\{0,3,6\}$ are an arithmetic ladder (step 3 = the three channels; $6 = |R^+(A_3)|$, the $C_6$ hexagon), and the $\pi$-power test confirms it: $2c_3^3 = 1/(256\pi^3)$ carries $\pi^3$, i.e. *three* boundary $c_3$-insertions, not a single bulk coefficient. **Typing:** the boundary structure and the gauge-curvature origin of the quadratic term are heat-kernel-forced [E]; the exact coefficient value

is [C] (the seam $F$-normalisation — retyped as the embedding index $k_Y$=5/3 by the fifth step below, v470); the cubic $\alpha^3$ Maxwell moment (a topological/Chern level) and the full "this *is* the seam $U(1)$ $\zeta$-determinant" claim stay [O] — exactly the `EM.WARD.01` residual, unchanged. So the quadratic/curvature half of the EM-Ward origin is now derived; the cubic-moment half remains the honest open part. *A third honest step (`[verification/v434_alpha_quillen_betafunction.py]`) settles those residuals and shows they are not three independent problems.* The counterterm's *matter* factor is heat-kernel-forced too: by the $\beta = a_4$ theorem the one-loop $U(1)_Y$ $\beta$ *is* the $a_4$ coefficient, and the carrier hypercharge content gives $\sum \frac{2}{3}Y_f^2 + \frac{1}{3}Y_s^2 = 41/6$ (SM) $\Rightarrow b_1 = 41/10$ (GUT, v159/v246), so $8b_1c_3^6 = b_1\,(\mathrm{rank}\,E_8)\,c_3^6$ is an $a_4$ content coefficient times a boundary geometry — the same *kind* of object as the Calderón $a_4$. The exact value then turns only on the seam $F$-normalisation, so the "actual seam $a_4$" residual *is* the $F$-normalisation [C] (one obligation, not two); and the cubic factors $a^3 - 2c_3^3a^2 = a^2(a - 2c_3^3)$, exhibiting $a^3$ as the first moment of the Chern density $a^2$ — a level whose from-first-principles origin stays [O]. The EM-Ward residual is thus *one* [C] + *one* [O], not three. *A fourth honest step (`[verification/v435_alpha_quillen_chernlevel.py]`) sharpens that single remaining [O] — residual (2), the cubic $\alpha^3$ Chern level — without closing it.* A $\pi$-power / metric-independence test partitions the closure: the three coefficients carry $\pi$-powers exactly $\{0, 3, 6\}$ ($k_0 = 1$; $-2c_3^3 = -1/(256\pi^3)$; $-8b_1c_3^6 = -41/(327680\pi^6)$), so the Maxwell $a^3$ is the *unique* $\pi^0/c_3^0$ (metric-*independent*) term. A heat-kernel boundary coefficient always carries the $(4\pi)^{-d/2}$ measure; a metric-independent term carries none — the signature of a *topological* term. So the test confirms, now from a test rather than a claim, that $a^3$ is a level and not a Seeley–DeWitt curvature integral (the other two terms are heat-kernel boundary data). [E] *Conditional* on the Chern–Simons/$\eta$ reading of the Maxwell moment (the `EM1` interpretation, v48), large-gauge invariance quantises the seam $U(1)$ level to $\mathbb{Z}$, so $k_0 = 1$ is the *unit* (minimal) level — an integer, not a tunable real. [C] What stays [O] is the `EM1` lemma itself — the from-first-principles Chern–Simons-boundary proof that $\partial_\tau \log Z_{\mathrm{Maxwell}} = a^3$ — external math (type A, v384). *A fifth honest step (`[verification/v470_alpha_inflow_level.py]`) upgrades both leftovers, still without closing the target.* [C] *The level is computed, not minimal:* under the same `EM1` reading, the $a^3$ coefficient $k_0 = 1$ *equals* the FHS bulk Chern invariant $|C| = 1$ of the very p+ip collar that realises S3 of `SEAM.EQUIV.01` (v367/v461; control $C(M{=}3) = 0$) — "why exactly 1" is answered by a computation plus the cited inflow identification (TKNN / Avron–Seiler–Simon quantisation; Callan–Harvey inflow; the APS/Witten $\eta$=CS reading of $\delta \log\det$; Quillen/Bismut–Freed determinant-line integrality), *replacing* the minimality assumption of the fourth step. [E] *The $F$-normalisation is the embedding index:* $k_Y = \mathrm{tr}_{\bar{5}}(Y^2)/\,\mathrm{tr}(T_3^2) = (5/6)/(1/2) = 5/3$ (Ginsparg 1987), so $(3/5) \times \frac{41}{6} = \frac{41}{10} = b_1$ exactly — the "GUT 3/5" *is* the inverse embedding index, and once the seam net is $(E_8)_1$ at level 1 the $U(1)_Y$ normalisation $\langle J(z)J(w)\rangle = k/(z{-}w)^2$ is *fixed* by it (level-1 current-algebra rigidity, no continuous dial): the seam $F$-normalisation [C] has *zero independent content* — a face of `SEAM.EQUIV.01`, per the v382 typing. [C] *One phase, two responses:* the collar phase carries $c_- = 8$ (gravitational, feeding `SEAM.EQUIV.01`/S3, v456) *and* $C = 1$ ($U(1)$, feeding `EM1`) — so discharging the one realisation input feeds both named targets at once. *A sixth honest step (`[verification/v472_quillen_detline_moduli.py]`) exhibits that bridge lemma at the finite level.* `ALPHA.QUILLEN.EXACT.01` names the level as the curvature of the *determinant line over the $U(1)$ moduli* (Quillen/Bismut–Freed curvature, Dai–Freed section) — and the fifth step computed the Chern invariant only over the Bloch Brillouin zone, the translation-invariant shortcut. The sixth step computes the Quillen-shaped object itself: the same collar Hamiltonian in real space with twisted boundary conditions — the twist torus $(\theta_x, \theta_y)$ *is* the moduli space of flat $U(1)$ connections — and the Fukui–Hatsugai–Suzuki curvature of the determinant line of the occupied frame over that torus integrates to $2\pi \cdot 1$ exactly ($10^{-9}$), size-independently ($L = 4, 6$), with clean controls (trivial collar $\to 0$, orientation flip $\to -1$), the twist-moduli integer equal to the Bloch-BZ integer for all three phases (Niu–Thouless–Wu), and the Fermi gap open over the whole torus (the Dai–Freed section has no zero). So "$\delta \log\det$(seam model) = the inflow response" *holds verbatim at the finite/model level*, where $\log\det$ of the occupied frame replaces $\log\det_\zeta$. [E] (computation) / [C] (the EM1 reading). What stays [O] is the *continuum* identification "$\delta \log\det_\zeta$(seam) =

the inflow response" on the abstract seam (the `SEAM.EQUIV.01` face); `ALPHA.QUILLEN.EXACT.01` stays [O], and $\alpha_\star^{-1}$ stays [E] regardless. *A seventh step unifies this target with the $\varphi_0^{\mathrm{ret}}$-puncture target (`[verification/v484_seam_contact_unit.py]`).* The $\{0,3,6\}$ $c_3$-ladder above and the puncture's per-mark weight $c_3$ (v483) are *one* rule — the seam contact unit — and it is not a new unknown: "$c_3$ per insertion" *is* the KMS seam unit $2\pi = 1/(4c_3)$ with $\frac{1}{4} = 1/|\mu_4|$ (v239), i.e. one bare boundary propagator ($\frac{1}{2\pi} = 4c_3$) orbit-averaged over the four marks. [E] derived on the seam circle for the finite cycle sector (the bare Green function takes integer multiples of $c_3 \ln 2$ at the $\mu_4$ separations; the log-det contact expansion is exactly graded in $c_3$ per insertion), with the grammar already suite-wide (the $\Lambda$ prefactor $\frac{3}{4\pi^2} = 48c_3^2$, v60, carries the same $\Omega_{\mathrm{adm}} = 48$ at two insertions). [C]/[O] Consequence: `ALPHA.QUILLEN.EXACT.01` and `HYP.PHI0.PUNCTURE.01` *merge* into one remaining analytic step — the diagonal zeta-renormalisation at the marks plus the state-multiplicity matching ($41 = 10b_1$ here, $48 = \Omega_{\mathrm{adm}}$ there). *An eighth step settles that remaining step at the computable level (`[verification/v485_contact_diagonal_closed.py]`).* [E] The renormalised coincident-point Green function on a circle of circumference $\ell$ is $G_{\mathrm{reg}}(0;\ell) = \frac{1}{\pi}\ln(\ell/2\pi)$ exactly — it *vanishes* precisely at $\ell = 2\pi = 1/(4c_3)$, the KMS seam circumference (v239): the seam scale is the unique zero-self-energy scheme point, and the log is the pure *scale* channel (matching the fact that $F_{U(1)}$ carries exactly *one* log term while the $\varphi_0^{\mathrm{ret}}$ split is log-free). [E] With the diagonal zero the mark contact calculus *resums in closed form* (the BFK/gluing route, v151): $\det(I - uC) = (1-4u)(1+2u)^2$ with $u = \varepsilon c_3 \ln 2$ — all orders, the linear term absent because $\mathrm{Tr}\, C = 0$. [E] And the two multiplicities are *one* state set under two response weights: the same $48 = 3 \times 16$ admissible states counted flat give the $\varphi_0^{\mathrm{ret}}$ puncture count $48 = \Omega_{\mathrm{adm}}$, and $Y^2$/loop/GUT-weighted give the alpha budget $41 = 10b_1 = 40 + 1$ ($\mathrm{Tr}_{16}\, Y^2 = \frac{10}{3}$ exact, $\nu^c = 0$; Ginsparg, v470). [C]/[O] Net: every finite/computable piece of the merged target is proven; the single remaining [O] is the identification of the abstract seam $\zeta$-det with this glued mark determinant — *exactly* the v382 typing (a face of `SEAM.EQUIV.01`), now substantiated computationally rather than asserted. *A ninth step executes both named faces of the target at the computable level (`[verification/v974_alpha_faces_computed.py]`), typing what remains.* [E] *The diagonal-zeta face is executed:* the BFK Mayer–Vietoris determinant formula (v151) now *runs* on the seam circle cut at the four $\mu_4$ marks — gluing constant exactly $2^{-N}$ (symbolic, $N = 1,\ldots,4$, general interval lengths), massless reassembly $\det'_\zeta(\Delta_{S^1}) = \ell^2$, jump matrix = the *inverse* mark Green matrix $R(m) = (G_m|_{\mathrm{marks}})^{-1}$; at the KMS scale $\ell = 2\pi = 1/(4c_3)$ this gives $R_0 = 16c_3\,\mathrm{circ}(2,-1,0,-1)$, $\det' R_0 = 2^{16}c_3^3$ and glued product $(2\pi)^2$ — the v485 diagonal/scale channel reproduced *exactly* via the KMS unit. Against the v484 mark matrix ($-4c_3 \ln 2 \cdot \mathrm{circ}(0,1,2,1)$) there is *no* global proportionality — the two kernels sit in *swapped* $\mu_4$ channels (gauge zero mode vs. sign rep) — but the correspondence is channel-wise exact (same $\mu_4$ Fourier eigenbasis, |spec| multisets $\{0,2,2,4\}$, doublet ratio $4/\ln 2$ with $c_3$ cancelling, $\mathrm{Tr}\, B^k = 4^k + 2\cdot 2^k$ mirroring $\mathrm{Tr}\, C^k = 4^k + 2(-2)^k$), so this face's residual *sharpens* to the channel-swap question. [E] *The EM1/topological face carries a computed pump witness:* on the v367 collar, pumped charge = edge spectral flow = Streda count = Chern = 1 per flux quantum (hybrid-Wannier winding residual 0, Streda $\Delta N = C\, n_\phi$ exact integers, the conjugation mutant flips all four readings simultaneously, $M$=3 controls all zero) — so the cited TKNN/Callan–Harvey step "boundary level = bulk response" of the fifth step now has a computed lattice witness; [C] the identification of the complex-copy $U(1)$ with the physical Majorana collar stays where it was, with the 16-copy bookkeeping ($c_-$=8, $k_Y$=5/3, $(1/k_Y)\frac{41}{6} = \frac{41}{10} = b_1$) re-derived exactly. [E] *The finite Quillen identity is two-sided:* over the $U(1)$-twist torus the FHS/Berry curvature integral of the determinant line *and* the metric-side integral (Gaussian overlap norm, Poincaré–Lelong divisor evaluation) both equal +1, the difference an exact form ($3 \times 10^{-16}$), transition-function winding = Chern. [C] Typed conditional: *if* the seam state is quasi-free (measured, not forced in the strong form — `SEAM.STATE.RPMIXING.01`), *then* Quillen's identity applies literally to the seam determinant line. [E] The computed witnesses are carried in the ledger as `ALPHA.QUILLEN.FACES.01`; the remaining open content of `ALPHA.QUILLEN.EXACT.01` is thereby typed as exactly {rigidity premise} + {channel-swap question} + {continuum Bismut–Freed identification} — the target stays [O], and $\alpha_\star^{-1}$ stays [E] regardless (re-anchored: Quillen split

residual $10^{-58}$ at 50 digits, $8b_1 = \frac{164}{5}$ exact). [E] *Channel-swap update (2026-08-28):* the swap question is resolved at the *finite* level by $\mu_4$-character grading — the deck swap $r \to r{+}2$ maps the jump |spectrum| $[0,2,4,2]$ onto the mark pattern $|[4,-2,0,-2]|$ exactly (the odd swap fails), the graded $\det'$ difference is $c_3$-free $= \log(\ln 2/4)$, and the channelwise unit ratio is uniform $4/\ln 2$ ([`verification/v985_quillen_channel_swap.py`]); the open residual contracts to {rigidity premise} + {continuum Bismut–Freed identification}, and the target stays [O].

The full ablation — which inputs are load-bearing and which only set the last digit — is in the Ablation appendix below.

*Inversion test (this round).* Running the closure backwards — inserting the *measured* $\alpha$ and solving for the integer budget $M = \sum L + N_\Phi$ — returns $M = 41.0000001\ldots$, i.e. the external measurement reconstructs the internal discrete budget $41 = 40 + 1$ to $10^{-7}$. This is stronger than "we hit $\alpha$": the equation reads the discrete 41 out of nature. [E]

## 7.2 The decuple bridge (new, exact headline)

Eliminating $\alpha_\star^{-1} = g_{\rm car}\, A_{\rm EW} - \delta_{\rm ph}$ between the electroweak and the cosmological-constant lines gives a direct relation between the two scales:

**Decuple bridge: $\Lambda$ is the tenth power of the stripped EW scale [E]**

$$\boxed{\frac{\rho_\Lambda}{\bar{M}_{\rm Pl}^4} = \frac{3}{4\pi^2}\, e^{2\delta_{\rm ph}} \left[\frac{v_{\rm geo}}{5\,\beta_{\rm rad}^2\,\bar{M}_{\rm Pl}}\right]^{10}} \tag{29}$$

where the exponent $10 = 2g_{\rm car}$ and $5 = g_{\rm car}$. Verified to a relative residual $\sim 10^{-159}$.

*Proof.* From $v_{\rm geo}/\bar{M}_{\rm Pl} = g_{\rm car}\beta_{\rm rad}^2 e^{-A_{\rm EW}}$, $e^{-A_{\rm EW}} = (v_{\rm geo}/\bar{M}_{\rm Pl})/(g_{\rm car}\beta_{\rm rad}^2)$. From $\rho_\Lambda/\bar{M}_{\rm Pl}^4 = \frac{3}{4\pi^2} e^{-2\alpha_\star^{-1}}$ and $\alpha_\star^{-1} = g_{\rm car}A_{\rm EW} - \delta_{\rm ph}$, $e^{-2\alpha_\star^{-1}} = e^{2\delta_{\rm ph}} e^{-2g_{\rm car}A_{\rm EW}} = e^{2\delta_{\rm ph}}(e^{-A_{\rm EW}})^{2g_{\rm car}}$. Substituting and using $2g_{\rm car} = 10$, $g_{\rm car} = 5$ gives the boxed form. □

In words: the electroweak scale is not directly the $\Lambda$ scale, but once the electroweak ratio is stripped of its small $\beta_{\rm rad}$ prefactor, $\Lambda$ is its tenth power, and the "ten" is $2g_{\rm car}$, not a fitted exponent. This is the direct algebraic compression of the action ladder.

## 7.3 The full power tower: $x^1, x^5, x^{10}$ on the carrier graph $K_5$

The decuple bridge is one rung of a tower. Writing $x := e^{-A_{\rm EW}} = v_{\rm geo}/(5\beta_{\rm rad}^2\bar{M}_{\rm Pl})$ for the stripped electroweak unit, all three cosmological scales are powers of the *same* $x$ with binomial exponents (verified exactly):

$$\boxed{\frac{v_{\rm geo}}{5\beta_{\rm rad}^2\bar{M}_{\rm Pl}} = x^{\binom{5}{0}} = x, \qquad \frac{H_0}{\bar{M}_{\rm Pl}} = x^{\binom{5}{1}} R_{\rm H} = x^5 R_{\rm H}, \qquad \frac{\rho_\Lambda}{\bar{M}_{\rm Pl}^4} = x^{\binom{5}{2}} R_\Lambda = x^{10} R_\Lambda} \tag{30}$$

with residues $R_{\rm H} = e^{\delta_{\rm ph}}/(2\pi\sqrt{\Omega_\Lambda})$ and $R_\Lambda = \frac{3}{4\pi^2} e^{2\delta_{\rm ph}}$. [E]. The exponents are the complete-graph counts of the five-slot carrier $K_5$:

$$H \leftrightarrow \text{vertices of } K_5 \ (\,\# = \tbinom{5}{1} = 5\,), \qquad \Lambda \leftrightarrow \text{edges of } K_5 \ (\,\# = \tbinom{5}{2} = 10\,).$$

So the cosmological constant is the product of one stripped-electroweak unit over every carrier *pair* ($K_5$ edge product), and the Hubble scale the product over every carrier *slot* ($K_5$ vertex product). This is the cleanest single-variable statement of the action ladder.

## 7.4 Hubble is the square root of $\Lambda$ (corollary, exact)

On the comparison layer $\rho_\Lambda/\bar{M}_{\rm Pl}^4 = 3\Omega_\Lambda(H_0/\bar{M}_{\rm Pl})^2$; equating to the TFPT line gives

$$\boxed{\frac{H_0}{\bar{M}_{\rm Pl}} = \frac{e^{-\alpha_\star^{-1}}}{2\pi\sqrt{\Omega_\Lambda}}}, \qquad \mathcal{A}_H = -\ln\frac{H_0}{\bar{M}_{\rm Pl}} = \alpha_\star^{-1} + \ln\!\left(2\pi\sqrt{\Omega_\Lambda}\right) = 138.68. \tag{31}$$

So the Hubble rung is *not* an independent third hit: it is the square root of the vacuum energy plus a geometric cosmology factor. The genuine TFPT content is the single $e^{-\alpha_\star^{-1}}$. (The numerical coincidence $\mathcal{A}_H - \alpha_\star^{-1} \approx \pi^2/6$ is just $\ln(2\pi\sqrt{\Omega_\Lambda})$ and is $\Omega_\Lambda$-dependent; not a claim. [O])

### 7.5 The $\sim 123$ orders of the cosmological constant

$$\log_{10}\frac{M_{\rm Pl}^4}{\Lambda_{\rm IR}} = \underbrace{\frac{2\alpha_\star^{-1}}{\ln 10}}_{119.028} + \underbrace{\log_{10}\frac{1}{\delta_{\rm top}}}_{3.920} = 122.948 \approx 123. \tag{32}$$

The "119" is the action $2\alpha_\star^{-1}$; the "$\approx 4$" is the seam defect $\delta_{\rm top} = \Omega_{\rm adm} c_3^4$.

**High-precision cosmological constant: a $\sim 10^{-13}$ prediction [E]**

Because $\alpha_\star^{-1}$ is a *derived* number (the EM fixed point, known to 13 digits) and the residue $3/(4\pi^2)$ is exact, the dimensionless vacuum energy is predicted essentially exactly:

$$\boxed{\frac{\rho_\Lambda}{\bar{M}_{\rm Pl}^4} = \frac{3}{4\pi^2}\, e^{-2\alpha_\star^{-1}} = 7.12533 \times 10^{-121}} \qquad \text{[E] [verification/v7_gravity_cosmo.py]}, \tag{33}$$

i.e. a dark-energy scale $\rho_\Lambda^{1/4} = \left(\frac{3}{4\pi^2}\right)^{1/4} \bar{M}_{\rm Pl}\, e^{-\alpha_\star^{-1}/2} = 2.23747$ meV (reduced Planck mass $\bar{M}_{\rm Pl} = 2.435323 \times 10^{18}$ GeV). The *dimensionless* ratio carries no observational input — its uncertainty is that of $\alpha_\star^{-1}$, $\sim 10^{-13}$ relative; the absolute meV value inherits only the Planck-mass uncertainty ($\sim 2 \times 10^{-5}$). Both are *far* sharper than the measured vacuum energy (Planck $\rho_\Lambda^{1/4} \simeq 2.24$ meV, known to $\sim 1\%$ and entangled with the $H_0$ tension). So TFPT pins the cosmological constant several orders of magnitude more precisely than it is currently measured — a genuine, falsifiable forward prediction: any future sub-percent $\rho_\Lambda$ must land on 2.2375 meV.

### 7.6 The ladder is binomial; $2g_{\rm car} = \binom{g_{\rm car}}{2}$ singles out $g_{\rm car} = 5$

In units of the electroweak action $\alpha_\star^{-1}/g_{\rm car}$, the three rungs are the first three binomial coefficients of the five-slot carrier:

$$\frac{A_{\rm EW}}{\alpha_\star^{-1}/g_{\rm car}} : \frac{\mathcal{A}_H}{\alpha_\star^{-1}/g_{\rm car}} : \frac{\mathcal{A}_\Lambda}{\alpha_\star^{-1}/g_{\rm car}} = 1 : g_{\rm car} : 2g_{\rm car} = \binom{g_{\rm car}}{0} : \binom{g_{\rm car}}{1} : \binom{g_{\rm car}}{2} = 1 : 5 : 10. \tag{34}$$

The first two are automatic ($\binom{g_{\rm car}}{0} = 1$, $\binom{g_{\rm car}}{1} = g_{\rm car}$). The non-trivial content is the cosmological-constant charge:

$$\boxed{2g_{\rm car} = \binom{g_{\rm car}}{2} \quad \Longleftrightarrow \quad g_{\rm car} = 5} \qquad \text{[E] [verification/v2_carrier_pascal.py]}, \tag{35}$$

since $\binom{g_{\rm car}}{2} = \frac{g_{\rm car}(g_{\rm car}-1)}{2} = 2g_{\rm car} \iff g_{\rm car} = 5$. So the "double overlap" charge $2g_{\rm car}$ equals the number of carrier-slot *pairs* $\binom{g_{\rm car}}{2}$ *only* at $g_{\rm car} = 5$: the cosmological constant is suppressed by one action unit per carrier pair, and this combinatorial reading is exclusive to the physical rank. It is an independent overdetermination of $g_{\rm car} = 5$ from the action grammar (cf. the landscape section).

### 7.7 Discriminant action theorem: the ladder reads the hypercharge discriminant

The $g_{\rm car}$ in the action ladder is not merely a rank: it is the square root of the discriminant of the carrier (hypercharge) polynomial. For a general split $b + s = g$ the carrier polynomial is $bs\,Y^2 + (s-b)Y - 1 = 0$ with discriminant

$$\Delta_Y = (s-b)^2 + 4bs = (b+s)^2 = g_{\rm car}^2, \qquad \text{so for } 6Y^2 - Y - 1 = 0:\ \Delta_Y = 25,\ \sqrt{\Delta_Y} = 5 = g_{\rm car}. \tag{36}$$

Hence the action ladder is, exactly,

$$\boxed{A_{\rm EW} = \frac{\alpha_\star^{-1}}{\sqrt{\Delta_Y}}, \qquad \mathcal{A}_H = \sqrt{\Delta_Y}\, A_{\rm EW}, \qquad \mathcal{A}_\Lambda = \binom{\sqrt{\Delta_Y}}{2} A_{\rm EW}} \qquad \text{[E]}. \tag{37}$$

The electroweak hierarchy therefore reads the *discriminant of the hypercharge polynomial*, tying the Standard-Model hypercharge derivation directly to the EW-hierarchy and Λ problems.

### 7.8 A two-dimensional action lattice

All scales sit on the integer lattice spanned by two generators — the gauge unit $\alpha_\star^{-1}/g_{\rm car}$ and the flavor unit $L_0 = \ln(1/u)$ — as $S_x = a_x(\alpha_\star^{-1}/g_{\rm car}) + b_x L_0 + (\text{residue})$:

| scale | $a_x$ | $b_x$ | axis |
|---|---|---|---|
| $v_{\rm geo}/\bar{M}_{\rm Pl}$ | $1 = \binom{5}{0}$ | 0 | gauge |
| $H_0/\bar{M}_{\rm Pl}$ | $5 = \binom{5}{1}$ | 0 | gauge |
| $\rho_\Lambda/\bar{M}_{\rm Pl}^4$ | $10 = \binom{5}{2}$ | 0 | gauge |
| $m_e/\bar{M}_{\rm Pl}$ | 1 | 5 | flavor |
| $m_\mu/\bar{M}_{\rm Pl}$ | 1 | 3 | flavor |
| $m_\tau/\bar{M}_{\rm Pl}$ | 1 | 2 | flavor |

The cosmological/gauge sector runs along $b = 0$ with binomial charges; the charged-fermion sector runs along $a = 1$ with the integer flavor ladder $n_f = (5, 3, 2)$. This is the compact "periodic table" of TFPT scales: two integer labels per scale. [E] (charges) / [C] (residues).
A third (seam/gravity) generator completes the lattice: $L_c = \ln(1/c_3) = \ln(8\pi)$. The scalaron lives purely on it, $M_{\rm scal}/\bar{M}_{\rm Pl} = c_3^{7/2} = e^{-\frac{7}{2}L_c}$, and the seam defect is $\delta_{\rm top} = e^{-(4L_c - \ln 48)}$. So the three log-generators are $\{\alpha_\star^{-1}$ (hierarchy), $L_0 = \ln(1/u)$ (flavor), $L_c = \ln(1/c_3)$ (seam/gravity)$\}$. [E]

## 8 The Einstein normaliser as a corrected tree value

*Machine checks for this section:* [verification/v1_e8_glue.py]

The gravitational normaliser $\xi = c_3/u$ has a transparent exact form. Writing $u = \frac{1}{6\pi}\left[1 + \frac{9}{128\pi^3}\right]$,

$$\boxed{\xi = \frac{c_3}{u} = \frac{3}{4} \cdot \frac{1}{1 + \dfrac{9}{128\pi^3}} = 0.748303} \qquad \text{[E]}. \tag{38}$$

So $\xi_{\rm tree} = \frac{3}{4} = \dim \mathfrak{g}_{\rm SM}/\dim S^+ = 12/16$ is the exact leading value (the inverse of the $\frac{4}{3} = \dim S^+/\dim \mathfrak{g}_{\rm SM}$ family/gauge compression that sits in the seed $u_{\rm base} = \frac{4}{3}c_3$, so $\xi_{\rm tree}\, u_{\rm base} = c_3$), and the deviation is precisely the small topological seed correction $9/(128\pi^3)$, i.e. 0.226%. Gravity does not pick up a mysterious factor: first comes 3/4, then the seam defect shifts it minimally.

## 9 Λ-anchored metrology closure, and two anchors

*Machine checks for this section:* [verification/v60_lambda_metrology_branch.py]

The absolute SI scale is a calibration, not a compiler output. The dimensionless vacuum quotient lets one anchor it to the observed geometric curvature without circularity.

**Theorem 1** (Λ-anchored metrology closure)**.** With $\lambda_\Lambda := \rho_\Lambda/\bar{M}_{\rm Pl}^4 = \frac{3}{4\pi^2} e^{-2\alpha_\star^{-1}}$ and the observed geometric curvature $\Lambda_{\rm geom}^{\rm obs} > 0$,

$$\bar{M}_{\rm Pl}^2 = \frac{\Lambda_{\rm geom}^{\rm obs}}{\lambda_\Lambda}, \qquad G_N^{\rm TFPT} = \frac{c^3}{8\pi\hbar}\frac{\lambda_\Lambda}{\Lambda_{\rm geom}^{\rm obs}} = \frac{3c^3}{32\pi^3\hbar\,\Lambda_{\rm geom}^{\rm obs}}\, e^{-2\alpha_\star^{-1}}.$$

*Proof.* $\bar{M}_{\rm Pl}^2 = 1/(8\pi G_N)$ and $\Lambda_{\rm geom} = \rho_\Lambda/\bar{M}_{\rm Pl}^2$; the branch sets $\rho_\Lambda = \lambda_\Lambda \bar{M}_{\rm Pl}^4$, so $\Lambda_{\rm geom} = \lambda_\Lambda \bar{M}_{\rm Pl}^2$, giving $\bar{M}_{\rm Pl}^2$ and then $G_N = \lambda_\Lambda/(8\pi\Lambda_{\rm geom}^{\rm obs})$; restoring $c^3/\hbar$ closes it. □

**Non-circularity ([O], essential).** The anchor must be the curvature $\Lambda_{\rm geom}^{\rm obs}$ (units $\mathrm{m}^{-2}$), *not* the SI vacuum energy density $\rho_\Lambda^{\rm obs} = \Lambda_{\rm geom} c^4/(8\pi G_N)$, which already contains $G_N$. The curvature follows from $\Lambda_{\rm geom}^{\rm obs} = 3\Omega_\Lambda H_0^2/c^2$.

**Two independent anchors (crosscheck with teeth).** The metrology can be pinned in two ways that should agree:

1. *Electroweak anchor*: the dimensionless $G_N v_{\rm phys}^2 = 4.06717 \times 10^{-34}$ fixes $G_N$ once $v_{\rm phys}$ is calibrated non-gravitationally.
2. $\Lambda$*-curvature anchor* (this note): $G_N = \frac{3c^3}{32\pi^3 \hbar \Lambda_{\rm geom}^{\rm obs}} e^{-2\alpha_\star^{-1}}$.

With the example anchor $H_0 = 67.36\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_\Lambda = 0.6847$ ($\Lambda_{\rm geom}^{\rm obs} = 1.08914 \times 10^{-52}\,\mathrm{m^{-2}}$) the $\Lambda$ anchor gives

$$G_N^{\rm TFPT} = 6.65071 \times 10^{-11}\,\mathrm{m^3 kg^{-1} s^{-2}}, \qquad G_N^{\rm TFPT}/G_N^{\rm CODATA} - 1 = -3.5 \times 10^{-3}. \tag{39}$$

Closure algebra [E]; SI anchor and numerical match [C] (the cosmological anchor is model dependent; the Hubble tension applies). EW crosscheck: the $\Lambda$-pinned $\bar{M}_{\rm Pl} = 2.43964 \times 10^{18}\,\mathrm{GeV}$ gives $v_{\rm geo} = 242.6\,\mathrm{GeV}$, needing $Z_{\rm EW} = (246.22/242.61)^2 = 1.030$.

**Prefactor branch ([O]).** This note uses $\Lambda_{\rm IR}/M_{\rm Pl}^4 = \delta_{\rm top} e^{-2\alpha_\star^{-1}}$. A competing prefactor $2c_3 e^{-2\alpha_\star^{-1}}$ would differ by $2c_3/\delta_{\rm top} = 661.5$ and misscale $G_N$ by the same factor; the branch must be versioned.

**Mandatory $\Lambda$-layer typing ([O]).** Three distinct $\Lambda$ objects must never be conflated:

$$\underbrace{\frac{\rho_\Lambda}{\bar{M}_{\rm Pl}^4} = \frac{3}{4\pi^2} e^{-2\alpha_\star^{-1}}}_{\text{vacuum density quotient (dimensionless)}} \neq \underbrace{\frac{\Lambda_{\rm IR}}{M_{\rm Pl}^4} = \delta_{\rm top} e^{-2\alpha_\star^{-1}}}_{\text{seam-determinant / IR scale (dimensionless)}} \neq \underbrace{\Lambda_{\rm geom}^{\rm obs} = \frac{3\Omega_\Lambda H_0^2}{c^2}}_{\text{observed curvature } [\mathrm{m^{-2}}]} .$$

Only $\Lambda_{\rm geom}^{\rm obs}$ carries a dimension and is the non-circular anchor. Keeping the three typed protects the decuple bridge and the metrology closure from a spurious prefactor objection.

**Measurement-near (physical) decuple bridge.** Since $v_{\rm phys} = v_{\rm geo}\sqrt{Z_{\rm EW}}$ (charged-current residue, not a fit), the decuple bridge in the observable vev reads

$$\frac{\rho_\Lambda}{\bar{M}_{\rm Pl}^4} = \frac{3}{4\pi^2} e^{2\delta_{\rm ph}} \left[ \frac{v_{\rm phys}}{5\beta_{\rm rad}^2 \sqrt{Z_{\rm EW}}\, \bar{M}_{\rm Pl}} \right]^{10} . \tag{40}$$

This is the same identity, written in the low-energy charged-current scale rather than the geometric source scale; both forms should appear side by side. [E]/[C]

## 10 The landscape: $g_{\rm car} = 5$ is overdetermined

*Machine checks for this section:* [verification/v6_bootstrap.py] [verification/v14_carrier_uniqueness.py]

Varying the carrier rank and rebuilding every constant from the same machinery ($c_3$ fixed), the EM-closure value $\alpha^{-1}(g_{\rm car})$ is monotone and meets the observed value only at $g_{\rm car} = 5$:

| $g_{\rm car}$ | $N_{\rm fam}$ | $\dim S^+$ | $\Omega_{\rm adm}$ | $10b_1$ | $\alpha^{-1}(g_{\rm car})$ | verdict |
|---|---|---|---|---|---|---|
| 3 | 2 | 4 | 8 | 7 | 258.15 | wrong $\alpha$, wrong family count |
| 4 | 2.5 | 8 | 20 | 17 | 187.31 | non-integer families |
| **5** | **3** | **16** | **48** | **41** | **137.036** | **SM, 3 families, observed $\alpha$** |
| 6 | 3.5 | 32 | 112 | 97 | 101.30 | non-integer families |
| 7 | 4 | 64 | 256 | 225 | 75.61 | wrong $\alpha$ |

Several distinct requirements converge on $g_{\rm car} = 5$: integer families ($g_{\rm car}$ odd); three families $(g_{\rm car}+1)/2 = 3$; the observed $\alpha^{-1}$ on the monotone curve; the SM structure $(b, s) = (3, 2)$, $\dim S^+ = 16$; and $\Omega_{\rm adm} = 48$, $10b_1 = 41$. A clean algebraic shadow: $\frac{g(g+1)}{2} = 2^{g-1} - 1$ has $g = 5$ as its non-trivial solution ([C]).

## 11 The $E_8$ carrier atlas — stated unambiguously

*Machine checks for this section:* [verification/v5_e8_cascade.py]

In the series $E_8$ is a *downstream scale grammar*, not a primitive cause: the boundary kernel fixes $\alpha$, the carrier rank, families and admissibility without $E_8$ upstream. Three levels.

### 11.1 Level 1 — original $E_8$ data of the series ([E])

$$|R(E_8)| = 248 - 8 = 240 = 5 \cdot 48 = 5\,\Omega_{\rm adm} = \mathrm{lcm}(\Omega_{\rm adm}, D_{\rm start}) = \mathrm{lcm}(48, 60), \tag{41}$$

$$D_{\rm start} = g_{\rm car} \cdot \dim \mathfrak{g}_{\rm SM} = 5 \cdot 12 = 60, \qquad \gcd(\Omega_{\rm adm}, D_{\rm start}) = 12 = \dim \mathfrak{g}_{\rm SM}, \tag{42}$$

$$\frac{|R(E_8)|}{D_{\rm start}} = \frac{240}{60} = 4 = \frac{\Omega_{\rm adm}}{\dim \mathfrak{g}_{\rm SM}}, \qquad 2880 = D_{\rm start}\,\Omega_{\rm adm} = |R(E_8)| \cdot \dim \mathfrak{g}_{\rm SM}, \tag{43}$$

$$\kappa_{E_8} = \frac{\gamma_{\rm car}}{\ln(248/60)} = 0.58723, \qquad X(n, r, I_1) = \frac{\bar{M}_{\rm Pl}}{8} \sin^2\theta_{13} \Big(\frac{60 - 2n}{60}\Big)^{\kappa_{E_8}} e^{-(8-r)/64} e^{-12\pi I_1}. \tag{44}$$

**Coxeter/gauge typing of the lcm–gcd pair.** Writing $D_{\rm start} = 60 = 2h_{E_8}$ and $\Omega_{\rm adm} = 48 = 6\,\mathrm{rank}(E_8)$, the relations become $\gcd(2h_{E_8}, 6r_{E_8}) = \dim \mathfrak{g}_{\rm SM} = 12$ and $\mathrm{lcm}(2h_{E_8}, 6r_{E_8}) = |R(E_8)| = 240$: the local SM algebra is the gcd of the Coxeter double-period and the $\mathbb{Z}_6$ rank-occupancy, and the $E_8$ root count is their lcm. Likewise the EW exponent reads $I_1^{\rm EW} = \frac{41}{48} = \frac{h_{E_8}}{36} + \frac{1}{6r_{E_8}} = \frac{5}{6} + \frac{1}{48}$ (Coxeter principal part plus a rank singlet). [E] (arithmetic) / [C] (typing).

### 11.2 Level 2 — new exact counting identities ([E])

$$|R(E_8)| = 240 = 16 \cdot 15 = \dim S^+ \,(\dim S^+ - 1), \tag{45}$$

$$\dim E_8 = 248 = 8 \cdot 31 = (g_{\rm car} + N_{\rm fam})(2^{g_{\rm car}} - 1), \tag{46}$$

$$|R(E_8)| = 8\,(31 - 1) = \mathrm{rank}(E_8)\,(2^{g_{\rm car}} - 2), \qquad \mathrm{rank}(E_8) = 8 = g_{\rm car} + N_{\rm fam} = 5 + 3, \tag{47}$$

$$10\,b_1 = 41 = \frac{|R(E_8)|}{6} + 1, \qquad I_1^{\rm EW} = \frac{41}{48} = \gamma_{\rm car} + \frac{1}{\Omega_{\rm adm}}, \tag{48}$$

with $31 = 2^{g_{\rm car}} - 1$ the number of non-empty occupation words of the $[5, 4, 2]$ carrier code.

### 11.3 Level 2.5 — the even-flip transition atlas (a falsifiable program)

Identify $S^+$ with the even-weight binary code on five bits. A transition $x \to y \neq x$ has flip $F = x \oplus y$, even and nonempty, so $|F| \in \{2, 4\}$; per state there are $\binom{5}{2} + \binom{5}{4} = 10 + 5 = 15$ flips, giving

$$240 = 16 \cdot (10 + 5) = \underbrace{160}_{|F|=2} + \underbrace{80}_{|F|=4}, \tag{49}$$

and the 120 undirected flips match the 120 $E_8$ root *pairs* ([E], counting). This makes the atlas conjecture *testable*: a genuine root map must send each flip to a norm-2 vector in $\mathbb{R}^8$ with inner products in $\{-2, -1, 0, 1, 2\}$ reproducing the $E_8$ profile (per root: 1 self, 56 at $+1$, 126 orthogonal, 56 at $-1$, 1 anti).

**Honest obstruction ([O]).** The flip atlas splits $160 + 80$ by flip size; the standard $E_8$ ($D_8$) split is $112 + 128$. Since $160 + 80 \neq 112 + 128$, no flip-size-preserving map is an $E_8$ root system; a

metrization, if it exists, must mix flip sizes. Until a norm-2, $E_8$-profile embedding is exhibited, $E_8$ stays a *counting* atlas, not a derived root origin.

**Sharper two-stage audit (Coxeter before Weyl).** A full root-system isomorphism is hard; a much earlier, cheaper test is the Coxeter spectrum. *Stage A:* find a natural order-30 operator $C_{\mathrm{TFPT}}$ on the even-flip atlas (or its 8-dimensional primitive-character space) with characteristic polynomial $\chi_{C_{\mathrm{TFPT}}}(t) = \Phi_{30}(t)$. *Stage B (only if A holds):* the norm-2, $E_8$-profile, Weyl-closure metrization. Stage A can fail or carry far sooner than B, so it is the right first falsifier.

**Stage A is closed on the primitive-character space (**[`verification/v531_coxeter_tensor_stage_a.py`]**).** The explicit integer operator

$$T_{30} = \mathrm{Comp}(\Phi_5) \otimes \mathrm{Comp}(\Phi_6)$$

(the 4×4 carrier clock of v418 tensor the 2×2 family/CP hexagon) has *exact* order 30 ($T_{30}^{30} = I$, $T_{30}^{k} \neq I$ for $1 \leq k < 30$) and $\chi_{T_{30}} = \Phi_{30} = t^8 + t^7 - t^5 - t^4 - t^3 + t + 1$ exactly: by CRT ($\gcd(5,6) = 1$) its eigenvalues are the products (primitive 5th)×(primitive 6th) = exactly the eight primitive 30th roots — the carrier clock times the family clock *is* the Coxeter clock, as one integer operator. Its traces are the Ramanujan sums, $\mathrm{tr}(T_{30}^k) = c_{30}(k) = c_5(k)\, c_6(k)$ for all $k$, with the signed divisor table $(-1, 1, 2, 4, -2, -4, -8, 8)$ at $k = (1, 2, 3, 5, 6, 10, 15, 30)$ and the atom readout $(|\,\mathrm{tr}\, T^3|, |\,\mathrm{tr}\, T^5|, |\,\mathrm{tr}\, T^{15}|) = (2, 4, 8) = (|\mathbb{Z}_2|, |\mu_4|, \mathrm{rank}\, E_8)$ — eight integer traces type the clock *without* diagonalisation, a cheap kill test for any flip-operator candidate (honest scope: $\Phi_{15}$ passes the unsigned triple since $\Phi_{30}(t) = \Phi_{15}(-t)$; the *signed* table is the discriminator, and $\Phi_{16}, \Phi_{20}, \Phi_{24}$ die on the triple already). [E] *What stays open ([O]):* a canonical intertwiner from the concrete even-flip atlas onto this primitive space (`COX.FLIP.INTERTWINER`), and Stage B.

**Audit only — *non-load-bearing* speculation: the** dim 496 **heterotic count [O].** *This box feeds no derivation; it is an arithmetic coincidence kept for completeness only, under the no-free-pattern rule.* The even half $S^+$ and odd half $S^-$ of the exterior algebra each have dimension $2^{g_{\mathrm{car}}-1} = 16$ and each carry $16 \cdot 15 = 240$ transitions. Together the full carrier ($2^{g_{\mathrm{car}}} = 32$ states) gives

$$\dim S^+(2^{g_{\mathrm{car}}} - 1) = 16 \cdot 31 = 496 = \dim(E_8 \times E_8) = \dim SO(32) = \binom{32}{2} \qquad \text{(count)},$$

i.e. the two halves read as two $E_8$ root atlases, all $\binom{32}{2} = 496$ pairs as $SO(32)$ — the two anomaly-free heterotic gauge dimensions. The *dimension* equality is exact; that the carrier algebra *is* either gauge group is unproven and not used anywhere. [O] (audit-only). The same fingerprint recurs at checksum level: the dual products of the $E_8$ invariant degrees sum to $744 = 3 \cdot 248$ (the $j$-function constant), and the identical construction for $D_{16}$ gives $1488 = 3 \cdot 496 = 2 \cdot 744$ ([`verification/v532_e8_degree_modular_checksum.py`]) — the checksum measures the common heterotic rank-16/Coxeter-30 structure, *not* $E_8$ uniqueness. [E] (audit-only).

## 11.4 Level 2.7 — the carrier Coxeter theorem (the strongest new bridge)

Use the universal root-system identity $|R| = \mathrm{rank} \cdot h$ (Coxeter number $h$). With the carrier values $\mathrm{rank}(E_8) = g_{\mathrm{car}} + N_{\mathrm{fam}} = 8$ and $|R(E_8)| = \dim S^+(\dim S^+ - 1) = 240$,

$$\boxed{h_{E_8} = \frac{|R(E_8)|}{\mathrm{rank}(E_8)} = \frac{240}{8} = 30 = N_{\mathrm{fam}} \binom{g_{\mathrm{car}}}{2} = N_{\mathrm{fam}}\, A_\Lambda = 2 g_{\mathrm{car}} N_{\mathrm{fam}} = 2^{g_{\mathrm{car}}} - 2} \qquad \text{[E]}. \quad (50)$$

The Coxeter number of $E_8$ is exactly three families times the cosmological pair action: each family contributes one Λ-decuple ($A_\Lambda = \binom{5}{2} = 10$) to the Coxeter period. Equivalently the action ladder is a Coxeter ladder,

$$A_H = \frac{h_{E_8}}{2N_{\mathrm{fam}}} = 5, \qquad A_\Lambda = \frac{h_{E_8}}{N_{\mathrm{fam}}} = 10, \quad (51)$$

so Hubble reads half a Coxeter period per family and Λ a full Coxeter period per family. [C]

**Rank–Coxeter coupling.** Equivalently $\dim S^+ = 2\,\mathrm{rank}(E_8) = 16$ and $h_{E_8} = 2(\dim S^+ - 1) = 2 \cdot 15 = 30$, so $\dim E_8 = (g_{\mathrm{car}} + N_{\mathrm{fam}})(h+1) = 8 \cdot 31$. The two counting forms $16 \cdot 15 = 8 \cdot 30 = 240$ coincide because $\dim S^+ = 2r$. Dropping $\nu^c$ sends $\dim S^+ \to 15$, which breaks *both* $\dim S^+ = 2r$ and $\dim S^+(\dim S^+ - 1) = 240$: $\nu^c$ is the state that closes the rank–Coxeter coupling, not an optional singlet. **[E]**/**[C]**

### 11.5 Level 2.8 — positive-root moment: $|R^+(E_8)| = \mathrm{Tr}_{S^+} X^2 = 5!$

The number of positive $E_8$ roots equals the master hypercharge moment:

$$\boxed{|R^+(E_8)| = 120 = \mathrm{Tr}_{S^+} X^2 = 5!} \qquad \textbf{[E]}, \tag{52}$$

so the second hypercharge moment of one family *is* the positive-root half-system. The whole integer chain then reads off $|R^+|$:

$$\sum_{f,j} L_{f,j} = \frac{|R^+(E_8)|}{3} = 40, \qquad 10b_1 = 1 + \frac{|R^+(E_8)|}{3} = 41, \qquad \delta_2 = 4!\,|R^+(E_8)|\,c_3^8. \tag{53}$$

The $\mathbb{Z}_6$ slice is then $|R(E_8)|/6 = \frac{|R^+|}{3} = 40 = \Omega_{\mathrm{adm}}\gamma$: one $\mathbb{Z}_6$ phase slice of the $E_8$ root atlas is exactly the full flavor transport budget. **[E]** (arithmetic) / **[C]** (interpretation).

### 11.6 Level 2.9 — the 2·3·5 Coxeter cyclotomic compiler (meta-theorem)

The three hard carrier atoms — weak rank 2, color rank / family count 3, carrier rank $g_{\mathrm{car}} = 5$ — generate the $E_8$ data *cyclotomically*, without taking $E_8$ as input. Set

$$\boxed{h := 2 \cdot 3 \cdot g_{\mathrm{car}} = 30, \qquad r := \varphi(h) = \varphi(30) = 8 = g_{\mathrm{car}} + N_{\mathrm{fam}},} \tag{54}$$

where $h$ is the Coxeter number and $r = \varphi(2{\cdot}3{\cdot}5) = 8$ the rank. Then the standard $|R| = rh$, $\dim = r(h{+}1)$ give

$$|R(E_8)| = rh = 8 \cdot 30 = 240, \qquad \dim E_8 = r(h+1) = 8 \cdot 31 = 248, \qquad h+1 = 2^{g_{\mathrm{car}}} - 1 = 31. \tag{55}$$

Moreover the $E_8$ *exponents* are exactly the primitive residues modulo $h = 30$:

$$\mathrm{Exp}(E_8) = (\mathbb{Z}/30\mathbb{Z})^\times = \{1, 7, 11, 13, 17, 19, 23, 29\}, \qquad \sum_{m \in (\mathbb{Z}/30)^\times} m = 120 = \mathrm{Tr}_{S^+} X^2 = |R^+(E_8)|, \tag{56}$$

so the second hypercharge moment equals the sum of the $E_8$ exponents and the positive-root count. The $\mathbb{Z}_6 \times \mathbb{Z}_5 \simeq \mathbb{Z}_{30}$ tact has $\varphi(30) = 8$ primitive characters, whose phases $e^{2\pi i m/30}$ are the Coxeter eigenphases of $E_8$; its characteristic polynomial is the 30th cyclotomic polynomial

$$\chi_C(t) = \Phi_{30}(t) = t^8 + t^7 - t^5 - t^4 - t^3 + t + 1. \tag{57}$$

**Reading:** $E_8$ here is not primarily a root count but the *Coxeter cyclotomic readout of the* $2, 3, 5$ *carrier*. **[E]** (arithmetic) / **[C]** (interpretation).

### 11.7 Level 3 — conjectural geometric readings ([C])

**Conjecture 1** (Carrier transition atlas)**.** $R(E_8) \cong \{(a, b) \in S^+ \times S^+ : a \neq b\}$ as a structure-bearing transition set, since $16 \cdot 15 = 240$. *Open:* a structure-preserving map (roots, lengths, Weyl data), not only the count.

**Conjecture 2** (Least common refinement / rank-8 over the exhausted carrier)**.** $240 = \mathrm{lcm}(\Omega_{\mathrm{adm}}, D_{\mathrm{start}})$ reads $E_8$ as the least common refinement of the occupancy lattice and the $\mathbb{Z}_6$ phase lattice; $248 = (g_{\mathrm{car}} + N_{\mathrm{fam}})(2^{g_{\mathrm{car}}} - 1)$ reads it as a rank-8 structure over the 31 non-empty carrier words.

**Two routes to $E_8$ — keep their status separate.** Levels 1–2 are exact arithmetic. There are now *two* distinct routes to $E_8$, and they must not be conflated:

- **Glue route (closed, [E]):** $E_8 = (D_5 \oplus A_3) + \mu_4$-glue is built explicitly in Section 4 ($\mathrm{disc}(D_5) = \mathrm{disc}(A_3) = \mathbb{Z}_4$, glue index 4, 240 roots of norm 2, simple-root Gram det 1). This closes the former "blind 240-metrisation".
- **Direct flip-atlas route (open, [O], optional bonus):** the even-flip transition set $240 = 160 + 80$ is *not* a flip-size-preserving $E_8$ map, since $160 + 80 \neq 112 + 128$. A direct structure-preserving metrisation of the flip graph remains open — a nice-to-have, not load-bearing, because the glue route already delivers $E_8$.

So $E_8$ is closed via the glue; only the direct flip atlas is open.

## 12 $\mathbb{Z}_6$ monodromy, flavor and CP readouts

*Machine checks for this section:* [verification/v88_cp_phase_audit.py]

A single $\mathbb{Z}_6$ motif recurs in the quotient $G_{\mathrm{phys}} = (SU(3)\times SU(2)\times U(1)_Y)/\mathbb{Z}_6$, the transport cusps $\{1, (2/3)^6, (1/3)^6\}$, $\delta_{\mathrm{ph}} = z_\star^{1/6}$, and the CP phases

$$\delta_{\mathrm{CKM}}^{(0)} = \frac{\pi}{3}, \quad \delta_{\mathrm{CKM}} = \frac{\pi}{3} + 3\lambda_C^2 = 1.19823\,\mathrm{rad}, \quad \delta_{\mathrm{CP}}^{\mathrm{PMNS}} = \frac{4\pi}{3} = 240^\circ. \tag{58}$$

From the documented inputs $s_{12} = \lambda_C$, $s_{23} = u/(1+\lambda_C)$, $s_{13} = \lambda_C^3/3$ one gets the new derived readouts $|V_{td}| = 0.00908$, $|V_{ts}| = 0.04264$, $J_{\mathrm{CKM}} = 3.33\times 10^{-5}$. On the neutrino side, with $\sin^2\theta_{13} = e^{-5/6}u = 0.02311$, $\sin^2\theta_{23} = 0.4557$, $\delta_{\mathrm{CP}} = 4\pi/3$ and the solar candidate $\sin^2\theta_{12} = \frac{1}{3} - \frac{u}{2} = 0.30675$, a near-maximal $|J_{\mathrm{CP}}^\nu| \approx 0.030$ (sharp readout for DUNE/Hyper-Kamiokande). [C]

### 12.1 CP as triple seed variance; a closed observable quadrilateral

The Cabibbo angle is the Bernoulli width of the seed. The *exact* CKM phase is the family transport holonomy (document 3), $\delta_{\mathrm{CKM}} = \arg \mathrm{Tr}\, \Pi_\triangle(\nabla_F^\star)$; on the small-area branch this has the compact readout form (with an $O(\lambda_C^4)$ remainder)

$$\lambda_C = \sqrt{u(1-u)}, \qquad \boxed{\delta_{\mathrm{CKM}} - \frac{\pi}{3} = N_{\mathrm{fam}}\, u(1-u) = 3\lambda_C^2} \qquad \text{[E]/[C] (readout; exact = holonomy).} \tag{59}$$

The coefficient 3 is the family number $N_{\mathrm{fam}}$: weak CP is the $\mathbb{Z}_6$ base angle $\pi/3$ plus $N_{\mathrm{fam}}$ times the seed (Bernoulli) variance. The $\mathbb{Z}_6$ base angle now has an $A_3$ root: since $\mathbb{Z}_6 = |R^+(A_3)| = 6$ (the positive-root hexagon of the family algebra), the CP phases are integer steps of the $A_3$ holonomy quantum $2\pi/|R^+(A_3)|$,

$$\boxed{\delta_{\mathrm{CKM}} = \frac{2\pi}{|R^+(A_3)|} + N_{\mathrm{fam}}\, u(1-u), \qquad \delta_{\mathrm{CP}}^{\mathrm{PMNS}} = 4\cdot\frac{2\pi}{|R^+(A_3)|} = \frac{4\pi}{3}} \qquad \text{[E]/[C].} \tag{60}$$

So weak CP = ($A_3$ positive-root phase step) + (family count × seed variance), while strong CP is the *determinant* sector, erased. The solar angle reads as a carrier root plus a half-seed, $\sin^2\theta_{12} = \frac{1}{3} - \frac{u}{2} = -y_- - \frac{u}{2}$ (since $-y_- = \frac{1}{3}$), so PMNS = carrier root + seed + $A_3$ holonomy phase. [C]Strong CP is killed by the determinant selector ($\arg\det M_u = \arg\det M_d = 0$, $\bar\theta = 0$), while weak CP survives as the seed-deformed $\mathbb{Z}_6$ holonomy: CP is not forbidden, only *determinant* CP is. [C] — the strong-CP closure lives on the admissible QFT branch and is *conditional* on the analytic hypotheses of the QFT closure (document 3, Part III: reflection positivity, temperedness, clustering, mass gap, OS reconstruction); its status is therefore weaker than the discrete lattice identities, even though the semantics "determinant CP forbidden, holonomy CP allowed" is robust.

Since $u = \Omega_b + \beta_{\rm rad}$ exactly, eliminating the seed closes the four UV-shadow observables into one cycle:

$$\lambda_C^2 = (\Omega_b + \beta_{\rm rad})(1 - \Omega_b - \beta_{\rm rad}), \quad \sin^2\theta_{13} = e^{-5/6}(\Omega_b + \beta_{\rm rad}), \quad \Omega_b = \left(1 - \tfrac{1}{4\pi}\right) e^{5/6} \sin^2\theta_{13}. \tag{61}$$

One measured member of $\{\lambda_C, \theta_{13}, \Omega_b, \beta_{\rm rad}\}$ fixes the other three. **[E]** (the four rows remain operationally distinct readout classes).

## 13 Mass hierarchy as a hexagon with one extra turn

*Machine checks for this section:* [verification/v18_quark_yukawa.py] [verification/v20_lepton_c_derivation.py]

The transport length packets are

$$L_u = (7,3,0), \qquad L_d = (7,5,2), \qquad L_e = (8,5,3), \qquad \sum_{f,j} L_{f,j} = 40 = \Omega_{\rm adm}\,\gamma_{\rm car} = 10b_1 - 1. \tag{62}$$

Modulo 6 they lie on related triples of the $C_6$ hexagon; the first generation carries one extra full turn ($u, d: 1+6$; $e: 2+6$), costing $\lambda_Y^6 = (u(1-u))^3 \approx 1.28 \times 10^{-4}$ and explaining why the first generation is so light — "one turn more", not a free fit. **[C]** *Flagged proximity ([O], not exact):* the full-turn cost is close to the topological seam defect, $\lambda_C^6/\delta_{\rm top} = 1.061$ ($\sim 6\%$), hinting that one flavor winding $\approx$ one seam-defect unit $\delta_{\rm top} = \Omega_{\rm adm} c_3^4$ — a structural smell, not a theorem. Writing the length matrix $L = R + W$ with $R = L \bmod 6$ the residue shadow and $W$ the full-winding part,

$$L = \begin{pmatrix} 7 & 3 & 0 \\ 7 & 5 & 2 \\ 8 & 5 & 3 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 3 & 0 \\ 1 & 5 & 2 \\ 2 & 5 & 3 \end{pmatrix}}_{R} + 6 \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}}_{W}, \qquad \sum R = 22,\ \sum W = 18 = 3 \cdot 6,\ \sum L = 40,$$

and the column sums are $(22, 13, 5)$ with $S_1 = 22 = \sum R$ and $S_2 + S_3 = 18 = \sum W$. So the first generation is light because it stores the *entire* residue budget plus the three full $C_6$ windings; generations 2 and 3 are the winding complement. **[E]** (arithmetic).

### 13.1 Lepton flavor decuple: the charged-lepton determinant carries $u^{10}$

The source masses on the $v_{\rm geo}$ branch are $(\hat m_e, \hat m_\mu, \hat m_\tau) = \frac{v_{\rm geo}\pi}{\sqrt{2}}\left(\frac{16}{7}u^5, \frac{4}{3}u^3, \frac{7}{6}u^2\right)$. Their product is a pure decuple in the seed:

$$\boxed{\frac{\hat m_e \hat m_\mu \hat m_\tau}{(v_{\rm geo}\pi/\sqrt{2})^3} = \left(\tfrac{16}{7} \cdot \tfrac{4}{3} \cdot \tfrac{7}{6}\right) u^{5+3+2} = \frac{32}{9}\, u^{10}} \qquad \textbf{[E]}, \tag{63}$$

with $5+3+2 = g_{\rm car} + b + s = 10 = \binom{5}{2}$. So the charged-lepton Yukawa powers $(n_e, n_\mu, n_\tau) = (5,3,2)$ *are* the three carrier numbers $(g_{\rm car}, b, s)$, and the lepton determinant carries the same decuple $(u^{10})$ that the cosmological constant carries in the gauge sector ($A_\Lambda = 10$): $\Lambda$ and the charged-lepton determinant are two realisations of the same degree-$\binom{5}{2}$ sector.

**Sum–product reading.** The lepton powers $(5,3,2)$ encode *both* decuple numbers at once: their sum is the $\Lambda$ action, their product is the $E_8$ Coxeter number,

$$5 + 3 + 2 = 10 = A_\Lambda, \qquad 5 \cdot 3 \cdot 2 = 30 = h_{E_8} = 2 \cdot 3 \cdot 5, \tag{64}$$

and $(5,3,2)$ is the prime factorisation of $h_{E_8} = 30$. So $\det M_\ell$, $\rho_\Lambda$ and the $E_8$ Coxeter period read the same carrier core additively ($= 10$) and multiplicatively ($= 30$). **[E]**/**[C]**

### 13.2 Why the top Yukawa is $\approx 1$: the $L = 0$ anchor (Froggatt–Nielsen ladder)

The transport length $L_f$ is a Froggatt–Nielsen-type charge with expansion parameter $\lambda_C$:

$$\boxed{y_f \simeq \lambda_C^{L_f} \times \Lambda_f \ (\Lambda_f = \mathcal{O}(1)), \qquad y_t \simeq 1 \text{ because } L_t = 0} \qquad \text{[E]/[C]}. \tag{65}$$

The top sits at $L_t = 0$ (the *unwound* carrier direction), so $y_t = \Lambda_t \simeq 1$ ($\sqrt{2}\, m_t/v = 0.991$); every other charged fermion is suppressed by integer powers of $\lambda_C = 0.2244$. Numerically (with $\mathcal{O}(1)$ sector prefactors $\Lambda_f \in [0.43, 1.07]$):

| $f$ | $t$ | $b$ | $\tau$ | $c$ | $\mu$ | $s$ | $u, d$ | $e$ |
|---|---|---|---|---|---|---|---|---|
| $L_f$ | 0 | 2 | 3 | 3 | 5 | 5 | 7 | 8 |
| $y_f/\lambda_C^{L_f}$ | 0.99 | 0.48 | 0.90 | 0.65 | 1.07 | 0.94 | $\sim$ 0.4–0.9 | 0.46 |

So the entire charged-fermion hierarchy is one $\lambda_C$-ladder anchored by the top at $L = 0$; the lightness of the first generation ($L = 7, 8$) and the heaviness of the top ($L = 0$) are the two ends of the same integer winding spectrum. This is the standard Froggatt–Nielsen mechanism with $\lambda_C$ as the flavon ratio and $L_f$ as the FN charge — a clean bridge between TFPT transport and an established flavor mechanism.

## 14 Inflation interface

*Machine checks for this section:* [verification/v7_gravity_cosmo.py] [verification/v86_nstar_reheating.py] [verification/v494_cosmo_running_mu_record.py]

The documented scalaron is Starobinsky $R^2$; with the pivot $N_\star = 57.1$:

$$\begin{aligned} n_s &= 1 - \frac{2}{N_\star} = 0.96497, \qquad r = \frac{12}{N_\star^2} = 0.00368, \\ \frac{M_{\rm scal}}{\bar{M}_{\rm Pl}} &= \sqrt{8\pi}\, c_3^4 = c_3^{7/2} = (8\pi)^{-7/2} \ \Rightarrow \ M_{\rm scal} \approx 3.06 \times 10^{13}\,\text{GeV}. \end{aligned} \tag{66}$$

$n_s$ matches Planck; $r = 0.0037$ is the falsifiable target for LiteBIRD/CMB-S4.

**Amplitude (status-clean).** The scalaron mass is a pure seam power,

$$\boxed{\frac{M_{\rm scal}^2}{\bar{M}_{\rm Pl}^2} = c_3^{r-1} = c_3^{\Omega_{\rm adm} - 10 b_1} = c_3^7} \qquad \text{[E]}, \tag{67}$$

an *exact* identity (the exponent $7 = 48 - 41$ is the post-flavor, post-Higgs occupancy deficit). The cosmology interface reduces the geometric gravity action to an FRW $R^2$/scalaron branch in which $M_{\rm scal}$ *is* the Starobinsky mass $M$; the identification is therefore derived in that sector, not posited. With the standard large-$N$ amplitude $A_s = \frac{N_\star^2}{24\pi^2}(M/\bar{M}_{\rm Pl})^2$ this gives

$$A_s = \frac{N_\star^2\, c_3^7}{24\pi^2} = \frac{N_\star^2\, \delta_2}{8640\pi} = 2.17 \times 10^{-9} \qquad \text{[C]/[O]}, \tag{68}$$

matching Planck ($\simeq 2.1 \times 10^{-9}$); $\delta_2 = 2880\, c_3^8$ is the second seam defect. The seam-power form is exact [E]; the amplitude is then determined *within* the $R^2$ sector, the only residual being the geometric-gravity (Hodge) closure on which that sector rests [C]. *Convention note:* $A_s \propto N_\star^2$, so the quoted $2.17 \times 10^{-9}$ uses the pivot $N_\star = 57.1$; the alternative seed pivot $N_\star = 3/u - 1 = 55.4$ gives $2.05 \times 10^{-9}$. A number should only be quoted with $N_\star$ fixed.

**Running (consistency relation).** The same attractor also fixes the *running* of the tilt parameter-free: $\alpha_s = {\rm d}n_s/{\rm d}\ln k = -2/N_\star^2 = -r/6$ and $\beta_s = -4/N_\star^3$ at leading order; the exact slow roll on the Starobinsky potential gives $\alpha_s = -7.09 \times 10^{-4}$, $\beta_s = -2.70 \times 10^{-5}$ at the reheating point $N_\star$=51.4 (leading-order band $[-8.0, -5.6] \times 10^{-4}$ over $N_\star \in [50, 60]$). Current data are consistent (Planck 2018 $-0.0045 \pm 0.0067$: $+0.57\sigma$; P-ACT-LB $+0.0062 \pm 0.0052$: $-1.33\sigma$), and since plateau potentials give $\alpha_s < 0$, any robust *positive* running would kill the branch. [C] [verification/v494_cosmo_running_mu_record.py]

## 15 Further simple bridges

*Machine checks for this section:* [verification/v74_compiler_micro_lemmas.py]

- **Dark-energy equation of state $w = -1$ (exact).** The TFPT $\Lambda$ is a genuine vacuum constant $\rho_\Lambda = \frac{3}{4\pi^2} e^{-2\alpha_\star^{-1}} \bar{M}_{\rm Pl}^4$, not a rolling field, so $w = -1$ identically (no quintessence). A robust measurement $w \neq -1$ would falsify the $\Lambda$-branch. [E] (structural).
- **Effective neutrino number $N_{\rm eff} \simeq 3$.** Three families ($N_{\rm fam} = 3$) give three light neutrinos and the standard $N_{\rm eff} \simeq 3.04$. [E].
- **Top Yukawa** $\simeq 1$ as the $L = 0$ anchor (above): a bridge to the Froggatt–Nielsen flavor mechanism.
- *Flagged (not clean):* $m_H/v = 0.509$ is near $\frac{1}{2}$ (1.7%); $m_W/m_Z = 0.881$ matches the running $\sqrt{1 - \sin^2\theta_W(M_Z)}$ (not the tree $\sqrt{5/8} = 0.79$). No clean closed bridge for the Higgs/$W$/$Z$ masses yet. [O]

## 16 Closures that the architecture rests on

*Machine checks for this section:* [verification/v36_spectral_action_g2.py] [verification/v4_flavor_matrix.py]

Five results close or sharpen the discrete core. They are *proved in this document set* — the $\varphi_0^{\rm ret}$-ladder, the H2 realisation and the strong-CP/dynamics gap in document 2 (`tfpt_2_standard_model`), the $E_6 \times A_2$ shadow in document 3 (`tfpt_3_e8_audit_bootstrap`) — and are collected here because the architecture above depends on them.

1. **The $\varphi_0^{\rm ret}$-ladder (whole spectrum, one formula).** Every fermion mass is $\hat{m}_{f,j} = \frac{v_{\rm geo}}{\sqrt{2}} \lambda_Y^{L_{f,j}} \Lambda_{f,j}$ with $\lambda_Y = \sqrt{\varphi_0^{\rm ret}(1 - \varphi_0^{\rm ret})}$, and the closed evaluation gives the universal form $\lambda_Y^L \Lambda = \pi\, c\, (\varphi_0^{\rm ret})^k$ — the nine $\Lambda$ are hexagonal-resolvent outputs (not free numbers); the *charged-lepton* ones are exact, the *quark* ones are reduced to the ($U_{\rm wall}$) selector. The $\varphi_0^{\rm ret}$-order ladder is $t(0) > \{b, c, \tau\}(2) > \{s, \mu\}(3) > \{u, d\}(4) > e(5)$. [E] (leptons exact, quark ratios closed v49/v71) / [O] (absolute quark scale)
2. **H2 reduced via the Mehta–Seshadri framework.** Mehta–Seshadri supplies the equivalence (stable parabolic $\leftrightarrow$ irreducible unitary); the TFPT-specific step — "stable = transport-minimal" and the geodesic $C_6$ word length — is conditional on the unitarity (U) of $\nabla_F^\star$ (document 3) and the parabolic-degree$\rightarrow C_6$ dictionary. H2 is the sharp realisation statement: the $D_4$-equivariant stable structure on $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ realises the branch with $\det R = h(D_5) = 8$, principal 2-minors $(2, 3, 5)$. [E] (given (U))
3. **Strong CP / dynamics via the explicit gap.** $\theta_{\rm eff} = 0$ needs only $\gamma_5$-Hermiticity + polar mass + sheet involution + reflection positivity (no mass gap). The full dynamics closes on the physical sector because the $C_6$ transport transfer operator has the *explicit* spectrum $\{(k/3)^6\}$, giving the mass gap $\Delta = 6 \log \frac{3}{2} = 2.43$ in closed form $\Rightarrow$ OS reconstruction $\Rightarrow$ measure on the admissible sector. [E] (phys. sector)
4. $E_6 \times A_2$ **flavor shadow.** The residue matrix sits inside an $E_8$ secondary branching: $248 = \|R\|_F^2 + \det R + 2(\mathbf{1}^\top R a) N_{\rm fam} = 78 + 8 + 2 \cdot 27 \cdot 3$, with $\|R\|_F^2 = 78 = \dim E_6$, $\det R = 8 = \dim A_2$. The seven $E_8$ slices form an audit raster (atlas note). [O]
5. **SM completeness.** The discrete structure and dimensionless core are closed; the dimensionful EW/QCD masses ($m_W, m_Z, m_H, \sin^2\theta_W, \alpha_s$) are schema-level via $\mathcal{R}_{\rm SM}$, with $m_H$ resting on the closed UV quartic $\lambda_\Phi = 1/(16\pi^2)$ (near-criticality). The *exact* solar angle $\theta_{12}$ (leading $1/N_{\rm fam} = \frac{1}{3}$) is *conditionally derived*, pending the seam-misalignment lemma — the sharpest remaining SM tension point. [C]

# 17 Status summary and open audit points

*Machine checks for this section:* [verification/v22_open_gates_audit.py]

| Statement | Status | Note |
|---|---|---|
| Dual-root dictionary ($Y$, cusps, $\gamma_{\rm car}$, $\varphi_{\rm base}$) | [E] | exact, from $\{1/2, -1/3\}$ |
| $10b_1 = 5^2 + 4^2 = \Omega_{\rm adm}\gamma + 1$ | [E] | Pythagorean carrier identity |
| $I_1^{\rm EW} = 41/48 = {\rm Var}(Y)\, b_1$ | [E] | exact |
| $\delta_2 = 4! \cdot 5!\, c_3^8$, ${\rm Tr}\, X^2 = 5!$ | [E] | factorial traces |
| $5/4 = g_{\rm car}/(g_{\rm car}-1)$ universal compression | [E] | exact |
| $u = \frac{4}{3}c_3 + 48c_3^4$; one-decoder identities | [E] | exact |
| $\xi = \frac{3}{4}/(1 + 9/128\pi^3)$ | [E] | corrected tree value |
| Ladder $A_{\rm EW} : \mathcal{A}_H : \mathcal{A}_\Lambda = 1 : g_{\rm car} : 2g_{\rm car}$ | [E]/[C] | Hubble rung is a corollary |
| Ladder $= \binom{g_{\rm car}}{0} : \binom{g_{\rm car}}{1} : \binom{g_{\rm car}}{2}$; $2g_{\rm car} = \binom{g_{\rm car}}{2} \Leftrightarrow g_{\rm car} = 5$ | [E] | new overdetermination of $g_{\rm car} = 5$ |
| 2D action lattice $S_x = a(\alpha_\star^{-1}/g_{\rm car}) + bL_0$ | [E]/[C] | two integer labels per scale |
| **Decuple bridge** $\rho_\Lambda/\bar{M}_{\rm Pl}^4 = \frac{3}{4\pi^2} e^{2\delta_{\rm ph}} [v/(5\beta_{\rm rad}^2 \bar{M}_{\rm Pl})]^{10}$ | [E] | exact, $10 = 2g_{\rm car} = \dim \Lambda^2 E$ |
| Family = ladder = Pascal row $\{1, 5, 10\}$; $16 = 1{+}5{+}10$ | [E] | central new identification |
| $g_{\rm car} = 5$ via $2^{g_{\rm car}-1} = \binom{g_{\rm car}}{0}{+}\binom{g_{\rm car}}{1}{+}\binom{g_{\rm car}}{2}$ | [E] | Pascal closure (unique) |
| ${\rm Tr}_{S^+}\, X^2 = 5!$ generates $40, 41, 240, \delta_2$ | [E] | master moment |
| $\lvert R(E_8)\rvert = 16 \cdot 5 \cdot 3$; $\dim E_8 = 240 + 8$ | [E] | $E_8$ compression |
| Even-flip atlas $240 = 160 + 80$ | [E]/[O] | metrization open ($\neq 112 + 128$); Coxeter Stage A closed on the character space (v531) |
| $\delta_{\rm CKM} - \pi/3 = 3u(1-u)$ | [E] | CP = triple seed variance |
| Observable quadrilateral ($u = \Omega_b + \beta_{\rm rad}$) | [E] | closed UV-shadow cycle |
| Power tower $v{=}x^1$, $H{=}x^5 R_H$, $\Lambda{=}x^{10} R_\Lambda$ | [E] | $K_5$ vertices/edges |
| $496 = 16 \cdot 31 = \dim(E_8{\times}E_8) = \dim SO(32)$ | [E]/[C] | heterotic-dimension count |
| $A_{\rm EW} = \alpha_\star^{-1}/\sqrt{\Delta_Y}$, $\Delta_Y = (b{+}s)^2 = 25$ | [E] | discriminant action theorem |
| $\hat{m}_e \hat{m}_\mu \hat{m}_\tau \propto \frac{32}{9} u^{10}$; $(5,3,2) = (g_{\rm car}, b, s)$ | [E] | lepton flavor decuple |
| $M_{\rm scal}/\bar{M}_{\rm Pl} = c_3^{7/2}$ | [E] | scalaron seam power |
| $\delta_{\rm CKM} - \pi/3 = N_{\rm fam}\, u(1-u)$ | [E] | weak CP = $N_{\rm fam}\times$ seed variance |
| $L = R + W$: $\sum R = 22$, $\sum W = 18$ | [E] | first generation = residue store |
| $\lvert R(E_8)\rvert/6 = \sum L = 40$ ($\mathbb{Z}_6$ slice) | [E] | one $\mathbb{Z}_6$ slice = flavor budget |
| $h_{E_8} = \lvert R\rvert/{\rm rank} = 30 = N_{\rm fam}\binom{5}{2} = N_{\rm fam} A_\Lambda$ | [E]/[C] | carrier Coxeter theorem |
| $\lvert R^+(E_8)\rvert = {\rm Tr}_{S^+}\, X^2 = 120 = 5!$ | [E] | positive-root moment |
| common decuple $\binom{5}{2}{=}A_\Lambda{=}u^{5+3+2}{=}h_{E_8}/N_{\rm fam}{=}10$ | [E]/[C] | one carrier pair-decuple |
| $\lambda_C^6/\delta_{\rm top} = 1.061$ (winding $\approx$ seam defect) | [O] | flagged proximity (not exact) |
| $y_f \simeq \lambda_C^{L_f}$, top $L_t = 0 \Rightarrow y_t \simeq 1$ | [E]/[C] | Froggatt–Nielsen ladder |
| $w = -1$ (true $\Lambda$); $N_{\rm eff} \simeq 3$ | [E] | dark energy / neutrino species |
| 3-generator lattice $\{\alpha_\star^{-1}, L_0, L_c\}$; $M_{\rm scal} = e^{-\frac{7}{2}L_c}$ | [E] | seam axis $L_c = \ln(1/c_3)$ |
| $\dim S^+ = 2\,{\rm rank}(E_8)$, $h_{E_8} = 2(\dim S^+ {-}1) = 2^g {-} 2$ | [E] | rank–Coxeter coupling ($\nu^c$ closes it) |
| $\gcd(2h, 6r) = \dim \mathfrak{g}_{\rm SM} = 12$, lcm $= 240$ | [E]/[C] | SM algebra = gcd; $E_8$ count = lcm |
| $I_1^{\rm EW} = h_{E_8}/36 + 1/(6r) = 41/48$ | [E]/[C] | Coxeter part + rank singlet |
| $M_{\rm scal}^2/\bar{M}_{\rm Pl}^2 = c_3^{r-1} = c_3^7$ exact; $A_s = N_\star^2 c_3^7/24\pi^2$ | [E]/[C]/[O] | seam power exact; $A_s$ conditional (Starobinsky) |
| $h_{E_8} = 2{\cdot}3{\cdot}5 = 30$, $r = \varphi(30) = 8$ | [E]/[C] | $2, 3, 5$ Coxeter cyclotomic compiler |
| ${\rm Exp}(E_8) = (\mathbb{Z}/30)^\times$, $\sum = 120$, $\chi_C = \Phi_{30}$ | [E]/[C] | exponents = primitive residues |
| lepton $(5,3,2)$: $\sum{=}10{=}A_\Lambda$, $\prod{=}30{=}h_{E_8}$ | [E]/[C] | sum=action, product=Coxeter |
| Hubble $H_0/\bar{M}_{\rm Pl} = e^{-\alpha_\star^{-1}}/(2\pi\sqrt{\Omega_\Lambda})$ | [E] | exact corollary |

| | | |
|---|---|---|
| $\Lambda$-anchored closure $G_N = \frac{3c^3}{32\pi^3\hbar\Lambda_{\rm geom}} e^{-2\alpha_\star^{-1}}$ | [E]/[C] | $G_N$ to $-0.35\%$ |
| $g_{\rm car} = 5$ overdetermined | [C] | monotone $\alpha^{-1}(g_{\rm car})$ |
| $\lvert R(E_8)\rvert = 240 = 16 \cdot 15$, $248 = 8 \cdot 31$ | [E] | counting identities |
| $E_8 \leftrightarrow$ carrier-transition map | [C] | needs structure map |
| $\delta_{\rm CP} = 4\pi/3$, $\theta_{12} = \frac{1}{3} - \frac{u}{2}$ | [C] | PMNS readouts |
| $\lvert V_{td}\rvert, \lvert V_{ts}\rvert, J_{\rm CKM}, J^{\nu}_{\rm CP}$; $n_s, r$ | [C] | derived readouts |

**Open audit points:** (1) *discharged* — the EM closure $F_{U(1)}(\alpha) = 0$ is now stated explicitly (Grammar III), with existence/uniqueness and root $\alpha_\star^{-1} = 137.0359992168$ ($2.9 \times 10^{-10}$, $1.9\sigma$ vs CODATA-2022); (2) keep $\gamma_{\rm car} = 5/6$ vs $\gamma_E$ distinct; (3) type $M_{\rm Pl}$ vs $\bar{M}_{\rm Pl}$ and "pure" ($\mathcal{A}_\Lambda = 274.07$) vs "total" ($-\ln(\delta_{\rm top} e^{-2\alpha_\star^{-1}}) = 283.10$, $-\ln(\rho_\Lambda/\bar{M}_{\rm Pl}^4) = 276.65$) actions consistently; (4) version the $\Lambda$ prefactor branch; (5) the scalaron-mass→ Starobinsky-$M$ identification is the *only* condition for $A_s = N_\star^2 c_3^7/(24\pi^2)$ (the seam power $M_{\rm scal}^2/\bar{M}_{\rm Pl}^2 = c_3^7$ itself is exact); (6) the $E_8$ *lattice* is now built explicitly as $(D_5 \oplus A_3) + \mu_4$-glue (Section 4), so the former "blind 240-metrisation" is closed. The flavor status (companion parabolic-weight note) now splits cleanly: the *anchor multiset* $\{1,1,2\}$ is derived (Schur–Horn + stability) [E]; the *sector assignment* $(1,1,2)$ follows from the $D_5$ cusp ordering [E]/[C]; the *first-generation* $+6$ *winding* is the hierarchy convention [C]; the six *non-anchor residues* are then forced (residue sets = $D_6$ orbit of $\{0,1,3\}$, minus anchor, monotone) [E]. The remaining geometric input at the time of writing was (H2), the specific $D_6$ cusp labelling on $\mathbb{P}^1 \setminus \mu_4$ — sharply characterised by the residue-matrix spectral selector ($\det R = h(D_5)$, principal minors $(2,3,5)$), with a naive cusp-position shortcut explicitly ruled out [C]. The explicit Riemann–Hilbert construction confirms the rigid $SU(3)$ system and its $A_3$ Coxeter rotation. *Status update:* the H2 dictionary has since been made *exact* (hexagon = sign-twisted family spectrum, addresses pinned, margins forced, v118–v122), the deck is *derived* (v141), and the residue is the R4′ selector establishment — two line pairings (v139/v142); what stays geometric is only the parabolic realisation premise (`GATE.QGEO`, research contracts).

**Honest negatives (reported to keep the discipline visible).** The TFPT lepton ladder gives a Koide ratio $Q = (\sum m_\ell)/(\sum \sqrt{m_\ell})^2 = 0.6645$, close to but *not* the empirical $2/3 = 0.66667$ ($-0.33\%$); TFPT is Koide-*like* but does not reproduce the precise relation. The baryon asymmetry $\eta_B \approx 6 \times 10^{-10}$ and the proton/electron ratio $m_p/m_e = 1836$ do *not* fall on a clean action charge or $L_0$ ladder; they remain outside the present grammar. These are stated so the strong items are not read as a uniform success.

## 18 Consolidated picture

**The three nested grammars.**

$$\textbf{Carrier:}\quad 3+2 \Rightarrow \{y_-, y_+\} = \{-\tfrac{1}{3}, \tfrac{1}{2}\} \Rightarrow Y,\ \dim S^+ = 16,\ N_{\rm fam} = 3,\ \Omega_{\rm adm} = 48,\ b_1 = \tfrac{41}{10},$$

$$\textbf{Seed:}\quad u = \tfrac{4}{3}c_3 + 48c_3^4 \Rightarrow \{\lambda_C,\ \beta_{\rm rad},\ \Omega_b,\ \sin^2\theta_{13}\},$$

$$\textbf{Action:}\quad \alpha_\star^{-1} \Rightarrow A_{\rm EW} = \tfrac{\alpha_\star^{-1}}{g_{\rm car}},\ \mathcal{A}_\Lambda = 2\alpha_\star^{-1},\ \mathcal{A}_H = \alpha_\star^{-1} \Rightarrow \frac{\rho_\Lambda}{\bar{M}_{\rm Pl}^4} = \frac{3}{4\pi^2} e^{2\delta_{\rm ph}} \Big[\frac{v_{\rm geo}}{5\beta_{\rm rad}^2 \bar{M}_{\rm Pl}}\Big]^{10}.$$

A small grammar generates many sectors: the Standard-Model packet, flavor, the electroweak hierarchy, the cosmological constant, the metrology, and the inflation/CMB interface. The strongest new statements are exact: the decuple bridge ties $\Lambda$ to the tenth power of the stripped electroweak scale with $10 = 2g_{\rm car}$; the Hubble scale is the square root of $\Lambda$; and the Einstein normaliser is the corrected tree value $\frac{3}{4}/(1 + 9/128\pi^3)$. None of this is claimed as a proof of a final theory; it is offered as a tightening of the existing architecture for your assessment.

**The compiler in one line:** TFPT = even carrier code + seed decoder + 2·3·5 Coxeter compiler. From the three atoms $(2, 3, g_{\rm car}{=}5)$:

$$h_{E_8} = 2{\cdot}3{\cdot}5 = 30, \qquad r_{E_8} = \varphi(30) = 8 = g_{\rm car}{+}N_{\rm fam},$$

$$|R(E_8)| = rh = 240, \qquad \dim E_8 = r(h{+}1) = 8{\cdot}31 = 248,$$

$$\mathrm{Exp}(E_8) = (\mathbb{Z}/30)^{\times}, \quad \textstyle\sum \mathrm{Exp} = 120 = \mathrm{Tr}_{S^+}\, X^2 = |R^+|, \quad \chi_C = \Phi_{30},$$

$$\dim S^+ = 1 + 10 + 5 = 16 = 2r, \quad A_{\rm EW}{:}\mathcal{A}_H{:}\mathcal{A}_\Lambda = 1{:}5{:}10, \quad \mathcal{A}_H = \tfrac{h}{6},\ \mathcal{A}_\Lambda = \tfrac{h}{3},\ N_{\rm fam} = \tfrac{h}{A_\Lambda} = 3,$$

$$\textstyle\sum L = \frac{|R^+|}{3} = 40, \quad 10b_1 = 1 + \frac{|R^+|}{3} = 41 = 1 + h\frac{\dim S^+}{\dim \mathfrak{g}_{\rm SM}},$$

$$\det M_\ell \propto u^{10}, \qquad \frac{\rho_\Lambda}{\bar{M}_{\rm Pl}^4} \propto \Big[\frac{v_{\rm geo}}{5\beta_{\rm rad}^2 \bar{M}_{\rm Pl}}\Big]^{10}, \qquad A_{\rm EW} = \frac{\alpha_\star^{-1}}{\sqrt{\Delta_Y}}, \quad \sqrt{\Delta_Y} = g_{\rm car} = 5.$$

**The common decuple (= 10):** $\binom{g_{\rm car}}{2}$ (carrier pairs) = $A_\Lambda$ (cosmological action) = $u^{5+3+2}$ (lepton determinant) = $h_{E_8}/N_{\rm fam}$ (Coxeter per family). $\Lambda$, the charged-lepton determinant, the $E_8$ Coxeter structure and the action ladder all read the *same* carrier pair-decuple. The open core is one question: can the even-flip atlas be metrised as a genuine $E_8$ root system (the 160+80 vs 112+128 obstruction)? Its Coxeter half (Stage A) is closed on the primitive-character space by the tensor clock $T_{30} = \mathrm{Comp}(\Phi_5) \otimes \mathrm{Comp}(\Phi_6)$ (v531); the atlas intertwiner and the metrization (Stage B) remain.

# Part II
# The Coxeter–cyclotomic compiler and the explicit $E_8$ construction

**Abstract**

This short companion compresses the TFPT bridge layer to a single discrete object. The claim is that the Standard-Model family, the cosmological action ladder, the $E_8$ counting data, the flavor budget, inflation and the cosmological constant are *not* independent coincidences but projections of one structure: the even-weight code $C^+$ on $g_{\rm car} = 5$ slots, together with its $\mathbb{Z}_{30}$ Coxeter spectrum and the single seam seed $c_3 = 1/(8\pi)$. The decisive new result is that $C^+$ is precisely the half-spin weight orbit of $D_5 = \mathfrak{so}_{10}$ ($\mathrm{Aut}(C^+) = W(D_5)$, order 1920), that the family geometry $\mathbb{P}^1 \setminus \mu_4$ is an $A_3 = \mathfrak{su}_4$, and that $E_8$ is then the standard *closure algebra* of $D_5 \times A_3$ — not an input. The fusion is exact: $\mathrm{disc}(D_5) = \mathrm{disc}(A_3) = \mathbb{Z}_4$, so the four family corners $\mu_4$ are precisely the glue code that closes $D_5 \oplus A_3$ into the unimodular $E_8$ lattice. The glue is metric: the discriminant-form norms $q(D_5) = \frac{5}{4}$ and $q(A_3) = \frac{3}{4}$ sum to the $E_8$ root norm 2 — and these are exactly the pre-existing TFPT constants $\delta_2/\delta_{\rm top}^2 = \frac{5}{4}$ and $\xi_{\rm tree} = \frac{3}{4}$. The $A_3$ side is explicit: the residue pairing on $H^1(\mathbb{P}^1 \setminus \mu_4)$ is the $A_3$ Cartan form, and the carrier corner rotation $z \mapsto iz$ acts as the $A_3$ Coxeter element (order 4, spectrum $\{i, -1, -i\}$). The lift is no longer a count: the 240 $E_8$ roots are *explicitly* built from $D_5 \oplus A_3$ plus the spinor ($\mu_4$) glue, all of norm 2, Cartan determinant 1. Seven statements carry it; every lattice identity below is checked exactly, and the analytic metric is computed and characterised (a $D_4$-equivariant deformation), so no *lattice* structural step is left open in the $D_5 \times A_3$ glue route (the upstream boundary kernel, EM closure and flavor Hecke saturation remain separate open problems). Master statement: *five carrier slots give $D_5$, four family corners give $A_3$, and $\mu_4$ glues $D_5 \times A_3$ into the $E_8$ atlas.*

**The compression.**

$$\mathrm{TFPT} = \underbrace{D_5 \text{ half-spin carrier}}_{C^+} + \underbrace{A_3 \text{ four-point family}}_{\mathbb{P}^1\setminus\mu_4} + c_3 \text{ seam seed},$$

$$E_8 = \text{closure of } D_5 \times A_3$$

with $c_3 = \dfrac{1}{8\pi}$, $g_{\rm car} = 5$, $h := 2 \cdot 3 \cdot g_{\rm car} = 30$, $r := \varphi(h) = \varphi(30) = 8$, $N_{\rm fam} = 3$, $N_\Phi = 1$, $\dim \mathfrak{g}_{\rm SM} = 12$. Equivalently $C^+ + \mathbb{Z}_{30}$ Coxeter compiler + $c_3$ seed; the $D_5 \times A_3$ reading is the structural lift of the same object.

## Theorem 1 — the even carrier code

Let $E$ be the rank-$g_{\rm car} = 5$ carrier, split $E = E_3 \oplus E_2$ (color $b = 3$, weak $s = 2$). The one-generation spinor packet is the even exterior algebra, equivalently the even-weight binary code

$$S^+ = \Lambda^0 E \oplus \Lambda^2 E \oplus \Lambda^4 E \;\cong\; C^+ = \{x \in \mathbb{F}_2^5 : |x| \text{ even}\}, \qquad \dim S^+ = |C^+| = 2^{g_{\rm car}-1} = 16. \quad (69)$$

The two carrier roots $y_+ = \frac{1}{2}$, $y_- = -\frac{1}{3}$ generate all hypercharges ($Y(Q_L) = y_+ + y_- = \frac{1}{6}$, $Y(u^c) = 2y_-$, $Y(e^c) = 2y_+$, $Y(\nu^c) = 0$) and the transport cusps $\{1, \frac{2}{3}, \frac{1}{3}\} = \{2y_+, -2y_-, 2(y_+ + y_-)\}$, with $\gamma_{\rm car} = \mathrm{Tr}_E\, Y^2 = 3y_-^2 + 2y_+^2 = \frac{5}{6}$. **[E]**

**Gauge-rank factorisation (`[verification/v79_review_identities.py]`).** The carrier polynomial factors into the two gauge ranks directly:

$$\boxed{6Y^2 - Y - 1 = (2Y - 1)(3Y + 1)}, \qquad Y = \tfrac{1}{2}\ (s{=}2,\ SU(2)), \quad Y = -\tfrac{1}{3}\ (b{=}3,\ SU(3)),$$

i.e. $(sY - 1)(bY + 1)$ — the polynomial splits into one linear representative of each Standard-Model non-abelian factor; the 5-slot carrier *forces* the split into $SU(3) \times SU(2)$.

**Hypercharge Lucas–Binet resonances [E] (mode reading [C], [`verification/v79_review_identities.py`])**

The integer carrier charges $(q_+, q_-) = (3, -2)$ (the primitive trace-free pair, $X^2-X-6 = (X-3)(X+2)$, Lean `unique_carrier_pair`) generate the Lucas $U$-sequence

$$D_n = \frac{3^n - (-2)^n}{5}: \qquad D_1, \ldots, D_6 = 1,\ 1,\ 7,\ 13,\ 55,\ 133,$$

which is, *in order*, $(N_\Phi,\ \cdot\ ,$ scalaron exponent 7, $\Delta_Q$=13, quark numerator 55, $\dim E_7$=133). In particular the quark ratio reads $c_u/c_d = D_5/(N_{\text{fam}}^2 D_4) = \frac{55}{117}$ — a single recurrence of the hypercharge generator, not a free integer. [E] for the arithmetic; the per-mode physical identification is a [C] cross-domain reading (only $n \le 6$, the carrier-slot range, is meaningful).

## Theorem 2 — Exterior–Action Duality

On $C^+$ the flip $F = x \oplus y$ is even, so $|F| \in \{0, 2, 4\}$. The three relations $R_0, R_2, R_4$ form a symmetric association scheme (the even Hamming scheme on five bits) with

$$\boxed{\begin{array}{l}\text{valencies } (1, 10, 5) = (\dim \Lambda^0 E, \dim \Lambda^2 E, \dim \Lambda^4 E),\\ \text{spectral multiplicities } (1, 5, 10) = A_{\text{EW}}{:}\mathcal{A}_H{:}\mathcal{A}_\Lambda\end{array}} \quad \text{[E]}. \tag{70}$$

The eigenvalues (computed directly) are $A_2 : \{10^{(1)}, 2^{(5)}, (-2)^{(10)}\}$ and $A_4 : \{5^{(1)}, (-3)^{(5)}, 1^{(10)}\}$, with $A_2 + A_4 = K_{16} - \mathbf{1}$. The two graphs are named objects:

$$\begin{array}{l} R_2(C^+) = \text{folded/complement, SRG}(16, 10, 6, 6) \text{ (the Clebsch complement);}\\ \boxed{R_4(C^+) = \text{Clebsch graph, SRG}(16, 5, 0, 2)} \quad \text{[E]}. \end{array} \tag{71}$$

**Remark.** The Standard-Model family is the *valency* side $(1, 10, 5)$ of one scheme; the cosmological action ladder is its *spectral* side $(1, 5, 10)$. The action ladder is therefore not merely "also binomial" — it is the spectral dual of the even exterior family. [C]

**This duality is the MacWilliams transform ([`verification/v79_review_identities.py`]).** $C^+$ is the even-weight code $[5, 4, 2]$; its dual $(C^+)^\perp$ is the repetition code $[5, 1, 5] = \{00000, 11111\}$ (the uniform/vacuum word). The MacWilliams identity sends the weight enumerator of the dual to that of $C^+$, namely $(1, 10, 5)$ — so the valency↔spectral duality *is* the coding-theoretic UV/IR duality, and the carrier grading $\Lambda^{\text{even}}(5) = 1{+}10{+}5$ is the MacWilliams image of the repetition vacuum. [E] (code identity) / [C] (holographic reading).

**The full Bose–Mesner algebra** (machine-checkable) fixes the duality beyond eigenvalues:

$$A_2^2 = 10\,\mathbf{1} + 6A_2 + 6A_4, \quad A_2A_4 = 3A_2 + 4A_4, \quad A_4^2 = 5\,\mathbf{1} + 2A_2, \tag{72}$$

with eigenmatrix and multiplicity orthogonality

$$P = \begin{pmatrix} 1 & 10 & 5 \\ 1 & 2 & -3 \\ 1 & -2 & 1 \end{pmatrix}, \quad \det P = -64, \qquad \boxed{P^\top \operatorname{diag}(1, 5, 10)\, P = \operatorname{diag}(16, 160, 80)} \quad \text{[E]}, \tag{73}$$

so the flip split $240 = 160{+}80$ ($|F| = 2, 4$) appears directly in the scheme's eigenmatrix orthogonality.

**Carrier $= D_5$ (verified).** The 16 vertices of $C^+$ are exactly the weights of the $D_5 = \mathfrak{so}_{10}$ half-spinor $\frac{1}{2}(\pm 1, \ldots, \pm 1)$ with an even number of minus signs. The flip-2 and flip-4 graphs are translation- and permutation-invariant, so the *affine* group $C^+ \rtimes S_5$ (translations $\rtimes$ coordinate permutations) acts by graph/scheme automorphisms; the group closure gives the automorphism group of the even Hamming scheme (equivalently the Clebsch graph),

$$\boxed{\operatorname{Aut}_{\text{aff}}(C^+) = C^+ \rtimes S_5, \qquad |\operatorname{Aut}_{\text{aff}}(C^+)| = 2^{g_{\text{car}}-1}\, g_{\text{car}}! = 16 \cdot 120 = 1920 = |W(D_5)|} \quad \text{[E]}, \tag{74}$$

and $1920 = r\,|R(E_8)| = 8 \cdot 240$. (The *linear* code automorphism group alone is only the coordinate group $S_5$ of order 120; the factor $2^{g_{\rm car}-1} = 16$ is the translation/scheme part. It is the affine/scheme group that equals $W(D_5)$.) Edges of the Clebsch graph: $|E(R_4)| = \frac{16\cdot 5}{2} = 40 = |R(D_5)| = \sum_{f,j} L_{f,j}$ (the entire flavor transport budget). So the five-slot even code is the half-spin weight structure of $\mathfrak{so}_{10}$, the classical home of one SM family with $\nu^c$.

## Theorem 3 — the Coxeter closure equation selects $g_{\rm car} = 5$

Requiring the even family to equal the action truncation through pair degree, $\dim S^+ = 2^{g-1} = \binom{g}{0} + \binom{g}{1} + \binom{g}{2}$, i.e.

$$\boxed{2^g - 2 = g(g+1) \iff g = 5} \qquad \textbf{[E]}, \tag{75}$$

the unique positive-integer solution. At $g = 5$ the two sides equal 30, so *carrier exhaustion* $2^g - 2$ and the *Coxeter period* $g(g+1)$ coincide only at five slots: $2^{g_{\rm car}} - 2 = 30 = h$ and $g_{\rm car}(g_{\rm car}+1) = 30$.

## Theorem 4 — the $\mathbb{Z}_{30}$ cyclotomic Coxeter compiler

From the three atoms $(2, 3, g_{\rm car})$, with $h = 2 \cdot 3 \cdot g_{\rm car} = 30$ and $r = \varphi(h) = 8$:

$$\boxed{|R(E_8)| = rh = 240, \qquad \dim E_8 = r(h+1) = 8 \cdot 31 = 248, \qquad h + 1 = 2^{g_{\rm car}} - 1 = 31} \qquad \textbf{[E]}. \tag{76}$$

The $E_8$ exponents are the primitive residues modulo 30,

$$\mathrm{Exp}(E_8) = (\mathbb{Z}/30\mathbb{Z})^\times = \{1, 7, 11, 13, 17, 19, 23, 29\}, \qquad \chi_C(t) = \Phi_{30}(t) = t^8 + t^7 - t^5 - t^4 - t^3 + t + 1, \tag{77}$$

and $|(\mathbb{Z}/30)^\times| = \varphi(30) = 8 = r$. The local SM algebra and the root atlas are the two arithmetic projections of $(r, h) = (8, 30)$:

$$\Omega_{\rm adm} = 6r = 48, \quad D_{\rm start} = 2h = 60, \quad \gcd(6r, 2h) = \dim \mathfrak{g}_{\rm SM} = 12, \quad \mathrm{lcm}(6r, 2h) = rh = 240. \tag{78}$$

## Theorem 5 — the root transition atlas and the positive-edge moment

The directed even-flips of $C^+$ are the complete graph $K_{16}$:

$$\boxed{\begin{aligned} |R^+(E_8)| = \binom{\dim S^+}{2} = \binom{16}{2} = 120 = \mathrm{Tr}_{S^+} X^2 = 5! = \sum_{m \in (\mathbb{Z}/30)^\times} m, \\ |R(E_8)| = 2\binom{16}{2} = 240 \end{aligned}} \qquad \textbf{[E]}. \tag{79}$$

($X = 6Y$.) So the positive $E_8$ roots are the unordered transitions of one family; the second hypercharge moment, the positive-root count and the sum of the $E_8$ exponents coincide at 120. The CP pairing $m \leftrightarrow 30 - m$ of the exponents gives four pairs, each of sum $h = 30$:

$$\boxed{\mathrm{Tr}_{S^+} X^2 = (N_{\rm fam} + N_\Phi)\, h = 4 \cdot 30 = 120 = \dim \mathfrak{so}_{16} = \underbrace{45}_{D_5} + \underbrace{15}_{A_3} + \underbrace{60}_{(10,6)}} \qquad \textbf{[E]}, \tag{80}$$

so the hypercharge moment is simultaneously four CP-paired Coxeter periods *and* the adjoint half $\mathfrak{so}_{16} = (45, 1) \oplus (1, 15) \oplus (10, 6)$ of the $E_8$ closure.

**How the root *origin* is now obtained — see Theorem 7.** A naive flip-size-preserving map $R_{\rm flip} \to \mathbb{R}^8$ cannot be $E_8$: the flip split $240 = 160 + 80$ ($|F| = 2, 4$) differs from the $D_8$

split $112 + 128$. The correct route is *not* a direct 240-metrization but the standard branching $E_8 \supset D_5 \times A_3$ (Theorem 7), which builds the root spaces from known $D_5$ and $A_3$ data. Until the $A_3$ lattice is derived from the four punctures (Theorem 7, [O]), $E_8$ remains a Coxeter atlas with a concrete, falsifiable lift.

## Theorem 6 — downstream readouts from $(h, r, c_3)$

All discrete budgets and the seam seed are functions of $(h, r, c_3) = (30, 8, 1/(8\pi))$:

$$\dim S^+ = 2r = 16, \qquad h = 2(\dim S^+ - 1) = 30, \tag{81}$$

$$10b_1 = 5r + 1 = 41 = 1 + h\tfrac{\dim S^+}{\dim \mathfrak{g}_{\rm SM}}, \qquad I_1^{\rm EW} = \frac{5r+1}{6r} = \frac{41}{48}, \tag{82}$$

$$u = \frac{r}{6}c_3 + 6r\,c_3^4 = \frac{4}{3}c_3 + 48c_3^4, \qquad \delta_2 = 4!\,\tbinom{16}{2}\,c_3^8 = 2880\,c_3^8, \tag{83}$$

$$\xi_{\rm tree} = \frac{6}{r} = \frac{\dim \mathfrak{g}_{\rm SM}}{\dim S^+} = \frac{3}{4}, \qquad \frac{M_{\rm scal}^2}{\bar{M}_{\rm Pl}^2} = c_3^{\Omega_{\rm adm} - 10b_1} = c_3^{r-1} = c_3^7. \tag{84}$$

**Two-graph product table.** With the carrier-pair graph $K_5$ ($|E(K_5)| = \binom{5}{2} = 10 = \mathcal{A}_\Lambda$) and the family-corner graph $K_4$ on $\mu_4$ ($|E(K_4)| = \binom{4}{2} = 6$, $|R(A_3)| = 2|E(K_4)| = 12$), every large budget is a product of the two graphs:

$$\boxed{40 = 10 \cdot 4, \quad 60 = 10 \cdot 6, \quad 120 = 10 \cdot 12, \quad 240 = 10 \cdot 24} \qquad \text{[E]}, \tag{85}$$

i.e. $(\sum L,\, D_{\rm start},\, |R^+(E_8)|,\, |R(E_8)|) = 10 \cdot (|\mu_4|, |E(K_4)|, |R(A_3)|, |S_4|)$. Two occupancy identities follow,

$$\begin{aligned} &\boxed{10b_1 = 1 + |\mu_4|\,|E(K_5)| = 1 + 4 \cdot 10 = 41}, \\ &\boxed{\Omega_{\rm adm} = N_{\rm fam}\dim S^+ = (N_{\rm fam}{+}N_\Phi)|R(A_3)| = 3 \cdot 16 = 4 \cdot 12 = 48} \qquad \text{[E]}, \end{aligned} \tag{86}$$

so the Higgs singleton is the fourth ($\mu_4$) direction that closes $N_{\rm fam} = 3$ to **4** of $A_3$. **Occupancy deficit ladder.** The admissible occupancy splits cleanly,

$$\boxed{\begin{aligned} &\Omega_{\rm adm} = \underbrace{40}_{\sum L\,(\text{flavor})} + \underbrace{1}_{\text{Higgs}} + \underbrace{7}_{\text{scalaron}} = 48, \qquad \Omega_{\rm adm} - \sum L = 8 = h(D_5) = r(E_8), \\ &\Omega_{\rm adm} - 10b_1 = 7 = h(D_5) - 1 \end{aligned}} \qquad \text{[E]}, \tag{87}$$

so the scalaron exponent $r - 1 = 7$ is exactly the post-flavor, post-Higgs occupancy remainder, $M_{\rm scal}^2/\bar{M}_{\rm Pl}^2 = c_3^{\Omega_{\rm adm} - 10b_1} = c_3^7$. With $h(D_5) = 8$ and $|R^+(A_3)| = 6$ the seed itself is a $D_5/A_3$ ratio–product, $u = \frac{h(D_5)}{|R^+(A_3)|}c_3 + h(D_5)|R^+(A_3)|c_3^4$, and $\xi = \frac{|R^+(A_3)|}{h(D_5)}\,(1 + |R^+(A_3)|^2 c_3^3)^{-1} = \frac{3}{4}(1 + 36c_3^3)^{-1}$. The seed readouts and the two decuples are

$$\lambda_C^2 = u(1{-}u), \quad \beta_{\rm rad} = 2c_3 u, \quad \sin^2\theta_{13} = e^{-5/6}u, \quad \boxed{\omega_b = \frac{\lambda_C}{\mathcal{A}_\Lambda} = \frac{\lambda_C}{10} = 0.02244}\ (\text{Planck } 0.02237), \tag{88}$$

$$\boxed{\begin{aligned} &\det M_\ell \propto u^{\,h/N_{\rm fam}} = u^{10}, \\ &\frac{\rho_\Lambda}{\bar{M}_{\rm Pl}^4} = \frac{3}{4\pi^2}e^{2\delta_{\rm ph}}\Big[\tfrac{v_{\rm geo}}{5\beta_{\rm rad}^2 \bar{M}_{\rm Pl}}\Big]^{h/N_{\rm fam}} \end{aligned}}, \tag{89}$$

with lepton powers $(5, 3, 2) = (g_{\rm car}, b, s)$: $\sum = 10 = \mathcal{A}_\Lambda$ (the $\Lambda$ decuple), $\prod = 30 = h$ (the Coxeter period). The first-generation transport sums read the non-primitive $\mathbb{Z}_{30}$ load,

$$(G_1, G_2, G_3) = (22, 13, 5) = (h - r,\ \dim \mathfrak{g}_{\rm SM} + 1,\ g_{\rm car}), \qquad h - r = 30 - \varphi(30) = 22, \tag{90}$$

**Flavor transport is a $D_5\times A_3$ object.** The transport packets $L_u{=}(7,3,0), L_d{=}(7,5,2), L_e{=}(8,5,3)$ split into a $D_5$ budget and an $A_3$ geometry:

$$\boxed{\textstyle\sum_{f,j} L_{f,j} = 40 = |R(D_5)| \text{ (Clebsch edges)}, \qquad L_{f,j} \bmod 6 \text{ lives on } C_6 = Z_6 = |R^+(A_3)|} \quad \text{[E]}, \tag{91}$$

the residues being the $D_6$-orbit of the perfect set $\{0,1,3\}\subset C_6$ (cyclic distances $\{1,2,3\}$): $u\equiv\{0,1,3\}$, $d\equiv 2-\{0,1,3\}$, $e\equiv 2+\{0,1,3\}$, with the universal first-generation $+6$ winding. So the carrier $D_5$ sets the total budget and the family $A_3$ (its 6-element positive-root hexagon) sets the per-entry distances. The scalaron mass is a pure seam power [E]; the amplitude is a *conditional* readout, kept [C]/[O] until the scalaron variable is formally identified with the Starobinsky $M$ parameter. That identification is now *heat-kernel grounded* (G2): the spectral-action Seeley–DeWitt coefficient $a_4|_{R^2}\sim R^2/72$ delivers exactly the Starobinsky $R^2$ term, with $M^2/\bar{M}_{\rm Pl}^2 = 6(4\pi)^2/f_0$ and the TFPT closure $c_3^7$ fixing the cutoff moment $f_0$ ([`verification/v36_spectral_action_g2.py`]):

$$\underbrace{\frac{M_{\rm scal}^2}{\bar{M}_{\rm Pl}^2} = c_3^{r-1} = c_3^7}_{\text{[E]}}, \qquad \underbrace{A_s = \frac{N_\star^2}{24\pi^2}\frac{M_{\rm scal}^2}{\bar{M}_{\rm Pl}^2} = \frac{N_\star^2 c_3^7}{24\pi^2} = 2.17\times 10^{-9}}_{\text{[C]/[O] (if } M_{\rm scal}=M_{\rm Star})}. \tag{92}$$

## Theorem 7 — the $D_5 \times A_3$ lift of the $E_8$ atlas

Since $A_3 = \mathfrak{su}_4 \cong \mathfrak{so}_6 = D_3$, one has the standard chain $E_8 \supset \mathfrak{so}_{16} \supset \mathfrak{so}_{10}\oplus\mathfrak{so}_6 = D_5\times A_3$, with the exact branching (verified dimensions and root/weight counts):

$$\boxed{\mathfrak{e}_8 = (45,1)\oplus(1,15)\oplus(10,6)\oplus(16,4)\oplus(\overline{16},\overline{4})},$$
$$\underbrace{45{+}15{+}60{+}64{+}64}_{=248} = \dim E_8, \qquad \underbrace{40{+}12{+}60{+}64{+}64}_{=240} = |R(E_8)|. \tag{93}$$

The adjoint **120** of $\mathfrak{so}_{16}$ is $(45,1)\oplus(1,15)\oplus(10,6)$ and its spinor **128** is $(16,4)\oplus(\overline{16},\overline{4})$. The TFPT dictionary makes each summand a known object:

| piece | value | TFPT reading |
|---|---|---|
| $(45,1)$ | $45 = 40+5$ | $D_5$ carrier adjoint: Clebsch edges $40=\|R(D_5)\|$ plus carrier rank $g_{\rm car}=5$ |
| $(1,15)$ | $15 = 12+3$ | $A_3$ family+Higgs adjoint: $\dim\mathfrak{g}_{\rm SM}=12$ plus family rank $N_{\rm fam}=3$ |
| $(10,6)$ | $10\cdot 6$ | $\Lambda$ decuple $\mathcal{A}_\Lambda=\binom{5}{2}=10$ times $Z_6$ transport $\binom{4}{2}=6$ |
| $(16,4)\oplus(\overline{16},\overline{4})$ | $2\cdot 64$ | matter half-spinor $S^+$ times the four corners $\mu_4$, plus conjugate sheet |

**Proof route (replaces the old blind 240-metrization).** *Stage 1 ($D_5$, [E]):* $C^+$ is the $D_5$ half-spin orbit, $R_4$ = Clebsch, Aut $= W(D_5)$, $|E(R_4)| = 40 = |R(D_5)|$ — Theorem 2. *Stage 2 ($A_3$, [E]):* the carrier geometry fixes $X_f^\circ = \mathbb{P}^1\setminus\mu_4$ with $H_1(X_f^\circ,\mathbb{Z})=\mathbb{Z}^3$ and a rigid $D_4\subset PGL_2$ generated by $z\mapsto iz$ and $z\mapsto 1/z$. The logarithmic forms $\omega_i = dz/(z-p_i)$ regular at infinity span $H^1$ by their differences, dual under the residue pairing to the puncture loops $\gamma_i$ with $\mathrm{Res}(\omega_i,\gamma_j)=\delta_{ij}$; hence the canonical (residue) metric on the puncture classes is $I_4$, and the twelve puncture-difference cycles $\gamma_i-\gamma_j$ are exactly the $A_3$ roots:

$$\boxed{\{\gamma_i-\gamma_j\}_{i\neq j}:\ \#=12=|R(A_3)|=\dim\mathfrak{g}_{\rm SM},\quad \mathrm{rank}=3=N_{\rm fam},\\ \mathrm{Gram}=\mathrm{Cartan}(A_3),\ \det=4} \quad \text{[E]}, \tag{94}$$

with $\dim A_3 = 15 = \dim S^+ - 1$, and the $A_3$ positive roots equal the exponent sum $1{+}2{+}3 = 6 = Z_6 = |E(K_4)|$. *The corner rotation realises the $A_3$ Coxeter element.* The carrier generator $z\mapsto iz$ permutes $\mu_4$ as the 4-cycle $(p_1p_2p_3p_4)\in S_4 = W(A_3)$ and acts on $H^1=\mathbb{C}^3$ as an isometry of the Cartan form with

$$\boxed{\mathrm{ord}=4=h(A_3)=|\mu_4|,\quad \mathrm{spec}=\{i,-1,-i\},\quad \chi(t)=\tfrac{t^4-1}{t-1}=\Phi_4(t)\Phi_2(t)} \quad \text{[E]}, \tag{95}$$

i.e. the eigenvalues are the $A_3$ exponents $\{1,2,3\}$ as fourth roots of unity — the exact analogue of the $E_8$ Coxeter element reading $\Phi_{30}$. The full $D_4 = \langle z \mapsto iz, z \mapsto 1/z \rangle$ embeds in $W(A_3) = S_4$ (order 8, index 3) by isometries. *Stage 3 (standard, [E]):* assemble $E_8$ from the $D_5 \times A_3$ branching — no 240-map required.

**The $\mu_4$ glue theorem (why the four corners appear everywhere).** The two factors have equal discriminant groups, and $E_8$ is unimodular:

$$\boxed{\operatorname{disc}(D_5) = \operatorname{disc}(A_3) = \mathbb{Z}_4, \qquad [\,E_8 : D_5 \oplus A_3\,] = \sqrt{\tfrac{\det D_5 \cdot \det A_3}{\det E_8}} = \sqrt{\tfrac{4\cdot 4}{1}} = 4 = |\mu_4|} \qquad \textbf{[E]}. \tag{96}$$

So $E_8 = (D_5 \oplus A_3) + \mu_4$-*glue*: the four family corners $\mu_4 = \mathbb{Z}_4$ are precisely the glue code that fuses the carrier and the family into the unimodular $E_8$ lattice. The glue pairs the $D_5$ spinor class ($= S^+$, the family) with the $A_3$ class — exactly the $(16,4) \oplus (\overline{16},\overline{4})$ spinor block. Hence $\mu_4$ plays three roles at once: family corners, $A_3$ origin, and $E_8$ glue.

**Discriminant-norm glue lemma (and the TFPT $5/4, 3/4$ identification).** The two glue generators carry the discriminant quadratic form $q \in \mathbb{Q}/2\mathbb{Z}$:

$$\boxed{\begin{aligned} &q(D_5) = \tfrac{n}{4}\big|_{n=5} = \tfrac{5}{4} \text{ (spinor class)}, \quad q(A_3) = \tfrac{n}{n+1}\big|_{n=3} = \tfrac{3}{4} \text{ (corner class)}, \\ &q(D_5) + q(A_3) = 2 \equiv 0 \pmod 2, \qquad q(D_5) - q(A_3) = \tfrac{1}{2} \end{aligned}} \qquad \textbf{[E]}. \tag{97}$$

The diagonal $\langle (s,g) \rangle \cong \mathbb{Z}_4$ is isotropic ($k^2 \cdot 2 \equiv 0$ for all $k$), so the glue closes to an *even unimodular* lattice — this is $E_8$.

**The two glue norms are just $(g_{\text{car}}, N_{\text{fam}})$ over the glue index.** Because $\operatorname{rank} D_5 = g_{\text{car}} = 5$ (so $q(D_5) = \frac{n}{4}|_{n=5} = g_{\text{car}}/|\mu_4|$) and $\operatorname{rank} A_3 = N_{\text{fam}} = 3$ (so $q(A_3) = \frac{n}{n+1}|_{n=3} = N_{\text{fam}}/|\mu_4|$, with $n{+}1 = 4 = |\mu_4|$),

$$\boxed{q(D_5) = \frac{g_{\text{car}}}{|\mu_4|} = \tfrac{5}{4}, \qquad q(A_3) = \frac{N_{\text{fam}}}{|\mu_4|} = \tfrac{3}{4}}$$

and *all four arithmetic operations* on the pair are then forced compiler atoms:

$$\underbrace{q(D_5){+}q(A_3) = \tfrac{g_{\text{car}}+N_{\text{fam}}}{|\mu_4|} = 2}_{E_8 \text{ root norm}}, \qquad \underbrace{q(D_5){-}q(A_3) = \tfrac{|\mathbb{Z}_2|}{|\mu_4|} = \tfrac{1}{2}}_{\delta \text{ (lepton transport)}},$$

$$\underbrace{q(D_5)q(A_3) = \tfrac{g_{\text{car}}N_{\text{fam}}}{|\mu_4|^2} = \tfrac{15}{16}}_{\dim \mathfrak{su}(4)/\dim S^+}, \qquad \underbrace{\tfrac{q(D_5)}{q(A_3)} = \tfrac{g_{\text{car}}}{N_{\text{fam}}} = \tfrac{5}{3}}.$$

So the distinguished lepton transport value $\delta = \frac{1}{2} = (g_{\text{car}} - N_{\text{fam}})/|\mu_4| = |\mathbb{Z}_2|/|\mu_4|$ is the carrier-minus-family count over the glue index — a fully atomic origin, not a chosen rational (it also equals the harmonic-frame holonomy diagonal modulus, `[verification/v41_leg_assignment.py]`); and (sum, ratio) invert uniquely back to $(\frac{5}{4}, \frac{3}{4})$. `[verification/v51_boundary_half_step.py]` The sum $\frac{5}{4}{+}\frac{3}{4} = 2$ is literally the $E_8$ spinor-root norm $\frac{1}{2}(\pm)^8$ split across $D_5 \times D_3$: $5 \cdot \frac{1}{4} + 3 \cdot \frac{1}{4} = 2$. Strikingly, both norms are pre-existing TFPT constants:

$$\boxed{\frac{\delta_2}{\delta_{\text{top}}^2} = \frac{2880\, c_3^8}{(48\, c_3^4)^2} = \frac{5}{4} = q(D_5), \qquad \xi_{\text{tree}} = \frac{\dim \mathfrak{g}_{\text{SM}}}{\dim S^+} = \frac{12}{16} = \frac{3}{4} = q(A_3)} \qquad \textbf{[E]}, \tag{98}$$

so $\delta_2/\delta_{\text{top}}^2 + \xi_{\text{tree}} = 2$: the two-loop/defect ratio is the $D_5$ spinor norm and the gravitational tree normalization is the $A_3$ corner norm, and their sum is the $E_8$ root norm.

**The whole integer skeleton from one pair $(g_{\rm car}, N_{\rm fam}) = (5,3)$ [E]** ( [verification/v53_compiler_core.py] )

Every load-bearing integer of the compiler flows from the single boundary pair $(g_{\rm car}, N_{\rm fam}) = (5,3)$ by sum, difference and squares:

$$\operatorname{rank} E_8 = g_{\rm car}+N_{\rm fam} = 8, \qquad |\mathbb{Z}_2| = g_{\rm car}-N_{\rm fam} = 2, \qquad |\mu_4| = \tfrac{g_{\rm car}+N_{\rm fam}}{2} = 4,$$

and $(N_{\rm fam}, |\mu_4|, g_{\rm car}) = (3,4,5)$ is *the* Pythagorean triple. This is not decorative: the grand-mass-volume exponent $\Delta_Y = g_{\rm car}^2 = 25$ ($\det M_{\rm SM} \sim (\varphi_0^{\rm ret})^{25}$; document 3) splits over the sectors as a *difference of squares*,

$$\Delta_Y = g_{\rm car}^2 = \underbrace{N_{\rm fam}^2}_{\text{down}=9} + \underbrace{(g_{\rm car}-N_{\rm fam})(g_{\rm car}+N_{\rm fam})}_{=|\mathbb{Z}_2|\cdot\operatorname{rank} E_8} = 9+16 = \dim S^+$$

i.e. the down sector carries $N_{\rm fam}^2$ and up+lepton carries $|\mathbb{Z}_2|\cdot\operatorname{rank} E_8 = \dim S^+$ (the $K$-row sums $(6,9,10)$). **The anchor is the Dirac square root of this skeleton:** $a = (1,1,2)$ is the *unique* 3-multiset whose elementary symmetric functions are $(|\mu_4|, g_{\rm car}, |\mathbb{Z}_2|) = (4,5,2)$, equivalently its characteristic polynomial is $\chi_a(t) = (t-1)^2(t-2) = t^3 - |\mu_4|\,t^2 + g_{\rm car}\,t - |\mathbb{Z}_2|$ — the compiler atoms *are* its char-poly coefficients. Sharper still ( [verification/v491_p2_partition_corollary.py]): the partition needs only $e_1 = 4$ — three *positive integers* summing to 4 already force $\{1,1,2\}$, so $g_{\rm car} = e_2 = 5$ and $|\mathbb{Z}_2| = e_3 = 2$ are *corollaries* of the four marks given the weight typing (integrality = Birkhoff–Grothendieck splitting exponents, positivity = $h^0$=0, sum = par-deg-0 "given (U)"); every dropped assumption is negative-controlled, and `AX.P2.01` stays a declared axiom — the residual is the typing, not the integer 5. That typing is itself now hardened ( [verification/v499_p2_weights_deligne_bg.py], `P2.TYPING.01`): given the four marks, rank 3, the cusp class and (U), Deligne + Birkhoff–Grothendieck + Mehta–Seshadri stability force $E = \mathcal{O}(-2)\oplus\mathcal{O}(-1)^2$ outright ($h^0$=0 is the positivity), leaving only the module identification and (U) as [C] residuals — and the module-identification rest is now *one* residual with the QGEO modulus rest, both reducing to the same order-4 clock = Coxeter/cusp-class carrier datum (v503, `QGEO.EMERGE.LIGHT.01`; markers unchanged); the clock-rigidity theorems then reduce that common residual to one alignment bit, with the clock's order fermionically forced, $U^2 = (-1)^F$ exactly ( [verification/v506_seam_clock_rigidity.py], `SEAM.CLOCK.RIGIDITY.01`); and the bit-origin theorems derive the deck's *class* (a half-period translation of the seam double) — only its position stays the [C] carrier input ( [verification/v507_seam_bit_origin.py], `SEAM.BIT.ORIGIN.01`); the freedom theorems then type that position half as *topology* (the covering deck is free on the seam circle, the edge class is excluded), so the residual is the square-modulus datum $\tau = i$ alone ( [verification/v510_seam_bit_freedom.py], `SEAM.BIT.FREEDOM.01`); and the flag-transitivity equivalence web restates that residual in discrete form — flag transitivity of the four marks ($V_4 \to D_4$) $\iff \tau = i$, with bare mark-transitivity automatic for every configuration and no local jet seeing the side bit ( [verification/v512_seam_tau_flag.py], `SEAM.TAU.FLAG.01`) — a residual that has since survived ten side-blind derivation attacks (v521/v525/v529) and carries a physical definition as the twist-class choice with a gauge-robust order parameter, in principle interferometrically readable (v528; it remains formal input) — and, since v534, the first *dynamical* selection datum: reflection positivity on the interacting mark family keeps exactly the $\delta = \pi/2$ member alive (toy-fenced).

**Later source reconciliation (2026-11-05).** For the actual $A_0^\star$ connection, the recovered September $C_4$ descent proves finite analytic monodromy: the original four-puncture image is $A_4$, extended to $S_4$ by the square rotation. An invariant positive Hermitian form therefore exists, with an explicit unitary signed-permutation realization. Thus mathematical monodromy and unitarizability of this supplied local system are already closed. Its physical module, normalized frame and metric identification retain their source conditions. The finite W-bridge closures of v1014/v1015 and the common explanation are collected in Section 2; the earlier square-modulus discussion records its weaker geometric route. So the theory's only transcendental is $\pi$ (through $c_3 = \frac{1}{8\pi}$, $8 = \operatorname{rank} E_8$, and the seed $\frac{1}{6\pi}$, $6 = |R^+(A_3)|$); everything else is this $(5,3)$-generated integer skeleton.

**Boundary Carrier Selection Theorem (Theorem A) [E]/[C]**

$E_8$ is not assumed: it is *selected* as the unique even-unimodular audit hull of the admissible boundary-carrier data. **Boundary side:** $\mu_4 = \{1, i, -1, -i\}$ gives $X_f = \mathbb{P}^1 \setminus \mu_4$ with $H_1 = \mathbb{Z}^3$, hence $A_3$ and $N_{\rm fam} = 3$. **Carrier side:** a minimal $D$-type carrier with a 16-dim half-spinor forces $D_5$ (only $n = 5$ gives $2^{n-1} = 16$; $S^+ = \Lambda^{\rm even}\mathbb{C}^5 = 1{+}10{+}5$). **Glue:** $\operatorname{disc}(D_5) = \operatorname{disc}(A_3) = \mathbb{Z}_4$ (isomorphic, cyclic,

anti-isometric), $q(D_5)+q(A_3)=2$ (the $E_8$ root norm), glue index $4=|\mu_4|$, and $E_8$ is the unique even unimodular rank-8 lattice. **Exclusion:** the four conditions

$$\text{(i) } \dim S^+{=}16, \quad \text{(ii) } N_{\text{fam}}{=}3, \quad \text{(iii) cyclic } \mathbb{Z}_4 \text{ glue}, \quad \text{(iv) } q_D{+}q_A{=}2$$

are met *only* by $D_5\oplus A_3$; $D_8$ (non-cyclic $\mathbb{Z}_2{\times}\mathbb{Z}_2$, no family geometry), $E_7{\oplus}A_1$ (1 family, no 16-spinor), $E_6{\oplus}A_2$ (2 families, $\mathbb{Z}_3$), $A_4{\oplus}A_4$ (no $D$-spinor, $\mathbb{Z}_5$) and $A_8$ (single factor) each fail at least one. **Typing:** the lattice/classification facts are **[E]**; the physical selection (reflection-positive seam $c_3=1/8\pi$ + minimal chiral carrier $\Rightarrow$ exactly these inputs) is the **[C]**/**[O]** selection axiom (P1 analytic interface + P2 carrier interface), not a zero-input derivation. [`verification/v47_selection_theorem.py`]

**Group-level closure ($\mu_4$ as the common centre).** The lattice glue lifts to a clean group statement, because both factors share the *same* centre and $\mu_4$ *is* the diagonal:

$$\boxed{Z(\mathrm{Spin}(10))=\mathbb{Z}_4, \quad Z(SU(4))=\mathbb{Z}_4, \quad |\mu_4|=4, \quad G_{\text{closure}}=\frac{\mathrm{Spin}(10)\times SU(4)}{\Delta\mathbb{Z}_4}\subset E_8} \quad \textbf{[E]}, \tag{99}$$

with rank $=5+3=8=\operatorname{rank}E_8$. Every adjoint block is diagonal-$\mathbb{Z}_4$ neutral (generator $\omega=i$; charges mod 4): $(45,1),(1,15):0$; $(10,6):2{+}2{=}0$; $(16,4):1{+}3{=}0$; $(\overline{16},\overline{4}):3{+}1{=}0$, so $45{+}15{+}60{+}64{+}64=248$ are genuine reps of the quotient. This promotes $\mu_4$ from a lattice glue index to the *common centre quotient* — the physically cleanest language for the gluing: the $D_5$ half-spinor 16 pairs with the $SU(4)$ fundamental 4 so the spinor charges $\pm1$ cancel the fundamental charges $\mp1$.

**Root-length completion.** Every $E_8$ block reaches norm 2:

| block | norm split | reading |
|---|---|---|
| $(45,1),(1,15)$ | $2,\ 2$ | $D_5$, $A_3$ root adjoints |
| $(10,6)$ | $1+1=2$ | $SO(10)$ vector $\oplus$ $SO(6)$ vector |
| $(16,4)\oplus(\overline{16},\overline{4})$ | $\frac{5}{4}+\frac{3}{4}=2$ | $D_5$ spinor norm + $A_3$ corner norm |

The old question "can the 240 flip atlas be metrized to $E_8$?" is now the sharp question "are the TFPT $D_5$ and $A_3$ metrics in standard normalization?" — and the $5/4, 3/4$ match says yes.

**Explicit construction (the lift is now built, not counted).** Embedding $D_5\oplus A_3=D_5\oplus D_3\subset D_8$ and adjoining the even spinor coset (the $\mu_4$ glue), the 240 roots are produced explicitly and verified:

$$\boxed{\{\pm e_i\pm e_j\}_{i<j}^{8}\ (112)\ \cup\ \{\tfrac{1}{2}(\pm)^8:\ \text{even}\}\ (128)\ =\ 240 \text{ roots, all norm 2, Cartan det} = 1} \quad \textbf{[E]}. \tag{100}$$

Splitting each root into its $D_5$ (first five) and $A_3$ (last three) parts gives exactly the branching blocks — $(2,0){:}40$, $(0,2){:}12$, $(1,1){:}60$, and $(\frac{5}{4},\frac{3}{4}){:}128$ — and the 128 spinor roots realise $\frac{5}{4}+\frac{3}{4}=2$ on the nose. All 240 vectors are integer combinations of one set of 8 simple roots, so the assembled lattice *is* $E_8$. The blind 240-metrization is therefore eliminated: $E_8$ is explicitly $(D_5\oplus A_3)+\mu_4$-glue.

**Remark** (the metric, computed)**.** This reframes $E_8$ from origin to closure: $E_8$ is the closure algebra of the $D_5$ carrier and the $A_3$ family geometry, glued by $\mu_4$. The root system, Cartan form, $W(A_3)=S_4$ and the corner-rotation $=A_3$ Coxeter element are exact (residue pairing + explicit carrier symmetry). The transcendental geometric metric $G_{ij}=-\log|p_i-p_j|$ on $\mu_4$ has been computed: it is only $D_4$-symmetric, with a regularisation-independent anisotropy $-\log 2$ on the root space, hence *not* proportional to Cartan$(A_3)$. It need not be: the $E_8$-relevant $A_3$ is the canonical *residue* pairing, and the geometric metric is a $D_4$-equivariant deformation that preserves the root system, Weyl group and Coxeter element. So the lattice data is exact **[E]**; the analytic metric is a *characterised* deformation, not an open gap.

**Uniform Coxeter law.** All three factors obey $|R| = h \cdot \text{rank}$, and the two TFPT Coxeter numbers are themselves carrier data:

$$\boxed{\begin{aligned} &h(D_5) = 8 = r(E_8) = \varphi(30), \quad h(A_3) = 4 = |\mu_4|, \\ &h(D_5)\,h(A_3) = 2^{g_{\text{car}}} = 32, \quad 2^{g_{\text{car}}} - 2 = 30 = h(E_8) \end{aligned}} \quad \text{[E]}, \tag{101}$$

so the Coxeter closure of Theorem 3 reappears as $h(E_8) = h(D_5)h(A_3) - 2$, and $h(D_5) + h(A_3) = 12 = \dim \mathfrak{g}_{\text{SM}} = |R(A_3)|$.

## Theorem 8 — the $(1,1,2)$ anchor microcode and the $\varphi_0$-ladder

The parabolic flavor bundle $E_\bullet = \mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ on $\mathbb{P}^1 \setminus \mu_4$ has splitting type $a = (1,1,2)$, whose power sums $p_n(a) = 2 + 2^n$ generate the whole compiler budget:

**The anchor power compiler.**

$$(p_1, \ldots, p_5) = (4, 6, 10, 18, 34) = (|\mu_4|,\ |R^+(A_3)|,\ \mathcal{A}_\Lambda,\ N_{\text{fam}}|R^+(A_3)|,\ \mathbb{Z}_6 \text{ lift}),$$

$$q(A_3) = \tfrac{p_2}{2p_1} = \tfrac{3}{4}, \quad q(D_5) = \tfrac{p_3}{2p_1} = \tfrac{5}{4}, \quad \textstyle\sum L = p_1 p_3 = 40,$$

$$\Omega_{\text{adm}} = 2p_1p_2 = 48, \qquad D_{\text{start}} = p_2 p_3 = 60, \qquad |R(E_8)| = 4p_2p_3 = 240.$$

The elementary symmetric data $\chi_a(t) = (t-1)^2(t-2)$ give $(e_1, e_2, e_3) = (4, 5, 2) = (|\mu_4|, g_{\text{car}}, |\mathbb{Z}_2|)$, so the anchor reconstructs the carrier rank, and $10b_1 = e_1^2 + e_2^2 = 4^2 + 5^2 = 41$. The seed itself is the two-term anchor form $\varphi_0^{\text{ret}} = \frac{2p_1}{p_2} c_3 + 2p_1p_2\, c_3^4 = \frac{4}{3} c_3 + 48 c_3^4$.

**The anchor affine normal form (2026-08-07) — all budgets from ONE recursion [E] ( [verification/v832_anchor_flavor_checksum.py] )**

The anchor power compiler above is one affine map. The power sums $p_n(a) = 2 + 2^n$ satisfy

$$\boxed{p_{n+1} = T(p_n), \qquad T(x) = 2x - 2} \qquad \text{identically in } n \text{ (sympy)},$$

with the shift $q_n = p_n - 2 = 2^n$ pure binary doubling. $T$ has the *unique* fixed point $2 = |\mathbb{Z}_2|$ (the sheet), and the $T$-orbit of 4 is exactly the compiler quintet $(4, 6, 10, 18, 34) = (|\mu_4|, |R^+(A_3)|, \mathcal{A}_\Lambda, N_{\text{fam}}|R^+(A_3)|, \mathbb{Z}_6 \text{ lift})$, with $p_0 = 3 = N_{\text{fam}}$ prepended by $T(3) = 4$. Every budget line is then a word in the orbit and the fixed point: $|R(E_8)| = p_1p_2p_3 = 240$; $\dim E_8 = 240 + q_3 = 248$, where the ladder identity $p_4 - p_3 = p_3 - 2 = 8 = h(D_5) = \text{rank}\, E_8$ makes the corpus form $240 + (p_4 - p_3)$ the *same* formula; $h(E_8) = p_2p_3/2 = 30$ (so $D_{\text{start}} = p_2p_3 = 60$); $|R(D_5)| = p_1p_3 = 40$; $\Omega_{\text{adm}} = 2p_1p_2 = 48$; and the elementary layer $(e_1, e_2, e_3) = (4, 5, 2)$ has the fixed point of $T$ as $e_3$, with $10b_1 = e_1^2 + e_2^2 = 41$. Controls: the wrong map $T'(x) = 2x - 1$ (fixed point 1, orbit misses the quintet) and the wrong anchor $a' = (1, 2, 2)$ (recursion offset $-1$, budget misses $(240, 48)$) both fire. A normal form of the existing compiler tables — a compression, not a new claim; 18/18 exact, verdict ANCHOR-AFFINE-EXACT.

**The $\varphi_0$-ladder.** With $\lambda_Y = \sqrt{\varphi_0^{\text{ret}}(1 - \varphi_0^{\text{ret}})}$, every fermion mass is the single closed form $\hat{m}_{f,j} = \frac{v_{\text{geo}}}{\sqrt{2}} \lambda_Y^{L_{f,j}} \Lambda_{f,j}$ with $\lambda_Y^L \Lambda = \pi\, c\, (\varphi_0^{\text{ret}})^k$, so all nine holonomies are resolvent outputs and the masses form the $\varphi_0^{\text{ret}}$-order ladder $t(0) > \{b, c, \tau\}(2) > \{s, \mu\}(3) > \{u, d\}(4) > e(5)$.

**Secondary $E_8$ atlas.** Beyond the main $D_5 \times A_3$ split, the residue matrix appears in the $E_6 \times A_2$ branching, $248 = \|R\|_F^2 + \det R + 2(\mathbf{1}^\top R a) N_{\text{fam}} = 78 + 8 + 2 \cdot 27 \cdot 3$, and the scalaron exponent in $E_7 \times A_1$, $112 = 7 \cdot 16$. The seven $E_8$ slices are charted in the companion note `e8_secondary_branching_atlas` as a falsifiable audit raster.

**The $E_6 \times A_2$ block is the $\tau = \omega$ face ( [verification/v407_dn_pairings_omega.py], SEAM.EQUIV.02).** $A_2$ is the Eisenstein lattice of the family CM point $\tau = \omega$ (Cartan $\det = 3 = N_{\text{fam}}$), and the $(d, n)$ selector's three pairings $\{n \cdot \mathbf{1},\ n \cdot a,\ n \cdot \sigma\} = \{2, 8, 121\}$ are exactly its family-slice atoms $\{|Z_2|,\ \dim A_2 = \det R = \text{rank}\, E_8,\ \|Pl(K)\|_1^2\}$, so deriving $(d, n)$ (the R4′ residue, v139) folds into the dual keystone (v405), leaving only "why $11 = \|Pl(K)\|_1$".

## Master statement and status

> Five carrier slots give $D_5$, four family corners give $A_3$, and $D_5 \times A_3$ gives the $E_8$ atlas.

Equivalently: $\mathcal{A}_\Lambda = \frac{h}{N_{\text{fam}}} = 10 = \binom{5}{2}$, $h = 30 = 2 \cdot 3 \cdot 5$, $(5,3,2):\ \sum = 10,\ \prod = 30$; $\Lambda$, $\det M_\ell$, the $E_8$ Coxeter structure and the action ladder read the same five-slot pair-decuple. The Standard model, flavor, $\Lambda$, inflation and $E_8$ are projections of the even carrier code $C^+$ ($= D_5$ half-spinor) on five slots together with the four-point family $A_3$.

| Statement | Status | Note |
|---|---|---|
| Even code $C^+$, $\dim S^+ = 16$, dual-root $Y$ | [E] | Theorem 1 |
| Exterior–Action duality $(1,10,5) \leftrightarrow (1,5,10)$ | [E] | scheme eigenvalues verified |
| $R_4$ = Clebsch $(16,5,0,2)$, $R_2 = (16,10,6,6)$; $\text{Aut}_{\text{aff}} = W(D_5) = 1920$ | [E] | carrier = $D_5$ half-spinor |
| $2^g - 2 = g(g+1) \Leftrightarrow g = 5$ | [E] | Coxeter closure (unique) |
| $h$=30=2·3·5, $r=\varphi(30)$=8, $\chi_C=\Phi_{30}$ | [E]/[C] | cyclotomic compiler |
| $\lvert R^+\rvert = \binom{16}{2} = \text{Tr}\, X^2 = 4h = 120$ | [E] | positive-edge moment |
| $40, 60, 120, 240 = 10 \cdot (4, 6, 12, 24)$; 41=1+4·10 | [E] | $K_5 \times K_4$ product table |
| $10b_1$=5$r$+1, $u$ totient, $M_{\text{scal}}^2 = c_3^{r-1}$, $\omega_b = \lambda_C/10$ | [E]/[C] | downstream readouts |
| $A_s = N_\star^2 c_3^7/24\pi^2$ | [C]/[O] | if $M_{\text{scal}} = M_{\text{Star}}$ |
| $\mathfrak{e}_8 = (45,1) \oplus (1,15) \oplus (10,6) \oplus (16,4) \oplus (\overline{16},\overline{4})$ | [E]/[C] | $D_5 \times A_3$ lift |
| 240 roots built from $D_5 \oplus A_3 + \mu_4$ glue, det 1 | [E] | explicit $E_8$ construction |
| flavor $\sum L$=40=$\lvert R(D_5)\rvert$, entries on $C_6 = \lvert R^+(A_3)\rvert$ | [E] | transport = $D_5 \times A_3$ |
| $A_3$ from $\mathbb{P}^1 \setminus \mu_4$: residue pairing = Cartan, det 4 | [E] | Stage 2 (Theorem 7) |
| corner $z \mapsto iz = A_3$ Coxeter el.: $\{i, -1, -i\}$, $\Phi_4\Phi_2$ | [E] | order 4 = $h(A_3) = \lvert\mu_4\rvert$ |
| $\text{disc}(D_5) = \text{disc}(A_3) = \mathbb{Z}_4$, glue = $\mu_4$ | [E] | $E_8 = (D_5 \oplus A_3) + \mu_4$ |
| $q(D_5) = \frac{5}{4} = \delta_2/\delta_{\text{top}}^2$, $q(A_3) = \frac{3}{4} = \xi_{\text{tree}}$, sum 2 | [E] | discriminant-norm glue |
| 120= $\dim \mathfrak{so}_{16}$; $\Omega_{\text{adm}}$=40+1+7 | [E] | adjoint half; deficit ladder |
| $h(D_5)$=8=$r(E_8)$, $h(A_3)$=4=$\lvert\mu_4\rvert$, $\lvert R\rvert = h \cdot r$ | [E] | uniform Coxeter law |
| Exp($E_8$) as 41/48 budget deficits | [O] | exponent ledger (fingerprint) |
| geometric metric $D_4$-deform (anisotropy $-\log 2$), residue = $A_3$ | [E] | metric computed/characterised |

*Scope.* This note compresses; it does not re-derive the upstream boundary kernel (the P1 hardening chain) or the carrier rigidity theorem (Lean), and it makes no claim of a final theory. The arithmetic and the $D_5$/Clebsch/$W(D_5)$, $A_3$/residue-Cartan/Coxeter, and $\mu_4$-glue identifications are exact, and the $E_8$ lattice is now built explicitly from $D_5 \oplus A_3$ plus the $\mu_4$ glue (240 roots, det 1). The only residual [C] is the analytic identification of the carrier geometry's $L^2$ cusp Hodge metric with the canonical residue metric up to scale, which does not affect the explicit root/lattice/Weyl construction. Honest non-fits (Koide $0.6645 \neq 2/3$, $\eta_B$, $m_p/m_e$) stay outside the compiler.

## Appendix: machine-checkable $E_8 = (D_5 \oplus A_3) + \mu_4$ construction

The lift is reproducible by a short script (all entries below verified exactly).

**Cartan matrices and discriminant groups.**

$$C(D_5) = \begin{pmatrix} 2 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & -1 \\ 0 & 0 & -1 & 2 & 0 \\ 0 & 0 & -1 & 0 & 2 \end{pmatrix}, \quad C(A_3) = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}; \quad \det C(D_5) = \det C(A_3) = 4.$$

Smith normal forms give $\mathrm{disc}(D_5) = \mathrm{disc}(A_3) = \mathbb{Z}_4$; the glue index is $[\,E_8 : D_5 \oplus A_3\,] = \sqrt{4 \cdot 4/1} = 4 = |\mu_4|$, with discriminant-form norms $q(D_5) = \frac{5}{4}$, $q(A_3) = \frac{3}{4}$, $q(D_5)+q(A_3) = 2 \equiv 0 \pmod 2$ (even-glue). Writing $\mu := |\mu_4| = 4$ the norms are $q(D_5) = \frac{\mu+1}{\mu}$, $q(A_3) = \frac{\mu-1}{\mu}$, so $q(D_5)q(A_3) = 1 - \frac{1}{\mu^2} = \frac{15}{16}$ and

$$\boxed{\dim S^+(1 - q(D_5)q(A_3)) = 16 \cdot \tfrac{1}{16} = 1} \qquad \textbf{[E]}, \tag{102}$$

i.e. the glue product fills all $15 = \dim S^+ - 1$ non-trivial half-spin states and the defect is exactly the singleton (the $\nu^c$/Higgs closure). And $E_8$ has a tidy $\mu_4$-cardinal form, $\dim E_8 = \binom{|\mu_4|^2}{2} + 2|\mu_4|^3 = \binom{16}{2} + 2 \cdot 64 = 120 + 128 = 248$ **[C]**.

**Explicit roots.** Split $\mathbb{R}^8 = \mathbb{R}^5_{(D_5)} \oplus \mathbb{R}^3_{(A_3)}$. The 240 roots are

$$\{\pm e_i \pm e_j : 1 \le i < j \le 8\}\ (112)\ \cup\ \{\tfrac{1}{2}(\pm 1)^8 :\ \text{even \# of } -\}\ (128),$$

all of norm 2. Splitting each by $(\|\cdot\|^2_{D_5}, \|\cdot\|^2_{A_3})$ gives the branching blocks

$$(2,0)\colon 40\ [D_5], \quad (0,2)\colon 12\ [A_3], \quad (1,1)\colon 60\ [(10,6)], \quad (\tfrac{5}{4},\tfrac{3}{4})\colon 128\ [(16,4) \oplus (\overline{16},\overline{4})],$$

so $\mathfrak{e}_8 = (45,1) \oplus (1,15) \oplus (10,6) \oplus (16,4) \oplus (\overline{16},\overline{4})$, $248 = 45{+}15{+}60{+}64{+}64$, $240 = 40{+}12{+}60{+}64{+}64$.

**Simple-root Gram (unimodular check).** With the Bourbaki simple roots $\alpha_1 = \frac{1}{2}(e_1 - e_2 - \ldots - e_7 + e_8)$, $\alpha_2 = e_1 + e_2$, and $\alpha_k = e_{k-1} - e_{k-2}$ for $k = 3, \ldots, 8$ (so $\alpha_3 = e_2 - e_1$, $\alpha_4 = e_3 - e_2$, $\ldots$, $\alpha_8 = e_7 - e_6$), the Gram matrix $G_{ij} = \langle \alpha_i, \alpha_j \rangle$ is the $E_8$ Cartan matrix with $\det G = 1$ and all $G_{ii} = 2$; every one of the 240 roots is an integer combination of the $\alpha_i$. Hence the assembled lattice is $E_8$, not merely a 240-count. **[E]** (The shifted index $e_{k-2} - e_{k-3}$ would duplicate $\alpha_3$ and make $G$ singular — use $e_{k-1} - e_{k-2}$.)

# Part III
# Appendix: ablation of the EM fixed point

The single sharpest reviewer test is: *was the closure* $F_{U(1)}(\alpha) = 0$ *engineered so that* $\alpha$ *comes out?* The honest answer is an ablation — vary each input and see which actually move the root. Three one-parameter scans (all others held at their compiler values) separate the *load-bearing* integers from the *fine* last-digit selector.

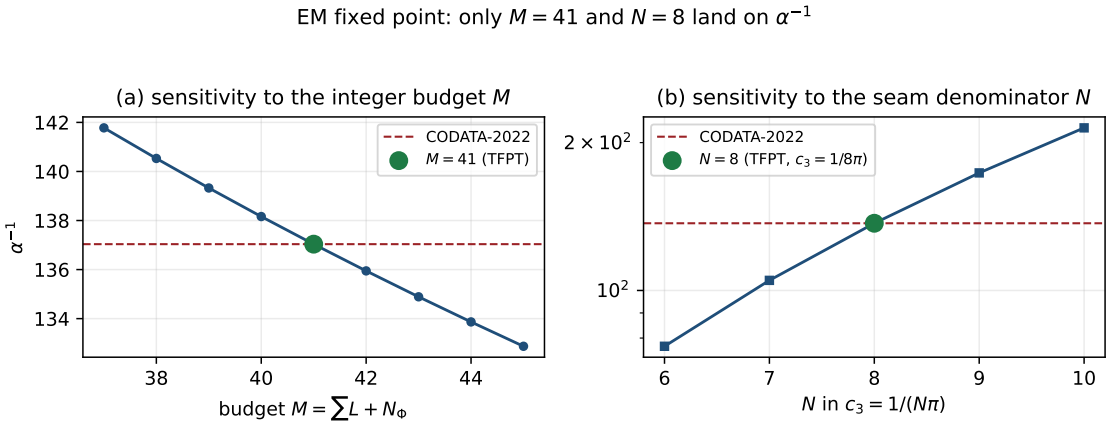



Figure 7: EM-closure ablation (data plot, `verification/make_figures.py`). **(a)** $\alpha_\star^{-1}$ vs the integer budget $M$: only $M = 41$ lands on CODATA-2022. **(b)** $\alpha_\star^{-1}$ vs the seam denominator $N$ in $c_3 = 1/(N\pi)$: only $N = 8$ lands. Both integers are fixed elsewhere (flavor budget; rank/$h(D_5)$), so the equation is not freely tunable to 137.036.

**(A1) The transport budget** $M = \sum L + N_\Phi$ (exponent held at $-\frac{5}{4}$):

| $M$ | 38 | 39 | 40 | **41** | 42 | 43 | 44 |
|---|---|---|---|---|---|---|---|
| $\alpha_\star^{-1}$ | 140.530 | 139.326 | 138.162 | **137.0359992** | 135.946 | 134.890 | 133.866 |

Each unit of $M$ shifts $\alpha_\star^{-1}$ by $\approx 1.09$, so *only* $M = 41$ lands on 137.036. This is a hard lock — and $41 = 10b_1 = \sum L + N_\Phi = 40 + 1$ is fixed by the compiler (flavor budget 40 + one Higgs). **Load-bearing.**

**(A2) The seam "8" in** $c_3 = 1/(N\pi)$ (here $\delta_{\text{top}} = 48c_3^4$ tracks $c_3$):

| $N$ | 6 | 7 | **8** | 9 | 10 |
|---|---|---|---|---|---|
| $\alpha_\star^{-1}$ | 76.97 | 104.85 | **137.036** | 173.53 | 214.34 |

Only $N = 8$ — i.e. $c_3 = 1/(8\pi) = 1/(\text{rank}\, E_8 \cdot \pi)$ — reproduces 137.036. The "8" is the same one fixed by the bootstrap ($h(D_5) = \text{rank}\, E_8 = \varphi(30)$) and the Gauss–Bonnet hardening. **Load-bearing.**

**(A3) The carrier discriminant exponent** (at $M = 41$, $c_3 = 1/(8\pi)$):

| exponent | $-\frac{1}{4}$ | $-\frac{3}{4}$ | $-\frac{5}{4}$ | $-\frac{7}{4}$ | $-\frac{9}{4}$ |
|---|---|---|---|---|---|
| $\alpha_\star^{-1}$ | 137.035995 | 137.035997 | **137.0359992** | 137.036001 | 137.036003 |

The exponent moves only the *seventh* digit; all candidates agree to six. Here $-\frac{5}{4} = q(D_5)$ is selected, but this is a *fine* selector, not a hard lock. **Fine (last-digit).**

**Ablation verdict — honest**

The EM fixed point is a **hard test of two integers** — the budget $M = 41$ and the seam denominator $8 = \operatorname{rank} E_8$, each decisive (a single-unit change misses by $\sim 1$ in $\alpha_\star^{-1}$) — and a **fine test of one norm**, the carrier discriminant exponent $-\frac{5}{4} = q(D_5)$ (seventh digit). The equation is therefore *not* freely tunable to 137.036: the two integers that hit it are exactly the compiler/bootstrap numbers, fixed elsewhere. What it does *not* do is pin the carrier norm strongly — that is honest, and it is the right level of claim. **[E]**

# Shared source and transfer update: 3 November 2026

**What this update establishes**

The original structured boundary source, its common action and the $S_{\mathrm{adm}}/Z_{\mathrm{rel}}/\mathrm{OS}/\Gamma_{\mathrm{ren}}$ chain are retained. Several complete source calls, field maps and transfer kernels have now been evaluated in their stated classes. These results include conditional charge and metric selection, physical Gauss-sector sources, vacuum and ground-source reconstruction, and a complete fixed-background Clifford kernel. The primitive spatial mirror-field insertion and its coefficient $H_{\mathrm{adm}}[F, 0]$ have not been evaluated by these constructions.

The reports A–L, R104–R113 and the named follow-up reports below are dated research results, not additional ledger grades. Their mathematical, numerical, software and conditional physical scopes remain separate. No claim status is promoted here. In particular, exact reproduction of the DET16 candidate is not a universal prerequisite for every physical TFPT completion.

## The common source and the corrected starting point (C, F, G)

P1 is a structured reflection-positive boundary kernel, not merely the number $c_3$. The operational seed already supplies its local algebra, state, time, reflection and collar operator. The original source route is

$$\text{boundary/carrier/admissibility} \longrightarrow S_{\mathrm{adm}}, Z_{\mathrm{rel}} \longrightarrow \text{full moments},$$
$$\text{full moments} \longrightarrow \text{OS/CAR representation} \longrightarrow \Gamma_{\mathrm{ren}} \longrightarrow \text{physical readouts}.$$

Each arrow retains its actual hypotheses and operator domains. A particular eightfold CAR/native-clock reconstruction is an available conditional route, not a global extra axiom. Algebraic internal clocks do not require a separately postulated laboratory gate preparation.

The action types also remain distinct. The primitive relative spectral seed, the bare input $S_{\mathrm{adm}}$ in the functional integral, $-\log Z$, the effective action and the master functional are not interchangeable. A proper-time profile measure and a geometric field-integration measure have different roles.

A conditional CAR reader can forget an overall factor. The complete joint source contains the weights of its integrated bosonic fields, so a field-dependent factor changes their marginal and mixed responses. Its loss in a reader is not a new freely adjustable global normalization. On the same actual integration basis the existing reweighting identity recovers the relative source from its bound weight quotient. This does not license replacing that quotient by a fitted one.

The full existing chart $U = (E, NE)$ carries $D$ and its bound $D_{\mathrm{ref}}$, their actual domains, reflection, grading, antilinear charge data, Berezin measure, bosonic weights and eta factor together. Applied to a given complete source, it gives the same unnormalized source call in new coordinates. A separate Gibbs reference-state representation is optional for this direct transport. The geometric spectral-action consumer is included in this whole-operator route; a finite coefficient reader is not equated with the full Dirac operator.

For physical BRST-closed insertions the original quotient has its stated gauge-fixing independence. This does not make arbitrary off-shell functionals gauge independent or make individual ghost determinants cancel at every background.

## Charge geometry and the actual clock completion (A, B)

For the four-punctured family surface,

$$H_1 = \mathbb{Z}^4/\mathbb{Z}(1,1,1,1), \qquad H^1 = \ker\left(\mathbb{Z}^4 \xrightarrow{\sum} \mathbb{Z}\right).$$

With the declared root normalization, the meridians form the $A_3^*$ side, their differences the $A_3$ root side. All four meridians have the same discriminant class. A cyclic permutation of the marked

points therefore acts trivially on that discriminant; it is not translation through the four glue classes. The earlier literal fourfold delta residue pairing does not by itself define the asserted positive metric.

On the fixed additive spinor–meridian extension, the whole positive real $W(D_5) \times C_4$-invariant metric has three parameters $x, a, b$. Integrality of the root pairings makes them positive integers. Unimodularity gives

$$x^5 a^2 b = 1, \qquad \text{hence} \qquad x = a = b = 1.$$

Scale, orthogonality and evenness follow within this joint completion class. The fixed charge extension, symmetries and integral self-dual completion are joint premises of this theorem. The later R102 result strengthens the standard $D_{n+1} \oplus A_{n-1}$ route: a primitive integral family insertion selects its known $n = 4$ completion under the specified pairing and locality assumptions. Neither theorem selects a physical insertion by matching a dimension.

The actual current constructor also supplies the mixed consequence: for the sixteen spinor signs and twelve ordered distinct meridian pairs, all 192 simple poles land on the corresponding family root. The 64 diagonal cases give their normalized double poles, with the original cocycle and adjoint convention.

For the actual later chart $M$ and native clock $\sigma$, the integral closure of $D_8$ and $RD_8$, $R = M^T \sigma M$, is the index-two extension $D_8 + \mathbb{Z}(q/2)$, with even integral Gram matrix and determinant one. A marked source word already produces the requisite spinor class. This is a concrete completion in the given source chart; the general clock-closure principle predates this calculation.

The marked bosonic DHR sector supplies the canonical local extension $B$ with conditional expectation $\mathcal{E} : B \to A$ and $\omega_B = \omega_A \circ \mathcal{E}$. Neutral final words preserve their old values while charged intermediate paths are retained. The given sector covariance, square and cocycle conditions determine the time lift. A passive background transport alone does not establish common physical admission of both algebras. This special route does not create a universal new CAR, clock or endpoint-normalization requirement.

## Variable fields, ordered sources and retained reference factors (D, E)

The original local source reader no longer needs constant Higgs backgrounds in its stated circle class. The smooth bounded variable one-form has the required Hilbert–Schmidt control, entire ordered $\det_2$ source, trace correction and all local response orders. The actual Galerkin calls converge to that same source operator. This is an analytic application to the existing coefficient family; it is not a probability law for the fluctuating Higgs field.

The actual Dyson source has separately been constructed. It is not replaced by the exponential of a smeared linear reader. The coherent two-face graph retains the crossing blocks and produces the positive history kernel and mixed CAR words. For the stated normal-variable graph, its polar operator fixes the transported partner word; the same untransported operator on both faces need not give the same answer. The existing unpaired echo amplitude is also evaluated with its actual ground-zero excitation convention and particle/hole signs.

The complete geometric, reference, eta and zero-source factors remain attached to their original callers. In particular the actual `v259` profile call uses its heat profile, not a Fock logarithm. No automatic general cancellation of that logarithm or of all source phases is inferred.

## Physical source existence, ground words and common geometry (H, I, J)

The given finite-range uniformly local Hamiltonian class has its thermodynamic real-time norm dynamics and ground/KMS states. Restrictions of one limiting state are compatible. Independently restarted small bare Gibbs weights need not be projective. Exact elimination carries the effective fiber weights, exterior determinants, higher vertices and memory, rather than imposing the original bare ansatz again at each cut.

Gauge invariance of an ambient Gibbs density is weaker than Gauss-sector support. With the

original complete interior stars preserved by the Hamiltonian, the source

$$\rho_{\Lambda,\mathrm{phys}} = \frac{P_\Lambda e^{-\beta H_\Lambda}}{\mathrm{Tr}(P_\Lambda e^{-\beta H_\Lambda})}$$

is positive and nonempty in the stated regular-link, free-boundary class. Its weak-* limit gives physical ground/KMS sources and the actual quotient representation. Broken seam or boundary stars are not added. The projected state can differ from the earlier ambient state.

The same source supplies all bounded ordered real and Euclidean time words. The thermal reflected word form is a Hilbert–Schmidt Gram form; product-norm bounds and the local entire-source majorant pass along the same limiting subnet. For the physical ground source, all bounded Euclidean words converge to the existing positive GNS semigroup words. The whole finite ground-projector trace has a diagonal Gibbs approximation without a uniform thermodynamic gap. It does not select an arbitrary vector in a degenerate ground space.

For an applicable complete vacuum OS source,

$$q[J] = \frac{Z_{\mathrm{rel}}[J]}{Z_{\mathrm{rel}}[0]} = \frac{Z_{\mathrm{adm}}[J]}{Z_{\mathrm{adm}}[0]}, \qquad \Omega = [1], \quad \|\Omega\| = 1, \quad H\Omega = 0.$$

Only the outer source-independent reference denominator cancels here; inner reference weights and absolute energy/eta factors remain. The actual full Euclidean covariance and appropriate reconstruction regularity are needed for the invariant vacuum and forward-cone conclusion. A bounded-net route retains its time-zero hypothesis. The existing neutral local $Q = 0$ vacuum/BW construction is preserved; its parabolic generator is not the unchanged gapped circle energy.

On the original geometric Lorentz branch, whole-source covariance and an invariant vacuum carry the common metric into the physical pole geometry through the Ward identity. The complete mixed kinetic tensor must be tested, including the time–space terms. If the same source supplies its stated uniform Hermitian error bounds $\epsilon, \eta$,

$$\sqrt{\eta^2 + 1 - \epsilon} - \eta \;\le\; \frac{v}{c} \;\le\; \sqrt{\eta^2 + 1 + \epsilon} + \eta, \qquad 0 \le \epsilon < 1.$$

This is a whole-class bound, not a fit of $\epsilon, \eta$ to the native static data. The coupled native static response, separate Wilson-slab consumer and typed velocity comparison are distinct calls. A Lieb–Robinson graph bound is not a Lorentz light cone. Thermal Hamiltonian time, vacuum time and a BW modular boost retain their actual identifications.

For the registered lattice route, the thermodynamic limit and controlled IR universality remain required. An $a \to 0$ limit is an optional strengthening there, not a universal additional gate.

## The complete transfer comparison and its field requirements (K, R104–R108)

The complete original word source gives its actual pairing and time kernel,

$$G(F, G) = q((\Theta F)G), \qquad K_t(F, G) = q((\Theta F)\theta_t G).$$

Its first derivative defines the generator form on the actual quotient. Equal full pairings and closed forms, with actual autonomous sewing, determine the whole transfer. A discrete source can instead be compared at its whole fixed step. In general $Pe^{-tH}P$ is not $e^{-tPHP}$; a nonreducing compression can lose the required sewing.

For the actual ordinary finite thermal trace, full joint-block reset insertions give

$$C_{ij}(t) = \frac{(e^{-(\beta-t)H})_{aa}(e^{-tH})_{ij}}{Z_\beta}, \qquad p_a = \omega_\beta(E_{aa}) > 0,$$

$$H_{ij} = -\frac{C'_{ij}(0)}{p_a} \quad (i \neq j), \qquad H_{ii} - H_{jj} = -\frac{C'_{ii}(0) - C'_{jj}(0)}{p_a}.$$

The actual unnormalized positive trace fixes the common scalar. Resetting only the mirror factor requires all actual rest tests to recover a rest-valued block. A removed physical sector is not inverted, and a charged inadmissible insertion is not declared observable. These are extraction formulas; they do not constitute an evaluated primitive field insertion. The off-diagonal qualification is essential.

The supplied DET16 factor is itself completely evaluable:

$$H_{\mathrm{DET}} = 2(I - P_\varphi), \quad |\varphi\rangle = \frac{|0\rangle + e^{i\varphi}|F\rangle}{\sqrt{2}}, \qquad e^{-\beta H_{\mathrm{DET}}} = aI + bP_\varphi,$$

$$a = e^{-2\beta}, \qquad b = 1 - a, \qquad Z_{\mathrm{DET}} = 65536a + b.$$

Its equal-time covariance is $\delta_{ij}/2$, but its higher density and sixteen-field responses are not quasifree for $\beta > 0$. R104 gives a matrix-free cubic renewal evaluation of every finite ordered CAR word. The conditional separated-source reduction retains the actual gauge weights and rest source; it does not solve the entire rest lattice.

R105 isolates the exact charge-sixteen generator block $H_{F0} = -e^{i\varphi} I_R$. With mirror number bandwidth $s$, $\langle F|H^j|0\rangle = 0$ for $js < 16$ on the stated regulator/core. At bandwidth two the first possible order is eight, whereas the literal DET transfer begins linearly. The first possible order is not a guarantee of a nonzero coefficient. No bandwidth-two premise is asserted for the whole original $S_{\mathrm{adm}}$.

The original even DET rotation carries the complete given source, gauge operators and source legs. R106 evaluates all sixteen field time laws and distinguishes the flat parent from the additive $\Delta N_a$ parent: a shared distinguished ground vector does not make their two-excitation energies equal. R107 supplies the rest-valued coefficient reader from an actual CAR field derivation, without demanding an autonomous cell or a new empty OS vacuum.

The original 96-dimensional Yukawa/Majorana field block retains all its given couplings and explicit family/antiparticle transitions. It is not a list of 96 independent couplings or a derived physical many-body mirror Hamiltonian.

K gives the exact internal $v474/v1002$ Clifford basis intertwiner, with determinant $+1$, so its exterior lift carries the top determinant without a new phase. This internal map does not supply spatial Weyl spin or a smearing/time map. The original vector source uses sixteen real Majoranas, hence eight complex modes in one packet. Native DET16 uses sixteen complex spinor annihilators. R108's specified double packet is tested by its actual central and physical-color actions, not by equal Fock dimension.

The precise common point-local Weyl assignment with scalar rest determinant coefficient is excluded: its fermionic internal epsilon contraction transforms as $\mathrm{Sym}^{16}\mathbb{C}^2$ (spin eight), not a rotation scalar. The state witness additionally requires the stated invariant fixed background and common $s$-wave smear. Orbital or rest spin, other field maps, larger charged sectors and effective IR descriptions are not excluded by this local theorem. The original open QWZ wall smearing remains a valid coupled field in its four-flavor scope.

## The complete given Clifford field and kernel (L)

Retain the normal-independent product collar and its actual smooth classical odd coefficient:

$$B = Z \otimes A - S \otimes \Phi, \quad [A, \gamma] = 0, \quad \{\gamma, \Phi\} = 0, \qquad h = A\gamma - i\gamma\Phi.$$

For $Ev = (P_+ v, iP_- v)$, where $P_\pm = (1 \pm \gamma)/2$, $BE = Eh$ on the transported $H^1$ domain and $h^2 = A^2 + \Phi^2 - i[A, \Phi]$ on $H^2$. No commutation of $A$ and $\Phi$ is imposed.

The oriented two-face covariance is

$$C = (1 + e^{2\ell h})^{-1}, \quad X = (2\cosh \ell h)^{-1}, \qquad P_h = \begin{pmatrix} 1 - C & X \\ X & C \end{pmatrix}.$$

The given Clifford quantization uses this full graph as its quantum covariance. This is a named source premise, not a derivation of every physical $S_{\mathrm{adm}}$ from a classical Cauchy projection.

On its existing thermal Hilbert–Schmidt representation, with genuine whole-source parity $\Gamma$, put $a_R = L_a$, $a_L = -\Gamma R_a$. The complete native map is

$$\iota c(T_{\text{faces}}(v_0, v_1, \bar{w}_0, \bar{w}_1)) = a_R^*(v_0) - a_L^*(v_1) + a_R(w_0) - a_L(w_1).$$

Here $a$ is the old signed charged field: $a(f) = b(Q_+ f) + d^*(\overline{Q_- f})$ with $Q_\pm$ the spectral signs of $h$, distinct from the grading $P_\pm$. Creators are linear and annihilators antilinear in their tests. The opposite real-structure sign on the second face is retained. The actual product reflection is implemented by the graded HS conjugation; the even reflected form is a positive HS Gram form, with the fixed twist for odd words.

The geometric field time and source time have the same parameter. The positive excitation generator is

$$E_{\text{exc}} = h|_{Q_+} \oplus (-\overline{h|_{Q_-}}), \qquad H = d\Gamma(E_{\text{exc}}) \geq 0.$$

The cyclic HS implementation has generator $L_H - R_H$, not $H$. For each finite compatible regulator the whole oriented kernel is

$$Q_{\text{raw}}(s) = \bigoplus_n \Lambda^n e^{-sh}, \qquad Q_{\text{heat}}(s) = e^{sE_{\text{sea}}} Q_{\text{raw}}(s) = e^{-sH}, \quad E_{\text{sea}} = \operatorname{Tr}(\mathbf{1}_{h<0} h),$$

$$(\eta | Q_{\text{heat}}(s) | \xi) = e^{sE_{\text{sea}}} \exp(\bar{\eta} e^{-sh} \xi).$$

All entries of the raw exterior lift are the corresponding minors. In infinite dimension one uses $e^{-sH}$ intrinsically; no separate infinite sea scalar is defined. The original scalar, reference and eta factors travel with the complete source. A general interacting caller is not declared to differ from this Gaussian fiber by a source-independent scalar.

There is also an exact obstruction to an arbitrary unitary whole-kernel identification with a separated flat DET16 factor. In the full unprojected source, heat summability gives, for $\Re z > 0$,

$$Z_{\text{free}}(z) = 2^m \prod_{\epsilon_j > 0} (1 + e^{-z\epsilon_j}) \neq 0.$$

The logarithmic product converges normally. A literal rank-one ground factor of dimension $d > 2$ and gap $\Delta > 0$ would imply

$$Z_{\text{free}}(z) = Z_R(z)\,[1 + (d-1)e^{-\Delta z}],$$

with holomorphic finite $Z_R$ on that half-plane. The bracket vanishes at $z_* = (\log(d-1) + i\pi)/\Delta$, a contradiction. An exact positive injective step suffices by its unique selfadjoint logarithm, up to a real energy offset for normalized steps.

The comparison concerns the same oriented positive physical heat operator and sewing. It does not concern arbitrary HS face refactorizations, relative quotient traces or merely the presence of a parent term. Actual Gauss/parity projection can break zero-freeness: three equal free modes restricted to even parity have $Z = 1 + 3e^{-2\epsilon z}$. Dynamical boson integration and other interacting sources are outside the free premise. The case $d = 2$ is the ordinary one-mode exception.

## Larger charged sources and the remaining actual insertion (R109, R110)

The larger integral odd spinor vertices have their explicit original spatial vectors, cocycle and all sixteen $V_0$ time weights. Their separately supplied collective $V_c$ is not substituted for $V_0$. A concrete original vertex-mode anticommutator is nonzero between two distinct components, so those components are not directly independent elementary CAR. This does not exclude other derived smearings, nonlocal maps or effective CAR realizations.

The separate declared 64-CAR/60-CCR interaction bank has

$$H_{\text{nat}} = \Delta N_b + \sum_A (g b_A^* P_A + \bar{g} P_A^* b_A).$$

Its original tensor gives $H_{\mathrm{nat}}[F,0]=0$ as a rest operator, but its sixteenth power has a nonzero complete environment block. The actual normalized witness has coefficient $16!\,8!\,\bar{g}^{16}$, and its leading Euclidean Dyson contribution is

$$8!\,\bar{g}^{16}\left(\frac{1-e^{-\Delta\tau}}{\Delta}\right)^{16}.$$

This is the first coupling coefficient with exact $\Delta$ dependence, not the complete nonperturbative propagator. The other 48 fermions and 60 bosons are retained. The compensated open transition and a vanishing single charged expectation under an invariant full weight are compatible. Neither is silently relabeled as a primitive scalar DET term.

## Dated continuation: the actual residual time form (R111)

The literal original positive-parent clause leaves its residual $R_0 = H_{\mathrm{tr,pos}}$ unspecified. Once the parent and the actual reducing physical/source map have been supplied, the original annihilating admissibility penalties vanish on their admitted range. The remaining binding equation is the whole source-time form minus the represented parent form. If one realization of the printed parent clause exists, its stated positive, common-vacuum and locality conditions also admit

$$R_s = R_0 + sH_{\mathrm{par}}, \qquad s \geq 0.$$

This comparison changes excited gaps and a fixed zero-time word's time derivative while retaining the same vacuum. It is an identifiability result about those parent hypotheses. It is not a family of different generators for one fully given source: that source's full time jet fixes the actual residual.

R111 also evaluates a marked response of the original normalized $C_6$ Green operator. For its declared $a > 2b > 0$, the three positive excitation rates are $-\log r_1, -\log r_2, -\log r_3$, where

$$r_1 = \frac{a-2b}{a-b}, \qquad r_2 = \frac{a-2b}{a+b}, \qquad r_3 = \frac{a-2b}{a+2b}.$$

The actual diagonal label probe has response $(2r_1^k + 2r_2^k + r_3^k)/36$ and a strictly positive two-step Hankel defect, so it is not a flat single-rate penalty. This uses the given internal transfer time; the probe is not identified as a physical field insertion or a four-dimensional mass/momentum measurement. The primitive joint spectrum remains uncomputed.

**Current physical status**

The common original source architecture and the calculated conditional field maps are available. The actual spatial mirror-field insertion, admissibility action and full fermion/Berezin call with dynamical bosonic weights must be identified on that same source/time/domain map. Its primitive coefficient $H_{\mathrm{adm}}[F,0]$ remains `not_computed`; a recorded JSON `null` is not the number zero.
The direct zero block of the given $H_{\mathrm{nat}}$, its nonzero sixteenth-order environment transition and this uncomputed primitive quantity are three different statements. Exact DET16 is a candidate comparison, not a universal physical axiom. A different source-derived viable mechanism must be assessed on its own controlled IR route. The complete physical TFPT and chiral-mirror claims are not closed by this update.

The four-mark geometry, $E_8$/clock structure and existing Yukawa/flavor/alpha readouts retain their original statuses and assumptions. Shared input data give correlated consequences, not additional independent empirical confirmations. R110 consolidates the source and provenance record; a technical inventory/hash audit is not a new semantic proof of every finding, a fresh full Python/Lean/Wolfram run or physical validation.

## Covariant action and its variational scope (R112)

On the existing conditional PS branch, the actual two-sided unitary orbit $U = uJuJ^{-1}$ forces the quadratic fluctuation with coefficient one in the known CCvS construction:

$$UDU^* = D + A_1 + \widetilde{A}_1 + A_2, \qquad A_2 = \widehat{u}[A_1, \widehat{u}^*], \quad \widehat{u} = JuJ^{-1}.$$

For the original right-quaternion subgroup its only nonzero correction is the Majorana block $(ww^T) \otimes M_R$ and its adjoint, where $w = q_R e_\nu - e_\nu$. The full original 96-dimensional operator, including all 72 real Yukawa and 12 real symmetric Majorana directions, is retained. The corrected operator is on the same unitary orbit; its full relative spectral action is invariant when reference, selectors, domains and eta data travel under the same admissible $U$. The SM first-order control has $A_2 = 0$. This fixes the existing orbit correction, not all invariant additions away from that orbit or a new physical PS gauge theory.

On the separate existing Clifford field chart, the original Green pairing gives the skew kernel $M_b$. If its *full off-shell* Euler operator is $M_b\Psi$, inverse variation fixes

$$S_F = \tfrac{1}{2}\Psi^T M_b \Psi + S_{\text{boundary}}, \qquad \delta_b S_F = \tfrac{1}{2}\Psi^T(\delta_b M_b)\Psi + \delta_b S_{\text{boundary}}.$$

The given open polarization recovers the whole original boundary transfer. The unintegrated zero-fermion source fixes the remaining bosonic functional if it is identified on that same chart. A stronger linear Berezin source identity likewise forces this conditional quadratic action; it is not supplied by bosonic spectral stationarity. Integrating the actual dynamical bosons retains their full cumulants and induced higher fermion vertices. Neither inverse variation nor the PS orbit identifies the entire primitive physical $S_{\text{adm}}$.

## The actual product pairing and the Craig comparison (CRAIG20261103B)

For the existing ordinary four-dimensional product branch with the whole original internal matrix $\Phi$, the total spin/finite real structure gives an antisymmetric Grassmann kernel on each total chiral fiber. Its mass block satisfies

$$M_{\text{mass},\epsilon}^T = -M_{\text{mass},\epsilon}, \qquad \|M_{\text{mass},\epsilon}\|_F^2 = 2\,\mathrm{Tr}_{96}\,\Phi^2, \quad \epsilon = \pm 1.$$

It is nonzero for nonzero Hermitian $\Phi$. The isolated finite KO6 pairing cannot be substituted for this full product pairing. At the specified free massless Weyl momentum $p = (\pi/2, 0, 0, 0)$ and lattice spacing one, however, the product kernel has singular values $\pi/2$, where Craig's allowed Weyl kernel has $\sqrt{2}$. Their squared defect $\pi^2/4 - 2$ excludes the direct fixed identification and any unitary basis repair of it. A derived nonunitary blocking, another regulator or a controlled continuum comparison remains outside this exclusion. The allowed Cayley fiber map also has its nontrivial Berezin Jacobian; all source legs, higher vertices, boson weights and reference phases must travel with a genuine source map. The actual Hessian of the full original $S_{\text{adm}}$ has not been evaluated by this product calculation.

## A joint Einstein and vacuum test (CONST20261103)

Take the explicitly conditional ordinary four-dimensional Dirac product with constant selfadjoint internal mass $M$, an equal-rank reference on the same geometry, and the single heat profile $F(u) = m_0 e^{-u}$, $m_0 > 0$. Assume its classical relative bulk trace alone supplies the vacuum and Einstein coefficients. With

$$T = \mathrm{Tr}_N\, e^{-M^2/\chi^2} - \mathrm{Tr}_N\, e^{-M_{\text{ref}}^2/\chi^2}, \qquad -N \le T \le N,$$

the complete constant-mass dependence gives

$$V = \frac{m_0 \chi^4 T}{4\pi^2}, \qquad \overline{M}^2 = \frac{m_0 \chi^2 T}{24\pi^2},$$

$$\frac{V}{\overline{M}^4} = \frac{144\pi^2}{m_0 T} \ge \frac{144\pi^2}{m_0 N} \quad (T > 0).$$

For $N = 96$, $m_0 = 1$, the bound is approximately 14.8044, incompatible with the existing dimensionless vacuum target of order $10^{-120}$. A vanishing $T$ also removes the Einstein term. No

constant Yukawa, Majorana or scale choice repairs this class. Other profiles can separate the two relative responses; dynamical bosons, quantum contributions and boundary terms are not excluded. The actual common source must produce its volume, curvature and gauge responses together with the same bound reference. The existing alpha/vacuum relation and its horizon rewrite remain conditional correlated readouts.

## Internal relaxation and physical cyclic time (GENESIS20261103)

The original family transfer has eigenvalues $1, 2/3, 1/3$. It relaxes in its two nonstationary directions and has no positive integer power equal to the identity. Under the additional identification $H_M = -\log M/\tau$, $\tau > 0$, the two positive energies are $\log(3/2)/\tau$ and $\log 3/\tau$. Their ratio is irrational by unique prime factorization, so this conditional Hamiltonian has no common exact real-time period. This is not the evaluated physical TFPT Hamiltonian. The external cyclic-universe proposal assumes commensurate energy differences and a low-entropy state; it supplies no original TFPT source or geometric collapse law. Physical recurrence requires the actual occupied energy differences to satisfy $(E_j - E_k)T \in 2\pi\mathbb{Z}$, together with geometry from that same source. Exact recurrence of the whole state also repeats every internal record. No general exclusion of cyclic TFPT cosmology follows.

**Whole color projection (CRAIG20261103C).** On the compatible existing CAR representation, average the three whole exterior lifts of the original color-center action. Its neutral sector has rank 21856 on the full sixteen-mode exterior space; the exterior lift of the neutral one-particle projector has rank only 16. The whole coherent transfer kernel is

$$K_{c,b}(\bar{v}, u) = \frac{1}{3}\sum_{k=0}^{2} \exp(\bar{v} z^k R_b u).$$

It retains neutral composite words with charged intermediate modes. Its higher boundary coefficients are selector effects, not newly derived primitive bulk vertices. The actual Craig sector sources must retain their respective operators, Weyl spaces and measures: averaging their operators or projectors fails the original GW and projector identities. This color-center projection is not the full SU(3) singlet projection and does not identify the other original admissibility selectors or the whole primitive quantum source.

**Joint marked source and selector contacts (R113).** For the same conditional whole Clifford caller, assume its actual semigroup time, justified short-time product rule, compatible finite CAR factor and Hermitian boson/rest coefficient operators. The resulting joint generator is

$$H = H_B \otimes I + \sum_{j,i} V_{ji} \otimes c_j^\dagger c_i.$$

Here each $V_{ji}$ is the original zero-time marked operator $Q_0[h_{ji}]$, not a normalized mean. The original Fock-vacuum matrix leg fixes $H_B$; a traced partition function does not replace that leg. The whole selector remains outside the boson integral, with its exact commutator contacts and allowed charged intermediate words. A nonreducing selector retains complementary-sector return memory; its compressed transfer cannot be replaced by evolution under $PHP$. Unbounded extensions need the actual common domains and strong reduction. Regular gauge/BRST directions preserve the fermion-coordinate ideal and therefore do not supply full Grassmann translations. These conditional consequences neither identify the primitive physical quantum source nor evaluate its joint spectrum.

## Source and reproducibility route

The dated report labels resolve through `docs/research/fundamental-plan/RESULTS.md` and the synthesis `docs/research/fundamental-plan/CONSOLIDATION-R110-20261103.md`. The main proof owners are:

- A and B: `experiments/theory-contracts/puncture-metric-joint-completion-20261102/PROOF.txt` and `experiments/theory-contracts/joint-source-clock-field-closure-20261102/PROOF.txt`.
- C: `experiments/theory-contracts/original-paper-source-reconciliation-20261102/AUDIT.txt`.
- D–L and R104–R111: `experiments/theory-contracts/original-local-source-variable-background-20261102/`. The exact theorem and scope owners include `VARIABLE-BACKGROUND-PROOF.txt`, `PAIRED-DYSON-PROOF.txt`, `FULL-SOURCE-CORRECTION.txt`, `REFERENCE-SOURCE-COMPLETION.txt`, `PHYSICAL-THERMAL-SOURCE.txt`, `SHARED-METRIC-SOURCE.txt`, `GROUND-SOURCE.txt`, `TRANSFER-EXTRACTION.txt`, `SEED-FIELD.txt`, `ZERO-FREE-JOINT.txt`, `ZERO-FREE-ORIGIN.txt`, `LARGER-CHARGED-SOURCE-EVALUATION.txt` and `SOURCE-GENERATOR-RESIDUAL-20261103.txt`.
- R112: `experiments/theory-contracts/original-local-source-variable-background-20261102/ADMISSIBLE-ACTION-DERIVATION-20261103.txt`.
- CRAIG20261103B: `experiments/theory-contracts/craig-original-source-bridge-20261103/ACTUAL-PRODUCT-PAIRING.txt`.
- CONST20261103: `experiments/theory-contracts/constant-triangulation-20261103/CONSTANT-TRIANGULATION.txt`.
- CRAIG20261103C: `experiments/theory-contracts/craig-original-source-bridge-20261103/COLOR-PROJECTED-KERNEL.txt`.
- R113: `experiments/theory-contracts/original-local-source-variable-background-20261102/PROJECTED-JOINT-SOURCE-GENERATOR-20261103.txt`.
- GENESIS20261103: `experiments/theory-contracts/cyclic-genesis-literature-20261103/GENESIS-CYCLIC.txt`.

The original registered constructors include:

- [`verification/v252_full_finite_triple.py`]
- [`verification/v1002_spin10_mirror_projector.py`]
- [`verification/v1027_signed_det_car_wall.py`]

Their existing scope is preserved. The source reports distinguish executed native controls from analytic arguments, given inputs and uncomputed primitive insertions. No new verification-module or ledger promotion is implied by citing them in this shared update.

## Appendix: computational verification

The load-bearing identities of this document (marked **[E]**, **[E]**, **[E]**) are re-derived from $\{c_3, g_{\text{car}}\}$ alone and machine-checked by the suite in the repository folder `verification/`; the inline tags `[verification/...]` point to the exact script. Relevant here:

- `v1_e8_glue.py` — the $\mu_4$ glue ($q(D_5)+q(A_3)=2$, disc $=\mathbb{Z}_4$, $240, 248$).
- `v2_carrier_pascal.py` — $g_{\text{car}}=5$, $16=1+5+10$, $N_{\text{fam}}, \Omega_{\text{adm}}, b_1$.
- `v3_em_alpha.py` — the EM fixed point $\alpha_\star^{-1}=137.0359992168$ (existence/uniqueness, ablation).
- `v4_flavor_matrix.py` — $R$ with $\det=8$, minors $\{2,3,5\}$, $\sum L=40$.
- `v7_gravity_cosmo.py` — scalaron $c_3^7$, $\Lambda/A_s$ readouts.

Run `python verification/run_all.py` (needs `mpmath, numpy, sympy`); it exits 0 iff all checks pass. Full map: `verification/README.md`.

## External works built upon

The TFPT discrete core (the two axioms, the $E_8$ glue, the integer skeleton) is self-contained and machine-checked. Its *bridges* to continuum/AQFT physics, to gravity and to the lattice/singularity picture build on established mathematics, listed here by topic; the body refers to these by author name.

**Modular theory and algebraic quantum field theory.**

- M. Takesaki, *Tomita's Theory of Modular Hilbert Algebras and its Applications*, Lecture Notes in Math. **128**, Springer (1970).
- J. J. Bisognano, E. H. Wichmann, *On the duality condition for a Hermitian scalar field / for quantum fields*, J. Math. Phys. **16** (1975) 985–1007; **17** (1976) 303–321.
- K. Osterwalder, R. Schrader, *Axioms for Euclidean Green's functions* I, II, Comm. Math. Phys. **31** (1973) 83–112; **42** (1975) 281–305.
- S. Doplicher, R. Haag, J. E. Roberts, *Local observables and particle statistics* I, II, Comm. Math. Phys. **23** (1971) 199–230; **35** (1974) 49–85.
- R. Haag, *Quantum field theories with composite particles and asymptotic conditions*, Phys. Rev. **112** (1958) 669–673; D. Ruelle, *On the asymptotic condition in quantum field theory*, Helv. Phys. Acta **35** (1962) 147–163.
- R. Longo, *Index of subfactors and statistics of quantum fields* I, II, Comm. Math. Phys. **126** (1989) 217–247; **130** (1990) 285–309.
- Y. Kawahigashi, R. Longo, M. Müger, *Multi-interval subfactors and modularity of representations in conformal field theory*, Comm. Math. Phys. **219** (2001) 631–669.


### Conformal nets, vertex operator algebras and lattice constructions.


- I. B. Frenkel, V. G. Kac, *Basic representations of affine Lie algebras and dual resonance models*, Invent. Math. **62** (1980) 23–66; G. Segal, *Unitary representations of some infinite-dimensional groups*, Comm. Math. Phys. **80** (1981) 301–342.
- T. J. Osborne, A. Stottmeister, *Conformal field theory from lattice fermions*, Comm. Math. Phys. **398** (2023) 219–289, `arXiv:2107.13834`, `doi:10.1007/s00220-022-04521-8`. (This is the seam scaling-limit theorem the TFPT continuum leg cites — the Koo–Saleur → Virasoro convergence, $c = \frac{1}{2}$, and the smeared WZW-current Theorem 5.1. The internal TFPT code-token "MMST" and the legacy claim-ids `SEAM.MMST.*` / filenames `v*_mmst_*` denote *this* Osborne–Stottmeister theorem.)
- V. Morinelli, G. Morsella, A. Stottmeister, Y. Tanimoto, *Scaling limits of lattice quantum fields by wavelets*, Comm. Math. Phys. **387** (2021) 299–360, `arXiv:2010.11121`, `doi:10.1007/s00220-021-04152-5`; and *Operator-algebraic renormalization and wavelets*, Phys. Rev. Lett. **127** (2021) 230601. (The operator-algebraic (wavelet/Wilson–Kadanoff) renormalization *framework* the Osborne–Stottmeister theorem above builds on.)
- M. S. Adamo, Y. Moriwaki, Y. Tanimoto, *Osterwalder–Schrader axioms for unitary full vertex operator algebras*, `arXiv:2407.18222` (2024).
- M. S. Adamo, L. Giorgetti, Y. Tanimoto, *Rational and non-rational two-dimensional conformal field theories arising from lattices*, `arXiv:2506.01008` (2025); *Representations of conformal nets associated with infinite-dimensional groups*, `arXiv:2508.07109` (2025).
- P. Naaijkens, D. Penneys, B. Wallick, *(commuting-projector reflection positivity / LTO-RP)*, `arXiv:2605.10693`.


### Lattices and quadratic forms.


- J. H. Conway, N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd ed., Grundlehren der math. Wiss. **290**, Springer (1999).
- C. L. Siegel, *Über die analytische Theorie der quadratischen Formen* (the Minkowski–Siegel mass formula), Ann. of Math. **36** (1935) 527–606.


### Singularities and geometry.


- E. Brieskorn, *Beispiele zur Differentialtopologie von Singularitäten*, Invent. Math. **2** (1966) 1–14.
- J. Milnor, *Singular Points of Complex Hypersurfaces*, Ann. of Math. Studies **61**, Princeton Univ. Press (1968).


### Spectral geometry, gravity and thermal time.


- A. H. Chamseddine, A. Connes, *The spectral action principle*, Comm. Math. Phys. **186** (1997) 731–750.
- A. Connes, C. Rovelli, *Von Neumann algebra automorphisms and time–thermodynamics relation in general covariant quantum theories*, Class. Quantum Grav. **11** (1994) 2899–2918.
- R. M. Wald, *Black hole entropy is the Noether charge*, Phys. Rev. D **48** (1993) R3427–R3431.

- H. Casini, M. Huerta, R. C. Myers, *Towards a derivation of holographic entanglement entropy*, JHEP **05** (2011) 036, `arXiv:1102.0440`.